[m*nds*ght]

An Interactive Aural Adventure

"Can binaural recording techniques be employed effectively to create a fully immersive, and interactive audio-only entertainment experience?"

November 2009
BA Recording Arts
RA303: Research Project
Luke Charman
Student 12992, Class AD1108.2
SAE Institute: London

Word Count: 14,189

Abstract

MiNDSiGHT is an interactive audiobook created for the Apple iPhone platform, which uses binaural dummy head recordings to tell an immersive, reactive, and absorbing narrative in full 3D through any pair of stereo headphones. The game features professional actors, a gripping script, puzzles based entirely within audio, and a novel, exciting experience.

The title has been designed to allow players to rely solely upon their ears, and the game's implementation means it is fully playable by both visually-impaired and fully sightless people.

This accompanying documentation details the conception, development, and implementation of the software itself, as well as thorough examinations into the technicalities, intricacies, and limitations of binaural audio, the psycho-acoustic methods by which sounds are localised, and an analysis into means by which emotion can be captured in an aural drama narrative.

Hands-on research into binaural recording was conducted at length when recording the game's scenes with the use of binaural microphones including a Neumann KU-100, arguably the finest such device available in the world today, and ear-mounted clip-on microphones to allow the player to experience lead character Ethan Lewis' world exactly as captured.

Upon evaluation, it became apparent that MiNDSiGHT provides an excellent gateway to providing binaural audio in commercial gaming titles to enhance their immersion, depth, and ability to emote. Believable characters, quality voice acting, and solid programming helped make MiNDSiGHT a success as an exploration, and helped prove the interactive audiobook a format with a stimulating future.

Table of Contents

ABSTRACT II

TABLE OF CONTENTS III

PREFACE V

ACKNOWLEDGEMENTS VI

THE PERILS OF PRIMARILY PICTORIAL PLAY 1

Portraying the Problem 2

Introduction 2
Investigating Immersivity in Audio 4
Demystifying the Multichannel Method 6
"These Ears Were Made For Listening" 8

Staging the Solution 10

Enter Immersion – Using Binaural Audio 10
Attracting an Audience – Absorbing Narratives 11
Defining Dimensions – 'Interactive Audiobooks' 13

LET YOUR EARS BE YOUR GUIDE 15

Directional Hearing 16

Head Related Transfer Function 17
Psycho-Acoustics & Localisation 20
Vertical & Extreme Localisation 21
Horizontal & Frequency-Based Localisation 23
Horizontal Localisation ..... **Error! Bookmark not defined.**
Front / Back Confusion 26

REPRODUCING REALISM 28

Binaural or Bust 29

Nature Vs. Nurture in Audio Engineering 30
The Neumann KU-100 30
First-Person Binaural Recording 33
Starkey Labs & The Cetera Algorithm 36
The Power of the Plug-In 38

DARKNESS, CAMERA, ACTION! 40

All the World's a Stage 41

Actors & Characters 41

Theatrical Conventions 44
Environments & Foley 45
Minimal Music 46
ATTESTED DEVELOPMENT 48
Making Your MiND Up 49
Apple's iPhone Platform 50
iPhone Implementation with RjDj 51
PureData Programming 53
Procedural Programming / Decisions 54
MINDSIGHT IN HINDSIGHT 60
Evaluating MiNDSiGHT 61
Comparisons and Competitors 62
Comparison A: Radio Dramas 62
Comparison B: iPhone Applications 63
The Opinions of Others 65
Test A: The Gamer Test 65
Test B: The Visually-Impaired Test 67
Evaluating the Final Product 69
Target A: Immersion 69
Target B: Emotion 70
Target C: Portability 71
Final Thoughts 72
MINDSIGHT SCRIPT 63
Introduction to the Script 64
MiNDSiGHT Script 65
FLOOR PLANS AND PROP LISTS 118
Floor Plans 119
Prop Spreadsheet 128
Commented Code 132
Digital Documentation 141
BIBLIOGRAPHY 142

Preface

The history of audio within interactive entertainment and computer games has constantly and persistently taken the silver medal, left to pale in comparison to the ongoing desire for photo-realistic graphics. Today's modern high-definition gaming consoles such as the Xbox 360 and Playstation 3, clamour to outstrip one another of the glory of graphical dominance, while leaving their powerful audio processors firmly in the back seat.[1] The aim of this study, and the creation of MiNDSiGHT, attempts to reclaim some lost ground for aurality; to prove that one's ears can be as powerful as one's eyes in receiving and processing stimuli which evoke fear, emotion, or immersion.

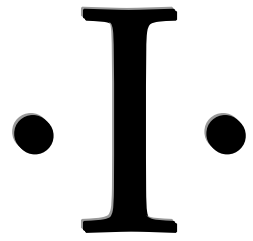
While the project is largely practical in nature, the scope of the research conducted into theoretical areas is broad, and covers both the topics of human audiology and psycho-acoustics, as well as analysing methods of reproducing reality in stereophony, and research into the history and operation of radio drama, from which MiNDSiGHT draws inspiration.

[1] Microsoft's Xbox 360 and Sony's Playstation 3 consoles both allow for the use of Dolby Digital surround sound, and offer 256 and 512 discrete audio channels respectively. However, in the announcement press releases for both consoles, upgrades to the audio processors are entirely unmentioned (TeamXbox.com, 2005, and Gamespot.com, 2005).

Acknowledgements

The author would like to thank the following people for their assistance, technical expertise, patience, and kind hearts, and for devoting their spare time to assisting with the creation of MiNDSiGHT:

- **Andy Farnell**, for acting as tutor over the project, firmly pointing the project towards the iPhone platform, and keeping my head from disappearing completely into the clouds.
- **Sally Shepard**, for graciously allowing me to make extensive use of her Neumann KU-100 binaural dummy head, and for proving that Rhode Island is known for more than just Family Guy.
- **Oliver Tipper**, for his unending patience, engineering expertise, and watchful eye on the VU meters in recording sessions.
- **Sophie Shelton**, for her invaluable sound designer's perspective on ambience and foley.
- My wonderful cast of characters: **Gemma Parker**, **Martin Horton**, **Razif Hashim**, **Matt Butchers**, and **Stacy Winter**, for their constant optimism, colourful re-writes, and endless patience in recording sessions.
- **Brooke Garner**, for bringing script-writing expertise to the process to ensure MiNDSiGHT spins an entertaining and emotional yarn.
- And, **Lauren Williams**, for keeping my brain running in the right direction throughout the entire process.



The Perils of Primarily Pictorial Play

An Introduction to MiNDSiGHT

Portraying the Problem

This initial section details the preliminary decisions that brought about the creation of MiNDSiGHT, as well as three distinct initial investigations into the possibility, and associated advantages and disadvantages, of providing both emotion and immersion through exclusively aural media, differing methods of immersing a listener in an aural world through multi-channel techniques including both binaural audio and more conventional 5.1 and 7.1 surround sound arrangements, and some of the differences found in storytelling through visual, aural, or written media formats.

Introduction

MiNDSiGHT's journey began on a colourless evening in April 2009, after a marathon session conquering both the lion's share of a bottle of Jack Daniels, and conquering the Dragon's Lair on an antiquated Sega Mega CD gaming system. Don Bluth's revolutionary interactive fantasy storybook incorporated hundreds of pre-recorded scenes, which the player navigated using both quick wits and even quicker button-presses on a directional keypad. The game took arcades by storm upon its release in the early 1980s, and since has been re-released over sixty times – a testament to the game's superior script, dialogue, gameplay, and level of immersivity. The game's introductory sequence proudly calls:

> *"Dragon's Lair: The fantasy adventure where you become a valiant knight, on a quest to rescue the fair princess from the clutches of an evil dragon. You control the actions of a daring adventurer, finding his way through the castle of a dark wizard, who has enchanted it with*

> *treacherous monsters and obstacles. In the mysterious caverns below the castle, your odyssey continues against the awesome forces that oppose your efforts to reach the Dragon's Lair. Lead on, adventurer. Your quest awaits!"*
>
> ***Dragon's Lair (Don Bluth, 1983)***

In order to conquer the Dragon's Lair, the player's only interactions were the four compass directions, up, down, left, and right, and a single function button which operated lead character Dirk the Daring's sword. Every action in the game needed to be neatly choreographed, and all potential outcomes to be catered for. The game's program code was remarkably simple, is able to happily run on a standard DVD player today, and consisted of a video player and an elaborate decision tree, that decided which video clips to play (and in which order) based on the button-presses made by the user.

The game was, in the simplest terms, an audio-visual re-imagining of the 'Choose Your Own Adventure' books published by Bantam Books and (more recently) ChooseCo since 1979, more than 250 million of which have been printed (Lodge, 2007). The marriage of an incredibly popular semi-interactive concept, and a burgeoning video-gaming craze propelled Dragon's Lair into the spotlight, engulfing the pocket money of unsuspecting arcade patrons throughout the 1980s.

> *"Dragon's Lair is this summer's hottest new toy: the first arcade game in the United States with a movie-quality image to go along with the action... The game has been devouring kids' coins at top speed since it appeared early in July. Said Robert Romano, 10, who waited all day in the crush at Castle Park without getting to play, "It's the most awesome game I've ever seen in my life."*
>
> ***Newsweek Magazine (August 8, 1983)***

The question of how to tie this fantastic concept to an audiological base was a simple one to answer: the creation of an interactive audiobook utilising the exact same technology that ran Dragon's Lair in 1983, some twenty-five years earlier. Removing the visual component from the mix ensures that the player's ears are tested to their absolute limits, and provides endless possibilities for both the idiosyncratic audio cues and emotive voice acting of radio dramas, as well as portability, since an audio-only product would run on a handheld device, negating the need for a screen. The potential for planting the seed of the narrative into the player's ears and allowing them to visualise their own adventure seemed vast, and would certainly prove the inherent power human ears possess to trigger any emotion, and with an equal level of power as any visual stimulus, provoke that emotion to its fullest and most immersive.

Investigating Immersivity in Audio

Radio dramas, at least in Britain, trace their roots back to the 1920s, and to Captain PP Eckersley, the first Chief Engineer of the BBC, who pioneered the concepts of radio drama: actor placement, audio props, and emotive delivery of dialogue designed to promote the appropriate balance between anonymity and personification.

> *"We did a wireless play. We chose the balcony scene from Cyrano: it is played, on the stage, in semi-darkness with virtually stationary players and so it seemed very suitable for broadcasting. [...] We sat round a kitchen table in the middle of the wooden hut, with its shelves and benches packed with prosaic apparatus, and said our passionate lines into the lip of our separate microphones."*
>
> ***'The Power Behind The Microphone' (PP Eckersley, 1941)***

Eckersley's ambitious and uncertain gamble paid off, and now, almost a century later, only slight revisions to his methodology have been made. The recording of a modern radio drama consists of the same methods, simply modernised – assembled actors around individual microphones, foley effects to provide convincing background to the dialogue, and a complete reliance on aurality to convey absolutely every emotion the drama carried, from fear, through humour, to devastation or elation. Eckersley often compared his "listening plays" to silent films, which were enjoying popularity at the time, noting that their "silence" is in fact a lie, since they rely on audio such as music and externally placed sound effects to augment the moving pictures.

> *"'Those were the days of the silent film', he said, 'and our "listening play" (as I dubbed it) would have to be the silent film's missing half, so to speak, telling a complete story by sound alone. Yet even the silent film did not, strictly speaking, rely on pictures only. [...] No, we must rely on dramatic speech and sound entirely - and it had never been done before."*
>
> ***'The Power Behind The Microphone' (PP Eckersley, 1941)***

The concept of relying solely on one of the five senses to convey emotion was, at least in the realms of story-telling, unheard of at this point. Renaissance composers may have delighted and astounded ears centuries before, but until the 1920s, drama had been purely a mixed media – reliant on stimulating the eyes, and the ears simultaneously. The removal of what is arguably mankind's most relied-upon sense presented (and continues to present) huge hurdles to those wishing to convey emotion or immersion directly into the ears. Stecker (2005) theorised that works "are *expressive* of sadness, *expressions* of sadness, or simply that they *express* sadness", and continued to carefully define differences between the varying conjugations

of the word[2] – which emphasises the necessity for subtlety and nuance, and the dangers of over-compensating for the listener's removed sense of sight. Nothing destroys a radio drama more than an ostentatious rustling of a doorknob, or a clichéd series of over-trodden footsteps inappropriately punctuating the drama. The art of good audio storytelling is the art of subtlety – that is, the art of letting the story tell itself, and allowing the listener to create their own accompanying visual imagery, much in the same way as they would when reading written fiction. This art is something which was constantly considered and referenced during the recording and mixing of MiNDSiGHT.

Demystifying the Multichannel Method

Microsoft's Xbox 360, Sony's Playstation 3, Nintendo's Wii, and virtually every home computer sound card manufacturer one can imagine in 2009 utilise multi-channel audio[3]. All formats of multi-dimensional film audio are also multi-channel. Radio, on the other hand, is still exclusively a stereo medium, and since the inception of DAB, this is no longer a technological limitation. Since music is still conventionally not mixed in anything more than two-channel stereo, radio dramas must still pay the price of being limited to two discrete channels.

---

[2] Stecker's book "Aesthetics and the Philosophy of Art: An Introduction" contains numerous invaluable insights into means by which emotion can be conveyed aurally, with varying degrees of directness. It was referenced numerous times when directing actors, and helped shape MiNDSiGHT's audiological direction.

[3] Microsoft's Xbox 360 is capable of outputting Dolby Digital 5.1 surround sound, Sony's Playstation 3 increases that to DTS 7.1 audio, and even Nintendo's Wii, a less powerful machine, is capable of upmixing to 5.1 via Dolby Pro Logic II (Marks and Novak, 2009, p.25).

Multi-channel audio, particularly today's 5.1 and 7.1 setups, are complex systems which require positioning with precision in well-arranged and acoustically-optimised environments to obtain optimal performance. These arrangements require the user to invest in expensive equipment, and to remain stationary while listening – two sizeable disadvantages which even to this day prevent wider adoption of such systems in radio and music production. MiNDSiGHT was always intended to be a portable experience – inspired by the unending daily commute, and while great advances have been made in recent years in the realms of creating 5.1 headsets,[4] the simple fact remains: humans have two ears, and absolutely all of our directional hearing is performed in relation to those two ears. The need to provide sounds from four, five, seven, or three-hundred-and-sixty different locations can be entirely negated with the use of one deviously simple technology which negates the need for any more than one speaker positioned in close proximity to each ear, and also negates the need for complex and time-consuming multichannel mixing: *binaural dummy head recording*.

MiNDSiGHT's entire ethos as an interactive experience is based around every piece of audio it uses being recorded in binaural stereo. Every bullet flying towards the player, every passer-by on a street, every ringing telephone, and every line of dialogue has been delivered into binaural microphones, then carefully mixed in nothing more than humble two-channel stereo to ensure that no matter where the game is played, provided the player is making use of an adequate pair of stereo headphones, they can

[4] Michael Aulia's review of Australian firm Cybersport's Cyber Snipa SONAR 5.1 headset (Aulia, 2009), and the promotional websites for Saitek's Cyborg 5.1 headset (Saitek, 2009) promise that such headsets are capable of delivering realistic simulations of 5.1 audio via six separate drivers to represent the six aspects of a conventional 5.1 setup.

close their eyes and place themselves firmly into Ethan Lewis' world, feeling his pressures as bullets fly at his head, and the words of those who inhabit his world filling their ears with an unprecedented level of realism in interactive entertainment.

"These Ears Were Made For Listening"

MiNDSiGHT has been designed from the ground up to provide aural stimulation – a true aural adventure for the listener. A primary goal of the game was, from its outset, to allow a player to finish the game blindfolded, using only their ears and their hands to interact with the game. Our ears are, in drama and interactive entertainment, too often used as a backup organ – designed to validate what our eyes have already seen, and how our bodies have already processed that information. A case in point: Activision's phenomenon "Guitar Hero". Masquerading as a game where your ears are given some task to perform, the game is, in actuality, played entirely by looking at coloured note indicators scrolling down the play-screen, and strumming the appropriately-coloured note on the included guitar peripheral – the game allows you to play the guitar without needing to use your ears at all. The game's aural element, using the played notes to allow the correct reproduction of famous songs, is simply a response to the visual stimulus. For a "music" game, Guitar Hero's music is surprisingly unimportant.

Interactive entertainment which *relies* upon the ears for success is a rarity, as is entertainment which requires information presented solely aurally to be processed by the brain. A player can effectively complete a song in Guitar

Hero, with their television on mute. Why do creators of entertainment neglect our ears in this way? Why are our ears not trusted to dictate to our minds the correct path to take, button to push, or enemy to fight? In a modern musical world where an artist's physical appearance dictates their success as much as (if not more than) their musical competence, the answer is clear: the clay from which our eyes are formed is much more malleable than the clay making up our ears.

However, that answer is one we should simply not accept; we as audio professionals, we as media consumers, and we as human beings. There is a reason a child's first word stays with its parent forever, and there are reasons why the famous speeches of Winston Churchill and Martin Luther King Jr. will forever resonate in our minds. Emotion can, arguably, be *heard* more deeply than it can be seen. Covering one's emotions with a sleek outer appearance is an all-too-common trend in modern political circles, but a wavering in someone's throat or an unbridled sense of optimism simply cannot be hidden. Our appearances and our words may be untrue, but hiding one's tone is a much more difficult process. Conveying emotion and immersion through sound were crucial aims of the MiNDSiGHT project, and without visual distraction, the game's audio was truly able to shine.

With these aims in mind, and the problems with conventional visual media and multi-channel audio systems laid out, the solution of creating an audio-only gaming experience which required absolutely no visual interaction from the player came naturally. This solution, MiNDSiGHT, would attempt to prove that the human ear is as invaluable as the human eye, and to justify the power of audio to the media-consuming public in a much grander way than ever before.

Staging the Solution

With the problems to conquer laid out, and the initial sketch of the solution, drawn, MiNDSiGHT began to take shape. This section details that process, detailing initial research into the binaural recording technique the game would utilise, convincing ways to present a script via purely aural means, defining the scope of the project as an interactive audiobook and declaring what MiNDSiGHT would, and more importantly would not attempt to do, and considering some more cultural implications of this new genre of interactive entertainment.

Enter Immersion – Using Binaural Audio

The most important question to be asked when undertaking any kind of audio project is: how will the audio be delivered? A mono mix, stereo, or multi-channel surround sound? Beyond technicality, will the audio be presented in the first person, and if not, where will the player be positioned in relation to the drama they are listening to?

The decision to use binaural recording and playback answered both of these questions. MiNDSiGHT is mixed in standard 2.0 stereo, and requires the player to use headphones to succeed in the game. In terms of positionality, since binaural audio delivers such a startlingly realistic and immersive aural environment, particularly to first-time listeners, it was quickly decided that the player would *become* the game's protagonist Ethan Lewis. His words would ring out as if they were the player's own, and those that inhabit his world would direct their words towards him as if he were directly involved

in the action. Telephone conversations would take place in one ear, just as one would hold a real mobile handset to their ear, explosions would ring out powerfully from every direction imaginable, and ambient sounds would all be recorded from this same perspective, allowing the player to shut their eyes and enter Ethan Lewis' world. Revisiting the key concept of immersive gameplay, this first person aural perspective provides far greater immersion and emotional connectivity than a third-person perspective ever could.

Attracting an Audience – Absorbing Narratives

The stage is set – MiNDSiGHT will, through technology and recording technique, place the player directly inside the game's world. Now – what would fill that stage? It was intended from the outset that the game would feature emotional content, and to have that content delivered by anything less than able actors would prove a detriment to the overall experience. It was for this reason that Masters students at the University of Essex's prestigious East 15 Acting School were interviewed, auditioned, and recruited to perform the script, which was written in collaboration with Brooke Garner, a freelance writer located in Brighton, England. External parties were recruited to ensure that the experience did not suffer at any corner – MiNDSiGHT needed to be gripping, on paper, and in-ear.

The success of the title hinged on delivering an immersive and emotive experience, and then effectively capturing that immersion and emotion binaurally. This meant that not only did the actors need to perform their roles with style, it also meant they needed to be completely briefed in the technicalities of binaural recording so that they would fully understand how

the ominous personified microphone staring at them was picking up more than just their words. They would also need to experience the technology before beginning to work with it, to ensure that they truly appreciated the method and were sure to follow the rules set out by Robert Stecker: do not over-act in radio drama. The power of binaural audio would prove immense, meaning that ambient sounds such as footsteps or breathing would require absolutely no additional force besides those one would normally exercise. The Neumann KU-100 microphone used has been said to "survey hapless musicians with all the inscrutability of a rather ghoulish critic" (Lawson, 2003) – inscrutability which multiplied by the fact that the head represented Ethan Lewis, meaning the supporting cast needed to effectively deliver their lines, with the utmost sincerity, to a chiselled lump of plastic and rubber, or to the actor portraying Ethan's chest as he held the dummy head in place during specific scenes.

The game's storyline was designed to feature a blind protagonist. This decision was made to ensure that the player's missing visual field would not encumber their immersion in the environment. A sighted player with their eyes closed, stepping into the shoes of a blind character, share a great many similarities. Just as the blind man must rely on his more finely-honed sense of hearing, so must the player adapt their hearing to trust it implicitly, rather than using it to confirm what their eyes have already witnessed. Crook (1998) states that radio dramas "have to work in the fifth dimension...the energy of the listener's imaginative participation." While the deeper meaning of this statement may prove elusive initially, the concept of a fifth dimension acting as a distilled sense of immersion was kept constantly in mind during the entire process of creating MiNDSiGHT.

Defining Dimensions – 'Interactive Audiobooks'

MiNDSiGHT is not a computer game. Classifying the title as an interactive audiobook allowed focus to remain on the narrative, and for the gameplay to become a response to, and a natural extension of it, rather than the other way around, as is the convention with the action-packed titles one may play on the Xbox 360 or Playstation 3. It was important to remember in the early stages, and throughout development and documentation, that the work was being completed for a Recording Arts qualification, not any kind of game design or development course. MiNDSiGHT is interactive, and audiological, but does not constitute a computer game in the conventional sense, despite obvious comparisons being drawn to one. However, this does not mean that analysis into the uses of audio in computer gaming cannot provide invaluable insight into methods of incorporating audio into interactive media.

Utilising the iPhone's accelerometer technology, all of MiNDSiGHT's user input is handled directly by pushing the device forwards, pulling back, or rolling left and right, much like a conventional flight simulator joystick (further details regarding the target platform can be found in Chapter V).

Since the iPhone's screen is not required for gameplay, and all input is handled via motion, MiNDSiGHT can be considered an EEE title (Entirely Eyeless Entertainment) – one's chance of successfully completing the story is at least equal with eyes open or closed. Furthermore, the advances of isolating one's sight and hearing has been demonstrated by researchers at Tel Aviv University, who discovered that brain activity is significantly higher when experiencing audio with the eyes closed (Hendler, 2009). Test subjects were quoted to have felt "more emotionally charged" with their

eyes closed, as their imaginations were allowed to run riot, and conjure up images far more immersive than the real world could offer.

> *"Small physical behaviours can radically alter the balance and colour of emotions. Not long ago in U.S. classrooms, teachers found that hyperactive students learned better while standing, rather than sitting at their desks. This latest study with scary music is "just an example of how a small manipulation in one's physical state such as eyes open or shut can change our mental experience,"."*
>
> ***Prof. Talma Hendler, 2009***

The game does not require any 'action buttons' or other input from the user besides these four basic directional cues – the movement of the iPhone handset allows them to make their in-game choices which will affect the outcome and guide Ethan Lewis through the narrative, and to physically pitch their device (and potentially, their bodies) in the directions required to navigate the world.

With the solution planned, detailed, researched, and organised, the practical elements of the application's design and development began in earnest. These are detailed at length in the relevant appendices, as well as Chapter V, which discusses the practical aspects of developing for the iPhone platform. In parallel to the recording sessions with the voiceover cast, focused research was conducted into aural biology, to ensure that the scenes were recorded with accurate reproduction in mind.

·II·

Let Your Ears Be Your Guide

Spacialisation and Localisation in Human Hearing

Directional Hearing

With the exception of the praying mantis,[5] virtually every vertebrate on Earth has two ears. This fact allows us to localise sounds in our environment – to determine exactly where they come from, with varying degrees of accuracy dependent on species. Møller (2006) states that binaural hearing "has had enormous importance for many species of vertebrate animals during evolution." – it is important to realise that while a helpful tool for humans, for creatures such as owls and bats, binaural and directional hearing forms their very livelihood – the hunt for food, and the evasion of predators.

Human localisation and spacialisation are crucial in positioning ourselves in our environment, and are arguably akin to our eyesight in terms of mental analysis of our surroundings. Remember the concept of MiNDSiGHT's aural presentation: *immersion*. Human ears provide total immersion – our eyes cannot see beyond a range of around 120 degrees without movement, whereas our ears can constantly provide us with a full 360 degrees of audio, with in-built object tracking, and more besides.

The entire concept of MiNDSiGHT was developed around our bodies' ability to process audio directionally. The scientific methods by which the head-related transfer function operates and allows humans to pinpoint sound sources in our environment were analysed in depth, and this research was considered throughout the recording and mixing processes, which are

[5] The mantis' single ear is tuned specifically to a specific ultrasonic frequency band, and is used to detect the echolocation of bats, thus negating the creature's need for positional hearing (Hess, 1971).

detailed in Chapter III. This chapter specifically discusses the technical and psycho-acoustic properties which allow human beings to place sounds in their aural field, and determine the location of sound sources through biological and scientific principles.

Head Related Transfer Function

Head-related transfer functions (HTRFs) compensate for the world around us. The remarkable technology built inside the human head, and more specifically built inside the ears, is capable of identifying our situation with startling accuracy. One need only examine the sweeping rise in adoption of surround sound and multi-channel audio beyond conventional stereo in television, film, and other modern media, to see exactly how much faith is being placed in our ears to position us inside fictitious worlds. While our limited visual field prohibits our eyes from seeing beyond the borders of the television screen without physically needing to turn our heads, our ears are all-encompassing, and provided a sound source is located within our audible range and is close enough to us, we will hear it.

HTRFs are, in essence, variables determined by the minute changes made in the frequency (IFD, interaural frequency difference), level (ILD, interaural level difference), and arrival time (ITD, interaural time difference) of a sound by each of the two ears – these three defining factors coexist and cooperate to provide virtual spherical coordinates for sound sources located within audible range of a listener (or receiver, such as a microphone): the HTRF "shows how the sound reaching the ear-drum is affected by the direction of the sound source relative to the head" (Gelfand, 2009). HTRFs have also been

defined (Blauert, 1997) as the *Free-Field* Transfer Function (FFTF), emphasising that the values gained from the transformation do not compensate for external factors directly – that is, the difference in IFD, ILD, and ITD are required as given variables, not calculated by the HTRF itself.

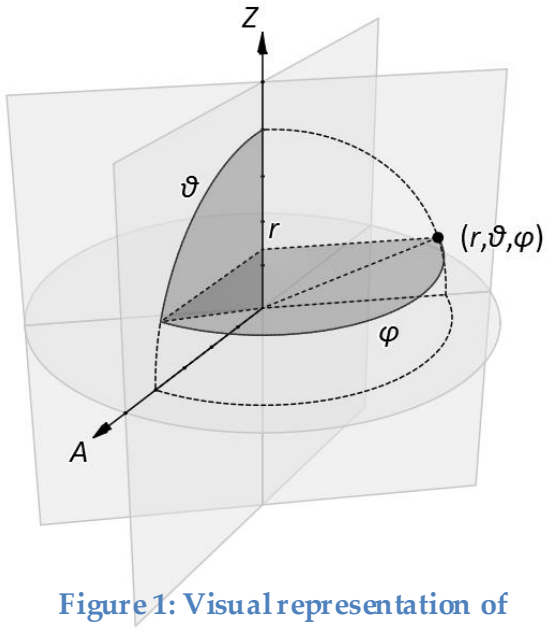


Figure 1: Visual representation of spherical coordinate system (Stolfi, 2009).

Figure 1 shows the designated directional bearings of spectral coordinates. The majority of the sound effects in MiNDSiGHT were placed using approximations, and were more 'felt' then placed according to mathematically-defined bearings and azimuths, but specific sounds, such as those found in the car accident scene and the roof shootout at the climax of the narrative, were placed with pinpoint accuracy. In this diagram, the radius (**r**), inclination (**ϑ**), and azimuth (**φ**) are measured from the centre (the **origin**, and in the case of audio localisation, the listener). The **Z** axis shows the zenith (the direction which is perpendicular to the plane by which reference will be made, in the case of audio, the vertical plane),[6] and the **A** axis represents the azimuth (the resultant plane when the zenith is offset by the inclination).

[6] Reference is conventionally made to the vertical plane when referring to positional audio (i.e. the vertical plane always remains stationary, pointing precisely upward) due to the fact that conventionally, localization of sound on a two-dimensional plane at the height of the ears of the listener is the standard. Speakers are placed at varying locations on a two-dimensional plane defined by the line drawn between the listener's ears, but their position on the perpendicular (vertical) axis remains constant between all sound sources (Hofman and Van Opstal, 1997).

MiNDSiGHT's aural presentation was designed to incorporate sounds on more than a two-dimensional plane. The ceiling fan in the office at the game's outset spins above Ethan and Alix's heads, and the traffic and street sounds heard at the end of the game attempt to convey a sense of elevation, really indicating to the listener that they are at a significantly higher altitude than the ambience below.

The formulae below show mathematically the manner by which sound is altered upon leaving its source and reaching the two ears (Duda, 2000):

$$X_L(f) = H_L(f)X_{ff}(f)$$
$$X_R(f) = H_R(f)X_{ff}(f)$$

Figure 2 displays the elements of these formulae graphically – the resultant Fourier transformation of sound received by each ear is equal to the distance travelled relative to the free-field pressure at the sound source. Duda's research also went on to perform various tests using KEMAR[7] to analyse the relationship between the division that occurs before sound sources are received at the left and right ears.

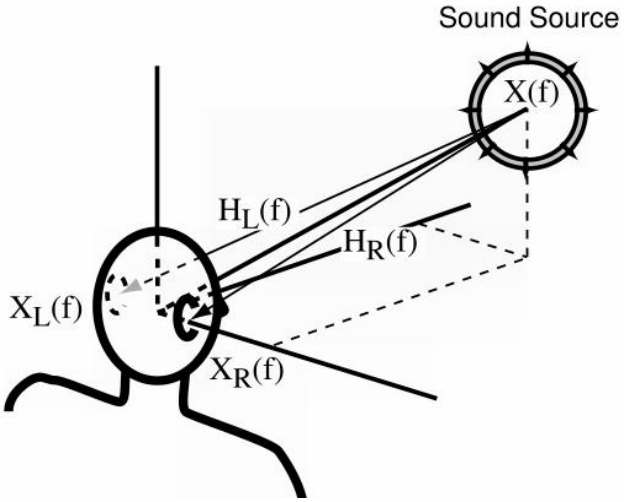


Figure 2: Visual representation of Head-Related Transfer Function (Duda, 2000).

[7] KEMAR is the Knowles Electronics Manikin for Acoustic Research, a device created by the Knowles Electronics corporation for the Massachusetts Institute of Technology. It is a binaural microphone (dummy head) which extends to a fully clothed torso and works in much the same way as 'Fritz', the Neumann KU-100 microphone used to record MiNDSiGHT's audio content.

Chapter III contains an in-depth study into re-creating the human localisation process through artificial means – namely the Neumann KU-100 dummy head (or Fritz, as he lovingly became known), while the remainder of this chapter continues to discuss natural processes of localisation.

Psycho-Acoustics & Localisation

The definitive study on human sound localisation was conducted by Middlebrooks and Green (1991), who detailed the manner by which humans position sound in three dimensions, as well as perceived distance of sound sources, and means of detecting motion in those sound sources. The thorough and respected study concludes by laying out the process by which "spatial cues are used for localizing a sound source in a free-field listening situation":

> *"The azimuth of the source is determined using differences in interaural time or interaural intensity, whichever is present. The elevation of the source is determined from spectral shape cues. The received sound spectrum as modified by the pinna, is in effect compared with a stored set of directional transfer functions. [...] The elevation that corresponds to the best-matching transfer function is selected as the locus of the sound"*
>
> ***'Sound Localization' (Middlebrooks and Green, 1991)***

In order to create cues in MiNDSiGHT which would fully immerse the listener in a virtual aural world, the latter part of this conclusion drawn by Middlebrooks and Green was particularly important. This "stored set of directional transfer instructions" became of particular interest when establishing the vertical arrangement of audio props in the game environment. This is an area where conventional surround sound systems (at

least in the present day) fail – the speakers used in home theatre systems are exclusively in a two-dimensional plane – height is a concept that does not exist in the Dolby Digital world. This fact was an early reason not to adopt any kind of multi-channel setup beyond stereo, and an issue often noted as a significant limitation of current multichannel setups.[8]

Vertical & Extreme Localisation

Middlebrooks and Green's exploration bucked the trend of exploring "the efficacy of some stimulus aspect (interaural level difference, interaural temporal difference, spectral shape, etc.) in revealing the locus of the source", instead preferring to delve deeper into the exact machinations by which two dimensional, and subsequently three-dimensional localisation occurs. Upon reading the results of one particular test (a graphical representation of which can be found in Figure 3 on the following page), it became apparent that while localisation varies from subject to subject, and sound to sound, it is a remarkably accurate response. One important note to take from Figure 3 is that the stimuli (represented with an asterisk) at the top and bottom of the sphere (which represents the three dimensions around a

[8] While technologies are in development to implement "height channels" (such as THX creator Tomlinson Holson's behemoth 10|2 system (Holson, 2009) which offers an upper left and upper right channel for perceived elevation), the price of a binaural system (the need to wear stereo headphones) is far outweighed by the benefits of what is, in effect, analogue positioning, not tied down by physical positions of loudspeakers, and not constrained by a sweet spot for optimal balance between those loudspeakers.

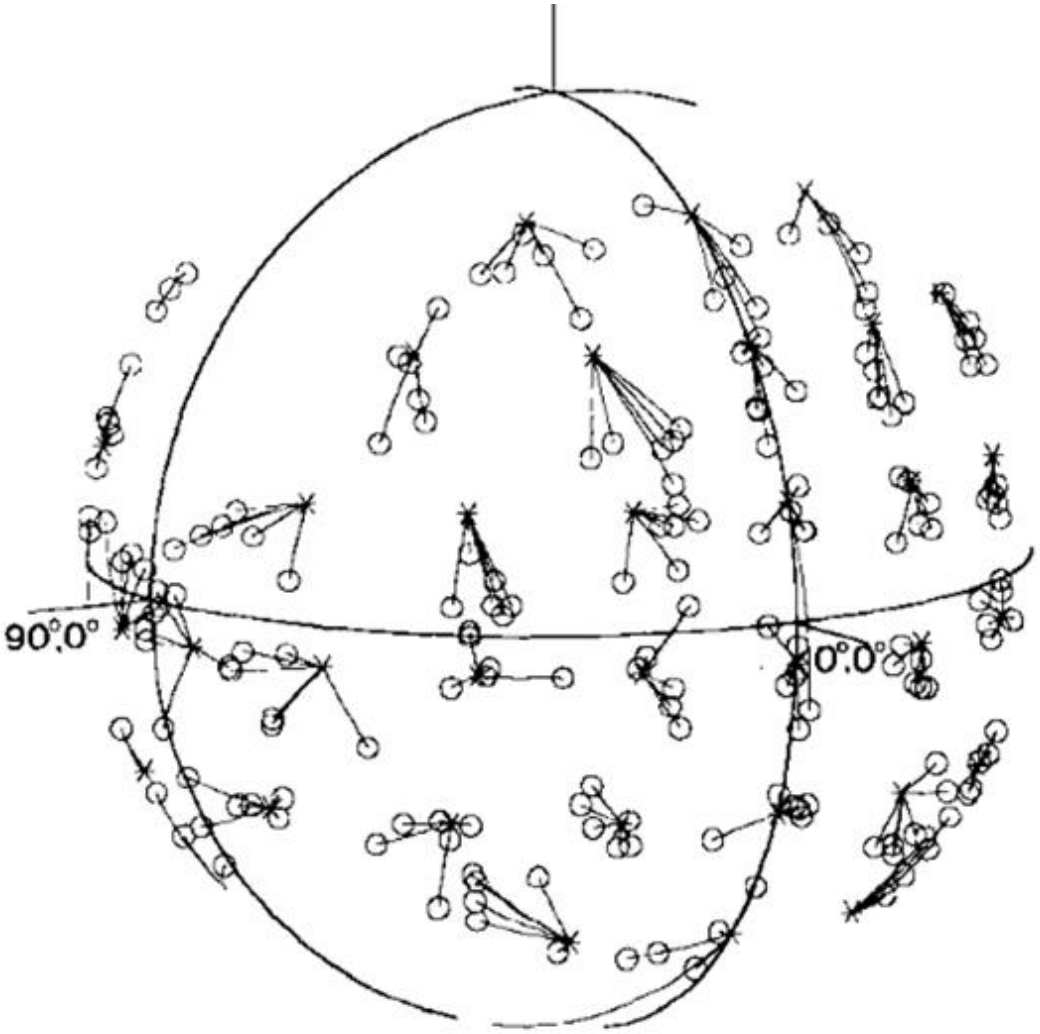


Figure 3: Localisation of sounds. Asterisks indicate the stimulus location, and circles represent five responses to that stimulus by a single subject.

subject's head) were not falsely tested as having originated from a more extreme location – that is, the highest asterisks (representing the stimuli at the highest elevation) were predominantly deemed to have originated from slightly below their actual origin. The reverse can also be said for the stimuli at the very lowest elevation. The conclusion which emerges from this pattern is that pushing sounds to the extremities of a simulated binaural field will not negatively affect the immersion of the scene. The human brain has an innate ability to normalise – that is, to subconsciously relocate sounds pushed to the extreme side of a mix (be it left or right in a conventional two-

channel stereo mix, or the top or bottom in this example) to a more equilibrious position to negate strain on one ear over another. Studies (Jesteadt, 1997) have noted that in patients exhibiting complete sensorineural[9] hearing loss in one ear, sound heard from the extremity of the operational ear (at a 270° angle from the front of the head) are shifted around 30° toward the median, which Jesteadt surmises occurs as a natural method of preserving the functionality of the remaining ear. This concept, it appears, also lends itself to the extremities in vertical positioning (that is, sounds originating from above the listener or below them), as well as horizontal extremities (sounds occurring fully to their left, or right).

Horizontal & Frequency-Based Localisation

With the ears positioned at the extreme horizontal axes of the head, John William Strutt's (the third Baron of Rayleigh's) groundbreaking commentary at the annual Sedgewick Lecture in 1907 still remains an authority on the natural baffle created by the human head when attempting to localise sound sources (Strutt 1907). Over a century ago, he theorised that ITDs and ILDs (or as he referred to them "interaural difference cues") would differ if the human head acted as a barrier, preventing a sound originating from the listener's right side from reaching their left ear via any direct route. These differences vary based upon the frequency of the sound, as well as the width and density of the particular head in question. The difference in level, understandably, is negligible at frequencies below around 1 KHz, where

[9] Sensorineural nearing loss is that which affects the cochlea or other components of the inner ear, or the nerve pathways which link the cochlea to the brain (also known as retro-cochlea).

wavelengths become long enough to pass the head without incident. In these situations, Rayleigh judged that "the only alternative to the intensity theory is to suppose that the judgment is founded upon the difference of phases at the two ears" – phase differences (Interaural Phase Differences, or IPDs) are used as the 'backup' for sounds which cannot be localised based on ILDs or ITDs alone.

Rayleigh concluded by presenting what has become known as the **duplex theory**, which states that for the most part, spatial information for high frequencies is primarily derived from ILDs, and for low frequencies derived from IPDs. Subsequent experiments into horizontal localisation over the past century (Stevens and Newman, 1936, Sandel, et al, 1955, Searle, et al, 1975, and Hofman, et al, 1998) have all reached similar and undisputable conclusions: the accuracy of localisation was dependant on the frequency of the stimulus. As Middlebrooks and Green state: "the accuracy of localization of low frequencies would be attributed to IPDs, and that at high frequencies to ILDs".

However, ambiguity arises from this separation of techniques used to localise sounds – where is the tipping point? Where is the frequency range at which the balance shifts, and ILDs take over from IPDs as the key source of localisation cues? To find the answer to this question, we need to examine one particular experiment conducted by Sandel, Teas, Feddersen, and Jeffrees in 1955, who subjected participants to a range of sine waves positioned at different locations around the pole of the head, then tasked them with positioning white noise at as close a location to their perception of the sine wave as possible. The results of the test were analysed by frequency, and the results suggested that the crossover occurs between 1.5 and 3 KHz. At this point, the wavelengths of the control sine waves were too high in

frequency to be able to provide any usable IPDs, but not high enough to provide usable ILDs.

This 'grey area' effectively meant that, for MiNDSiGHT, that while the fundamental frequencies of male and female human speech (which Baken and Orlikoff (2000) cite as being between 85 and 255 Hz) were easy to place in the virtual world for listeners, select conversations (primarily those conducted over telephones, walkie-talkies, and other speakers by the story's villain Dexter) with an intentionally missing fundamental[10] would struggle with placement. Understandably, this posed a significant problem, as a voice originating from a telephone held directly to one's ear is localised in a very particular manner. In order to combat this, the telephone conversation in the game's introductory scene was treated using a Haas filter (Ben Vesco's Haas Delay, vescoFx, 2009) to widen the stereo field, intentionally leaving space in the centre of the mix and pushing the audio to the extreme left and right. The left side of this field (the side to which the telephone is not placed) was then manually attenuated, which resulted in a much more realistic reproduction of holding the telephone to the head. More information about specific reproduction of troublesome localisations, and a more in-depth look at the Haas Stereo Enhancing Delay Plug-In, can be found in Chapter III.

Baron Rayleigh's work also extended to create a mathematical formula capable of calculating the ITD of a particular sound source, based on the difference between the two inputs (the ears), the speed of sound in the

[10] The fundamental under 255 Hz would not be audible on a conventional telephone (or walkie talkie or similar) due to the fact that these devices operate on a frequency range of between 300 and 3500 Hz. This means that the predominant frequency used (taking the place of the fundamental) would likely fall in the range between 1500 and 3000 Hz, where localization difficulties occur.

environment, and the angle at which the sound source is positioned relative to the head pointing forward. The formula used to determine this is as follows, and Figure 4 explains the equation in a graphical manner:

$$ITD = \frac{a}{c}(\theta + sin\theta) = -90° \leq 0 \leq +90°$$

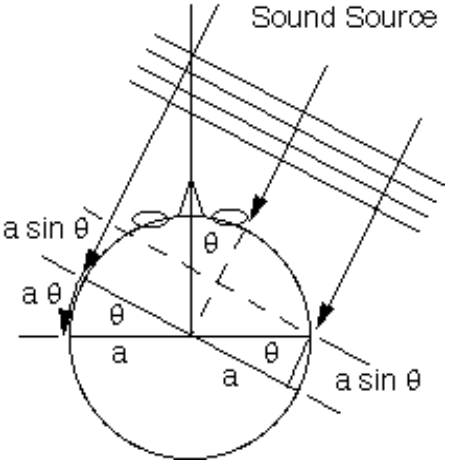


Figure 4: Diagram explaining ITD formula (CIPIC, 2005).

In this equation, theta (θ) is equal to the azimuth of the angle of the sound source in respect to the angle of the head, the radius of the head is represented by **a**, and **c** equals the speed of sound (344m/s). This calculation gives an average ITD of approximately 0.7ms for the size of an average human head – a clearly audible difference (CIPIC, 2005).

Front / Back Confusion

Another issue which became apparent during the various studies referenced herein was the notorious challenge listeners have with discerning whether or not sounds emanate from in front of, or behind them. Front to back depth often proves the most challenging dimension to localise for humans, particularly when compared to the relatively simple ITDs and ILDs used to position sounds horizontally and vertically.

Several studies have been conducted specifically into front/back confusion issues (Hill, 1997, Ngan, 2005), the former of which, Peter Hill's PhD thesis, came to the conclusion that head movement, which until now (in horizontal

and vertical localisation) has not been recommended, plays an important role in discerning the depth of sound sources.

> *"The rate of change of the IACC[11] with head rotation is an important probable means of resolving these [front/back] confusions. [...] It is believed that this is the method used by the hearing system to interpret the localisation cues contained within the ITDs."*
>
> ***'Front-back confusion...' (Peter Hill, 1997)***

MiNDSiGHT's reproduction through conventional stereo headphones requires absolutely no movement of the head on the part of the listener. However, it is expected that the turning of the head to attempt to face the simulated stimuli will occur impulsively by the majority of listeners, despite the obvious fact that relocation of the head will also cause direct relocation of the headphones, and thus, direct relocation of the sound sources. MiNDSiGHT is at its most aurally effective if the player is able to overcome their impulsive urge to turn their head, and since this is so, the scenes in which front/back localisation is a pre-requisite, such as the game's ending where bullets are fired at Ethan from varying directions, are specifically designed, and mixed, to minimise front/back confusion. Chapter III continues this discussion with more in-depth analysis of the mixing and recording processes which captured the world of MiNDSiGHT.

[11] Inter-Aural Cross-Correlation is a function determined by analysing the left and right inputs of a system (be it a human head or a binaural microphone), however in this example, the front and rear values were recorded to determine differences at the front and rear of the head.

·III·

Reproducing Realism

Recording with Binaural Audio

Binaural or Bust

Since its initial inception by German audiophiles at Berlin's Technische Universität in 1974, the 'Kunstkopf' (literally translated as 'art head') has revolutionised the way two-channel audio can be recorded, and subsequently reproduced. This chapter aims to explore binaural stereophony from two distinct angles:

- The use of binaural 'dummy head' microphones to recreate three-dimensional environments, and the inherent strengths, weaknesses, and pitfalls of such technology, including specific techniques used to record MiNDSiGHT's unique binaural puzzles.
- The use of binaural technology for audiological applications in the medical hearing industry – technology which helps restore the normalcy of binaural localisation to those with hearing loss, and thus requires incredibly accurate binaural input to create a realistic output.

The microphone used to record the vast majority of audio was Neumann's KU-100 dummy head. Our particular model became fondly known as Fritz due to his Germanic roots, and further analysis of this peerless microphone is carried out later in this chapter.

Major advances in binaural recording techniques have been made by medical audiologists, and Starkey Professional, arguably the world's most pioneering hearing aid developer, are behind numerous advances in digital hearing technologies such as the Cetera and Genetic algorithms, and devices which make use of them. Both of these systems are explored in the latter portion of this chapter.

Nature Vs. Nurture in Audio Engineering

The world of audio engineering is a double-edged sword; the sinister side carving through TV talent show tripe in an attempt to create perfection deep within the bowels of a mixing desk, while its dexter edge battles valiantly for the capture of uninhibited realism and perfection, with as little post-production necessary as possible. In order for MiNDSiGHT's technical production to succeed, it was vital to amalgamate these two differing philosophies mindfully, ensuring that the recorded scenes conveyed the sense of realism and immersion naturally, and only then utilising post-production only to *enhance* the inherent realism in the scenes.

Deciding exactly where to position MiNDSiGHT on the metaphorical fence dividing the natural and developmental aspects of audio development was a choice made consciously, during the project's initial inception. It was always the intention to rely upon the microphone; the Neumann KU-100 allows for capture of unparalleled levels of aural depth, and an intention was to allow the microphone's strengths to shine without the need for excessive editing.

The Neumann KU-100

First introduced in 1992, long before the influx of surround sound systems which have subsequently consumed the market, the Neumann KU-100 is still considered over fifteen years later to be the finest binaural microphone available. With the rich pedigree of Neumann's superior microphone design distilled into a dummy head, using a KU-100 has ensured that MiNDSiGHT's core philosophy – immersivity, can flourish.

A dummy head like the KU-100 is designed to resemble the human head as much as possible, utilising moulded silicon rubber ears which are weighted and modelled with startling realism. Moulded ear canals inside the head are meticulously crafted to parallel those in a human head, and the weight of the head itself is intended to be as authentic as possible.

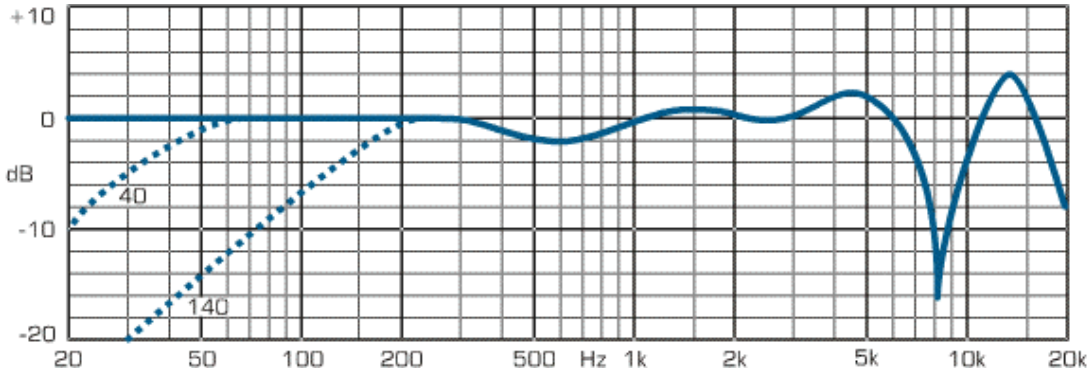


Figure 5: Frequency curve of the Neumann KU-100 (Microphone-data.com, 2009).

Figure 5 above shows the frequency response of the KU-100. While predominantly flat, particularly in the mid-range, the gradual bass roll-off over 20 to 200 Hz, the steep valley at 8 KHz and the subsequent peak at around 14 KHz all serve to flatten the natural frequency response of the human ear. Fletcher and Munson initially defined mankind's subjective loudness perception in 1933, and Figure 6 on the following page shows the standard Equal Loudness Contour for an average human ear. The Neumann KU-100's frequency response has been specifically tuned to attenuate areas where the human ear excels, and amplify areas that are naturally attenuated aurally. Due to this compensation for the limitations and strengths of human hearing, recording an aural environment with the KU-100 results in an exceptionally accurate reproduction of the original situation.

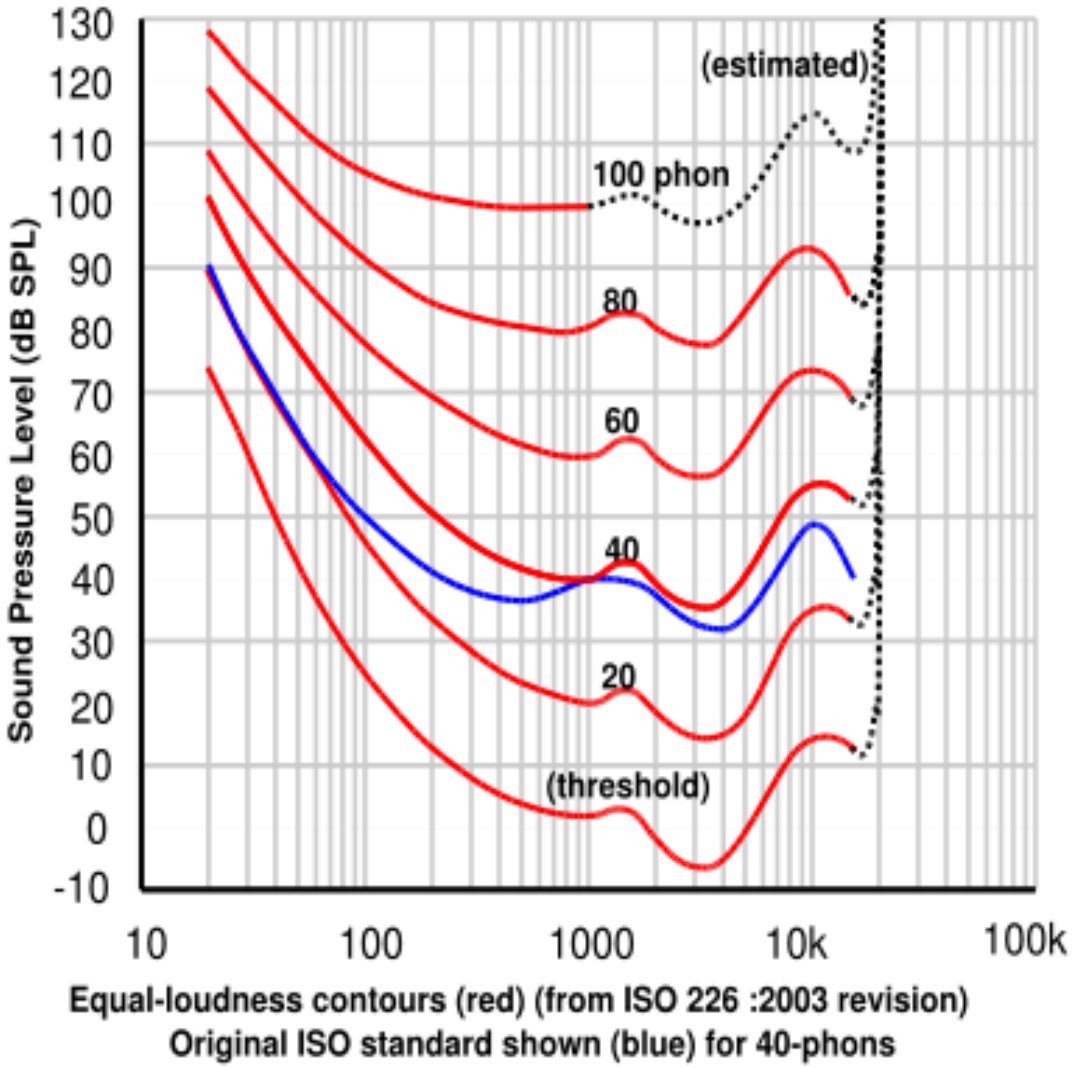


Figure 6: Equal-loudness Contours of the human auditory system (Toonz.ca, 2008)

MiNDSiGHT is played from a first-person perspective – the listener *becomes* Ethan Lewis, the protagonist. This made the task of Martin Horton, playing the role of Ethan, somewhat troublesome, as he would need to effectively project his voice so that it would originate from within the dummy head itself, just as a man's voice originates inside his throat. Numerous methods were employed to place Martin's words between the ears of the KU-100, and in the end a joint solution was devised.

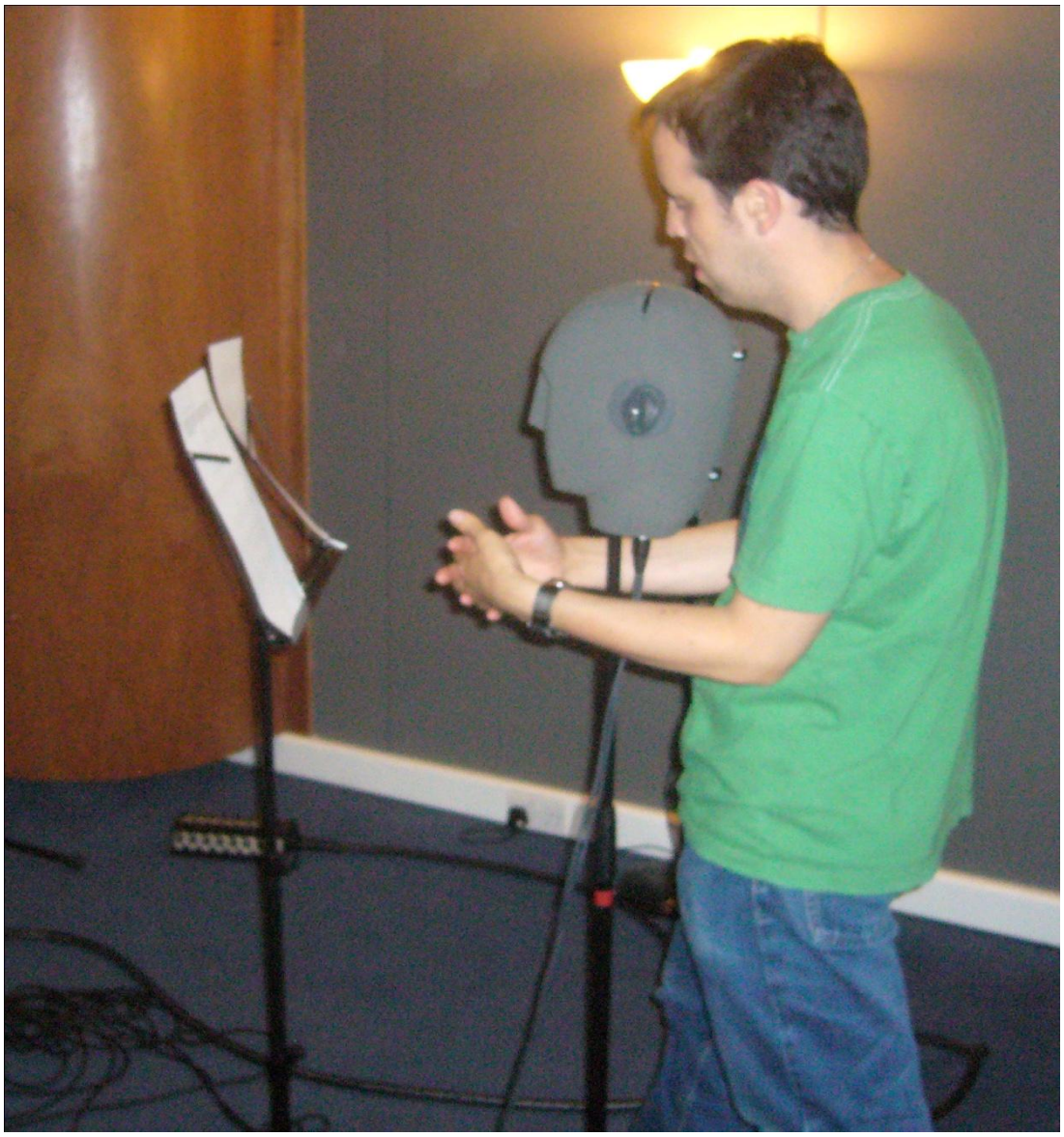


Figure 7: Martin Horton recording using the 'behind-the-head' method of first-person binaural localisation.

Figure 7 shows Martin performing the first of these techniques, the 'behind-the-head' technique. After experimenting with various placements of Martin's actual head circumnavigating the dummy head, it was determined that this particular placement provided the most authentic impression that

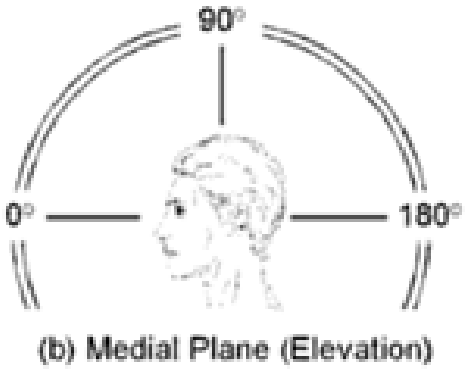


Figure 8: Medial Plane Bearings.

his words originated from within the listener's own head. This initially seemed strange considering the voice box is located at the base of the throat, but as Figure 7 shows, Martin was instructed to place his hands to form a primitive reflective surface to reflect some portion of his dialogue back into the lower region of the head's captive range. While this technique may seem fruitless, the difference between the two captured samples was audible, and so it was determined that Martin would speak into the head at an angle of approximately 110°, with his hands placed at the opposite pole of the head's axis (approximately 290°). This ensured that Martin's words were projected into the centre of the head, with a small proportion being reflected back into the throat region of the dummy head, which resulted in as natural of a first-person placement as possible with the KU-100.



Figure 9: The Core Sound High End Binaural Microphone system, with DPA microphones.

However, specific scenes which involved movement required a different solution. Martin could not move with the head in this position, and so an alternative was used – the Core Sound High End Binaural Microphone set, which comprises of a DPA4060 and DPA4061 microphone and are capable of being attached to a subject's ears. While these microphones were not comparable to the KU-100 in terms of quality, they provided a high quality solution to the problem of providing first-person

binaural placement from a moving sound source. Mixing these scenes in with those recorded with the KU-100 proved a challenge due to the sheer difference in fidelity between the two devices, not to mention the KU-100's moulded ears and ear canals. However, as Figure 10 demonstrates, real ear canals were used in conjunction with the Core Sound microphones, and although they were not able to be mounted as fully inside the ear canal as the KU-100's capsules are, Martin's own pinnae provided a provided a similar eventual result to the Neumann's moulded pinnae. Thanks to the similarities and the natural differences which occur when listening to sound while moving, minimal balancing between the two different microphones was required in post-production.

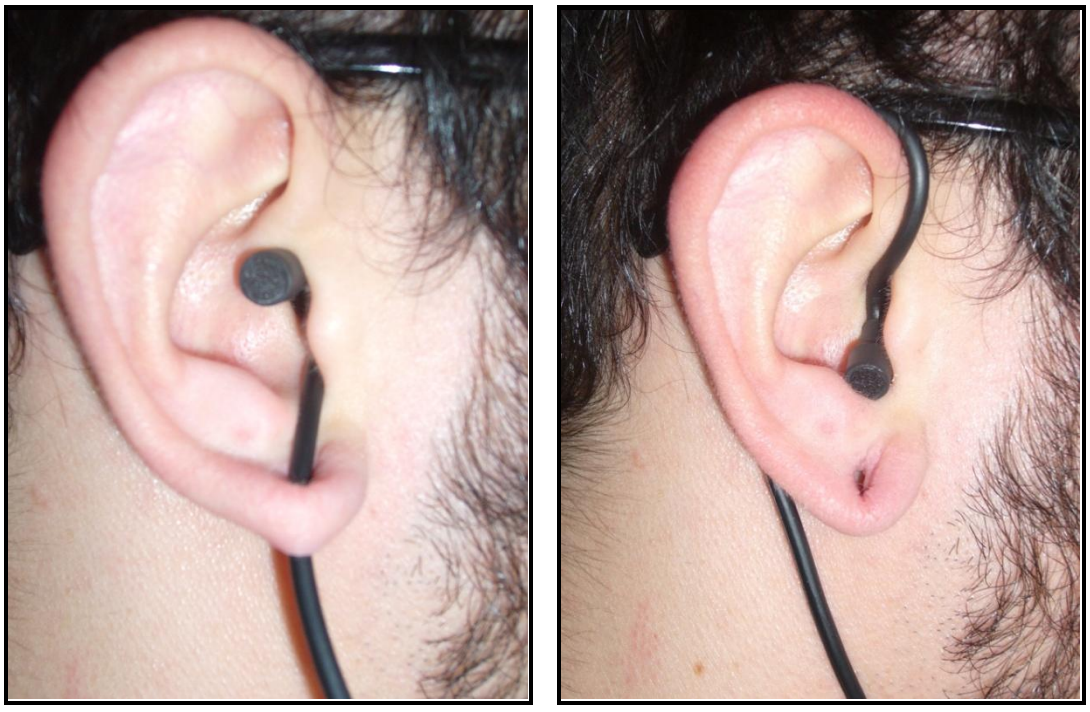


Figure 10: Developer Luke Charman wearing Core Sound binaural microphones inside the ear, mounted outward through an existing stretched ear piercing, or slightly downward, wrapped above the top of the pinna.

Starkey Labs & The Cetera Algorithm

Following on from a discussion of placing small binaural microphones in the ears, it is only natural to consider digital hearing aids; devices created not for entertainment, but for replacing malfunctioning or entirely non-functional ears in those exhibiting partial or total deafness. Starkey Professional revolutionised the quality of simulated localisation in digital hearing aids with the introduction of the Cetera system in 1999.

> *"The important tasks of identifying and defining sound - the real job of "hearing"- occurs in the brain. The job of the ears is to capture and send natural acoustic signals to the brain for processing."*
>
> ***Jerry Ruzicka, President of Starkey Labs (1999)***

Hearing aids equipped with Cetera technology were the first to reproduce the 'binaural squelch' of human hearing; the auditory system's innate ability to place additional emphasis on central sound sources and filter out external background noise (Ross, 2006). In the past decade, the Cetera algorithm has evolved into Starkey's Genetic Algorithm (GA) which, coupled with the Destiny line of hearing aids and their Acoustic Signature[12] technology, is capable of "immediately classifying the statistical properties of incoming sound sources" (Starkey Labs, 2006), meaning that ITDs and ILDs are actually captured and amassed by each aid, then processed to apply relevant filtering and balancing to accurately reproduce localisation for the listener. Starkey conducted numerous studies (Starkey Hearing Research Center,

[12] "A sophisticated, technology-driven system that detects and classifies listening environments and switches seamlessly between them, providing a totally natural hearing experience and clearer, better sound quality than ever before" (The Hearing Aid Center, 2008).

2004-2008) into the effectiveness of the GA technology, and one particularly pertinent paper (Baskent and Edwards, 2007) discusses the calibration of such devices, noting that they (either hearing aids or cochlear implants) are first set parametrically by exposing the user to a variety of stimuli from voices positioned at various bearings to ambient sound effects and a range of sinusoidal frequencies, and then the results of those exposures are used to dictate a set of rules for future localisation and auditory reproduction. The study goes on to state that "the final optimal solution is typically reached by evaluating only a fraction of all possible solutions", which is an accurate digitisation of the method the biological auditory system uses by stimulating the organ of Corti[13] in different regions of hair cells for different stimulus frequencies.

Within the realms of audio engineering, the infamous 'virtual barbershop' recording created by Starkey in 1999 as part of their promotional campaign to promote the unprecedented perceptual realism of the Cetera line of hearing aids[14] is a well known piece of binaural audio. The most important thing to realise when listening to the track for the first time is that the sounds therein have not been recorded; they have been rendered. The purpose of the algorithm is to pick up sounds from their original location, determine exactly where that location is about the user's head, then process the audio to re-introduce binaural qualities to it. With the technology a decade old, it has been analysed, re-appropriated, and re-shaped into audio plug-ins such as Soundhack's **+binaural** plug-in (discussed in Chapter IV), which was used to

[13] The organ of Corti is located within the inner ear of all mammals, and contains hairs of varying thicknesses and densities which respond to specific frequency bands.

[14] A 192 kbps version of the Starkey's Virtual Barbershop can be found on the accompanying DVD-ROM. In order to experience the binaural qualities of the audio, stereo headphones are required.

provide additional emphasis to particular elements of MiNDSiGHT. The Cetera algorithm paved the way for other digital systems to render and synthesise spatial audio, and while a major focus of this project was to require as little artificial rendering as possible, for the inevitable problem areas which did require synthetic modification, Starkey's medical work in the last century bore Soundhack's binaural synthesis applications, and allowed them to render as truthfully as they do.

The Power of the Plug-In

Starkey's work in the 1990s provided the groundwork for today's modern localisation and stereo enhancement plug-ins, and the two which were used within MiNDSiGHT's post-production were the previously-mentioned **+binaural** from SoundHack, and vescoFx's **Haas Stereo Enhancing Delay Plug-In**. Since some modification was necessary to certain scenes, these two RTAS units were utilised to fill in gaps in the recordings, and to emphasise particularly strong movements (or those which proved troublesome to capture) in the auditory field.

The Haas effect[15] dictates that if two sounds occur in extremely close temporal proximity to one another, they will be amalgamated by the brain, regardless of their location. Imagine two musicians playing similar instruments on a stage, one at the left, and one at the right. If they simultaneously hit a note within 35ms of one another, the brain effectively

[15] The Haas effect states that humans localise sound based on the first sound which arrives at their ear, if any subsequent sounds arrive within 35 milliseconds of the first. Any longer than this, and the brain will localise two separate sounds.

loses its ability to treat them as two separate sound sources. vescoFx's Haas enhancing plug-in is able to digitally split a signal, and send it to the left and right outputs of a stereo channel with a suitable delay to simulate varying intensities of the Haas effect. This effect was used to develop dialogue within a quick exchange between characters, as localisation of conversations taking place alternately at the left and right of the aural plane, particularly with large amounts of ambient sound, proved challenging to position.

The other plug-in used, **+binaural**, is more fully-featured and is able to replicate Starkey's algorithms and place a sound source at any location around the listener. The angle control in the lower-left of Figure 11 allows full 360° positioning of a sound, and the plug-in can also automate this positioning using an LFO to create natural movement effects. This particular effect was put to good use when creating the fly-by of the car, resulting in a realistic Doppler simulation which, coupled with Ethan's leap to avoid its movement, creates a complex and rich aural event.

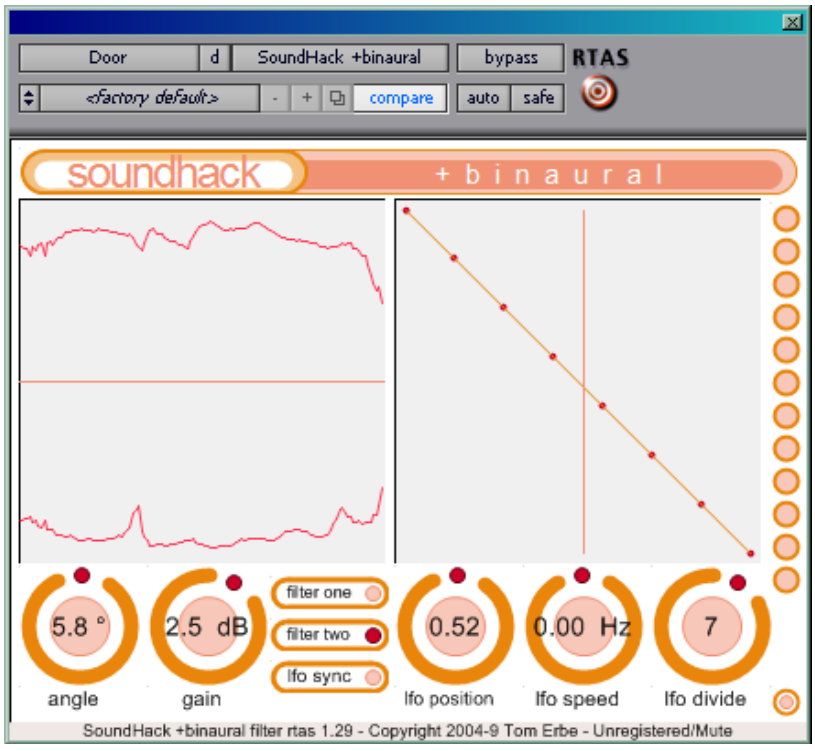


Figure 11: Screenshot of Soundhack's +binaural plug-in.

·IV·

Darkness, Camera, Action!

Creating the Binaural
World of MiNDSiGHT

All the World's a Stage...

The success of MiNDSiGHT as both an experiment and a commercially viable system hinges on more than just the technology used to capture the world – the content of that world can be considered equally important. The most accurate reproduction of a sub-par performance or a lacking storyline, is still a sub-par performance or a lacking storyline. This chapter discusses the creative work which went into writing MiNDSiGHT's script, including analyses of the characters (and conventions used therein), as well as examinations of each of the environments featured in the story, and how they were envisioned in three dimensions, planned, and populated with the characters themselves, as well as audio props, background and ambient sounds, and musical elements. MiNDSiGHT was intended to portray an audible world, and this chapter documents that intention.

Actors & Characters

There are a total of seven characters within MiNDSiGHT's narrative, and an additional narrator, who does not feature in the story but is the voice behind the audio instructions and credits scenes. These characters were all fleshed out to some extent, though since only Ethan and Alix appear in the majority of scenes, their personalities were developed much further. The supporting cameo cast were all at least given names and basic characteristics, and the actors were permitted to use their own judgment and flair in defining exactly who those characters were. The decision to allow the actors to re-write dialogue in order to better fit their adaptations of the roles proved a huge

success, as it meant that their empathy for those characters grew significantly, resulting in roles played more personally. The cast of characters featured in MiNDSiGHT are:

- **Ethan Lewis** *(played by Martin Horton)* – the story's main protagonist, a somewhat sarcastic private detective in his early thirties. Blind since birth, Ethan's hearing has developed to near super-human levels – culminating in the discovery of his ability to hear the thoughts of his family members in his head.
- **Alix Ingram** *(played by Gemma Parker)* – Ethan's hired help at his detective agency, aged in her early twenties. Trained in martial arts, and a bubbly character, Alix acts as Ethan's eyes, and the two share a close friendship.
- **Dexter** *(played by Razif Hashim)* – the story's villain, Dexter is an archetypal oriental Mafioso, with a penchant for staging elaborate kidnappings, and a murderous grudge against Ethan Lewis.
- **Juliet Lewis** *(played by Gemma Parker)* – Ethan's sister, who has been kidnapped by Dexter. She is later revealed to possess the same extra-sensory abilities as Ethan, as the two discover their MiNDSiGHT abilities.
- **Tim Lewis** *(Matt Butchers)* and **Susan Lewis** *(Stacy Winter)* -- Ethan's parents, scientists who appear only in a flashback scene, which reveals the origins of the family's mental abilities. The couple live, and raised their children, on a nuclear facility in Columbia, which caused the children to gain their unusual abilities.
- **Lars** *(Matt Butchers)* – an inept security guard posted to the main entrance to the Teracorps facility, who is beaten up by Alix in order for her and Ethan to gain access to Dexter's tower.

All of the roles are fulfilled by MA Radio & Television Acting students from the University of Essex's East 15 acting school. Recruiting post-graduate actors from a professional course ensured the highest possible quality without needing to remunerate, and the acting team also provided a lot of insight into their experience in radio dramas, adding a new layer of input and ideas to the recording stages of development.

> *"A character is developed in relation to other characters; he must act upon, react to, and in turn, be acted upon by others. He is revealed by encounter, decision, action, and reaction."*
>
> ***'The Elements of Screenwriting (Irwin. R. Blacker, 1987)***

Blacker's words here, and in the remainder of his fascinating book, were constantly referred to throughout the scripting process. Ethan Lewis' character evolves as the player does – in essence, the player steps into his shoes, and more literally, into his head. Ethan's words are delivered aurally as if coming *from* the dummy head microphone, and are filtered and processed to simulate how one hears their own voice when speaking – deeper, and seeming to originate from between the ears at an even spacing.

Some of Ethan's lines were recorded using a pair of binaural microphones mounted to ear clips, meaning that actor Martin's every movement and breath were captured exactly as his own ears captured them. The result, particularly in scenes such as the car accident, is an astounding amount of realism, designed to truly make the player forget their surroundings and fear for their life as they dive from the path of a speeding sports car.

Theatrical Conventions

Referring back to Captain P.P. Eckersley's book 'The Power Behind the Microphone', and his statement that radio drama must: "rely on dramatic speech and sound entirely", one question arose before beginning to record dialogue and foley for the project: **should the actors dictate their own movement within pre-defined parameters set forth in the script, or should each movement be specifically choreographed and rehearsed?** The answer to this question eventually provided itself in a very natural manner during recording sessions through simple trial and error. The actors were instructed to use their best judgment to navigate the environments, and if any particular takes lacked aural plausibility or sparkle, the actors were given more specific instructions. Siewart (2003) said that:

> *"...allowing actors the freedom to move may help some to explore how to say the character's words."*
>
> ***Drama Team Handbook (Alison Siewert, 2003)***

Completing the recordings in this guided, yet free manner, allowed the actors to personify their roles beyond accents and vocal delivery – to walk and breathe as they felt their characters would. This provided a huge amount of freedom, particularly in physically demanding scenes such as Alix and Ethan dragging Lars into the hallway in CIN-7, where their improvised grunts and groans as they carried their fellow actor across the room provided a sense of realism a script performed to the letter simply could not. Put simply, MiNDSiGHT was an experiment, designed to defy convention, and allowing for a suitable balance between improvisation and choreography provided a happy medium which shaped the immersive elements of the audio, while maintaining the pacing of the story.

Environments & Foley

Progressing on from the game's dialogue, the ambient sounds, environmental effects, and foley were all of great importance to enhancing the plausibility of the scenes. Separate floor plans were made for each of the environments featured in the game, including detailed plans of all sound sources within each environment.[16] Creating a road map for each scene allowed greater precision when mixing in ambient sounds – since some ambient effects were layered, and scenes were not all carried out in their respective surroundings (for example, the final scenes upon the roof of Teracorps were recorded in a back garden, then layered with an ambient recording taken upon the roof of a seven storey building in a busy area of London), which provided an encompassing sense of externality, coupled with the dialogue, which was mixed and processed appropriately to situate the characters in that location.

Initially, it was unclear exactly how much additional mixing could be performed on the scenes without sacrificing the plausibility of the binaural environments. However, it quickly became clear that the superb quality of the Neumann KU-100 dummy head allowed scenes recorded in different environments to be mixed together with no discernable difference, or drop in realism. All of the indoor scenes were shot inside SAE London's studio live rooms, then additional sounds such as footsteps on marble floors and echoing voices in the large rooms of the Teracorps tower were all added in post-production – layered with other samples of those footsteps recorded with the Neumann KU-100. This allowed for simplicity and ease of recording

[16] Each floor plan can be found in the relevant Appendix at the rear of the document.

the actors, who did not need to travel to all of the different locations needed in the story's varying scenes. Since each environment had a detailed floor plan layout, calculating reverbs and quantifying early reflections was a simple process, and each environment was given its own entirely unique aural idiosyncrasies dependent on those layouts.

The majority of foley effects were also added in post-production, and the vast majority of these were recorded with the KU-100, which negated the need for them to be artificially placed within the binaural environment. Select sounds were added artificially, and for this purpose, Soundhack's **+binaural** plug-in was used, which allows for manual positioning of sounds in a simulated binaural environment. This plug-in was also used to enhance the directional shifting of certain scenes such as the car accident, which required a full shifting from left to right – **+binaural** was able to place further emphasis on the positionality of the car in the stereo field, which greatly enhanced that particular scene's sense of immersion.

The use of external plug-ins, including both **+binaural** and additional reverb and equalisation settings, did not in any way negatively impact upon the realism of the scenes. In fact, the superior recording quality of the dialogue was able to constantly keep the player's attention, meaning that modifications to ambient or foley sounds were both subtle and appropriate.

Minimal Music

MiNDSiGHT also includes minimal musical cues designed to heighten the drama at specific intervals. While it may be said that background music can serve to lessen the immersive content, the fact remains that as consumers of

film and television content, we are conditioned to respond in a certain way to happy or sad music – a result of cultural conditioning rather than any inherent emotion stored in the music itself.[17]

The musical content used in MiNDSiGHT is minimal, and serves to differentiate between individual scenes, and assist the user in understanding when interactive portions of the adventure are approaching – music indicates a lack of interactivity, meaning that until the music stops, the user is able to relax somewhat, without the pressure that a quick-fire scene may suddenly impose itself upon them.

Palermo (2005) argues that a 'musical bed', used for underscoring, is an "extremely powerful dramatic device", effective use of which can "push dialogue to great dramatic heights". An excellent example of this technique utilised in MiNDSiGHT is the musical cue which rises over the game's introduction. The cue actually continues throughout the entire opening scene, being ducked to a much more subdued level during dialogue, then rising dramatically, yet delicately, as the kidnap of Ethan's sister is revealed to him in his telephone conversation with the villainous Dexter.

The power of music is reduced if it is overused, so the reduced quantity of musical content peppered lightly throughout the narrative serves only to heighten the drama, and provide both subtle instruction to the player on the level of interaction they are required to provide at that stage, as well as elegant punctuation for specific aspects of the drama.

---

[17] For a more in-depth analysis of man's pre-disposition towards registering certain musical stimuli with certain emotional responses, please consult the author's earlier paper "The Major Fall and the Minor Lift" (Charman, 2008).

·V·

Attested Development

iPhone Programming

Making Your MiND Up

From the outset, the MiNDSiGHT project was always intended to be a physical, and complete product. The use of binaural audio meant that provided the user was wearing headphones, they should be able to enjoy the game absolutely anywhere they may be. Inspired by endless commuting in and out of London, the decision to place the game on Apple's burgeoning handheld platform, the iPhone, came rapidly, and proved to be one of the most consequential decisions made throughout the entire process. Coupled with the decision to create the program code in Miller Puckette's PureData programming language (which originally came from Andy Farnell, whose Advanced Sound Design module provided an insight and technical expertise into the programming environment which would shape MiNDSiGHT), the implementation of the application's software was remarkably straightforward.

Put in the simplest of terms, MiNDSiGHT's program code is laid out as follows:

- Upon launch, the game's main menu scene plays, giving the user the choice to hear instructions, credits, or launch the game.
- Upon selecting a new game, the narrative scenes play, in order, from start to finish, until the first interactive scene occurs.
- At this point, the player chooses from a number of pre-determined outcomes.
- Their choice was either successful for Ethan Lewis, or unsuccessful. For either eventuality, a separate consequential scene plays.

- If successful, the narrative scenes continue playing in order until the next puzzle presents itself.
- If unsuccessful, a scene plays indicating that fact, and the player is allowed to try again until they succeed.

Since the gameplay mechanics are so structured, and list-like in operation, the PureData code was assembled very quickly, allowing the bulk of development time to be focused on creating immersion through the exploitation of actors, and subsequent mixing of their performances to make each scene's stereo audio file into the very best it could be.

MiNDSiGHT's PureData patching can be considered, analogously, as a series of ties linking each piece of audio together in various permutations, providing the player with branching paths through the game dependant on their actions. MiNDSiGHT is merely a story – the implementation of PureData code allows that story to be told.

Apple's iPhone Platform

While Apple's iPhone launched with a huge marketing push and instant buzz, it was not until the release of the App Store in June 2008 that the platform really exploded, becoming a sleek Mecca for modern software developers worldwide. Developers were able to create their own software for Apple's market-changing smartphone, and to find a home for that software amongst millions of potential customers. The portability of the device means that MiNDSiGHT fulfils its goal of being able to be enjoyed anywhere, and the device's conventional stereo headphone output, and

accelerometer, mean that the game's technological presentation and unique method of operation are a perfect fit for the iPhone.

> *"[The arrival of the iPhone was] as important as the introduction of VHS or the NES. As a games machine this device was so much better than anything else that has come before it. It enables handheld console quality and frictionless distribution - it's better than the DS and PSP because it is connected, you have a direct relationship with customers. Apple have trained 30 million people to download and install applications on their phone where ever they are. It is an amazing process we are witnessing. The iPhone has changed game making - and you have to respect and embrace this new lifecycle."*
>
> ***Neil Young (founder of iPhone publisher ng:moco), 2008***

Since all programming for any iPhone application must be conducted in Objective C via Apple's own design environment Xcode, it was necessary to source an intermediary solution which allowed the operation of Pure Data patches on Apple's hardware platform. Enter RjDj: the middle-man.

iPhone Implementation with RjDj

Released on the iPhone by Reality Jockey Ltd. in early 2008, RjDj is a "mind twisting hearing sensation" (RjDj.me, 2008) which allows users to create their own 'scenes' – PureData patches which can be uploaded to the iPhone and operated natively, making full use of the iPhone's audio processing hardware, inputs and outputs, as well as control methods such as the touch screen and accelerometer. The software is, in essence, a 'wrapper' for PureData code, which connects all of its inputs and outputs to their counterparts in Apple's development environment. For example, accelerometer data can be inputted into PureData via a specific variable

called **[accel]**, which constantly streams the numerical values from the accelerometer, in all three directions (pitch, roll, and yaw), into the PureData patch, meaning that this information can be used to trigger events in the created code.

The use of RjDj as a host platform meant that the development of MiNDSiGHT could be completed entirely within PureData – a language taught as part of the Recording Arts qualification. Ensuring development was carried out in a relevant package with such tailoring to audio meant avoiding wasting time in other programming environments and to spend far more time focusing on the game's immersive audio presentation.

RjDj also provide a distribution model for the scenes created by their users, meaning that MiNDSiGHT can be made available worldwide through RjDj's own scene downloading system, providing a distribution system for the project – scenes are able to be distributed for free, or made available as commercial ventures at a chosen price.

While the RjDj system does provide excellent compatibility between the iPhone handset and the PureData programming environment, one main issue which occurred related to the sample rate at which the application operates – 22 KHz, half that of conventional 44.1KHz audio. All of the recordings for MiNDSiGHT were created in 24-bit stereo at a sample rate of 48KHz, meaning that these files needed to be downsampled in order to play back at the correct rate inside MiNDSiGHT running within RjDj. This raised concerns over whether a reduction in sample rate would cause a reduction in the binaural technology's ability to immerse. Various studies were consulted regarding the optimisation of simulated three-dimensional audio, (Leung, Braasch, and So, 2006, Myers, 1989, Blauert, 1997) and the results concluded

that a sample rate of 22KHz would provide no considerable issues. Personal tests were also conducted when downsampling the scenes, and it was concluded by the author, as well as by peers, that the reduction in sample rate would not cause a loss of immersion in the finished product.

PureData Programming

Developed by Miller Smith Puckette in the 1990s, PureData is a fundamentally user-friendly programming environment designed for the composition of interactive computer music. This project does not aim to provide explanation into the use of the language – the practical aspects of MiNDSiGHT, including fully-commented programming code demonstrating the use of PureData to create the aspects of interactivity and structure found in the title can be found in the relevant appendix at the end of the document.

The decision to use PureData as the housing for MiNDSiGHT came from Andy Farnell, lecturer at the SAE Institute and one of the world's foremost PureData experts, whose Advanced Sound Design module was centred around the use of the language. PureData's advanced audio manipulation and processing facilities were not required in any depth for this particular project, since all of MiNDSiGHT's audio files are pre-recorded and pre-mixed using Digidesign's Pro Tools software, and mastered using the industry-standard SADiE application. However, the language was still capable of organising and implementing the project as a whole, and allowing each scene to be played in the correct order, and triggered by relevant events.

Procedural Programming / Decisions

MiNDSiGHT's program code began with a relatively straight-forward flow chart, as do the majority of software applications. The game launches straight to the Main Menu system, which allows the player to begin a new game, amongst other options.

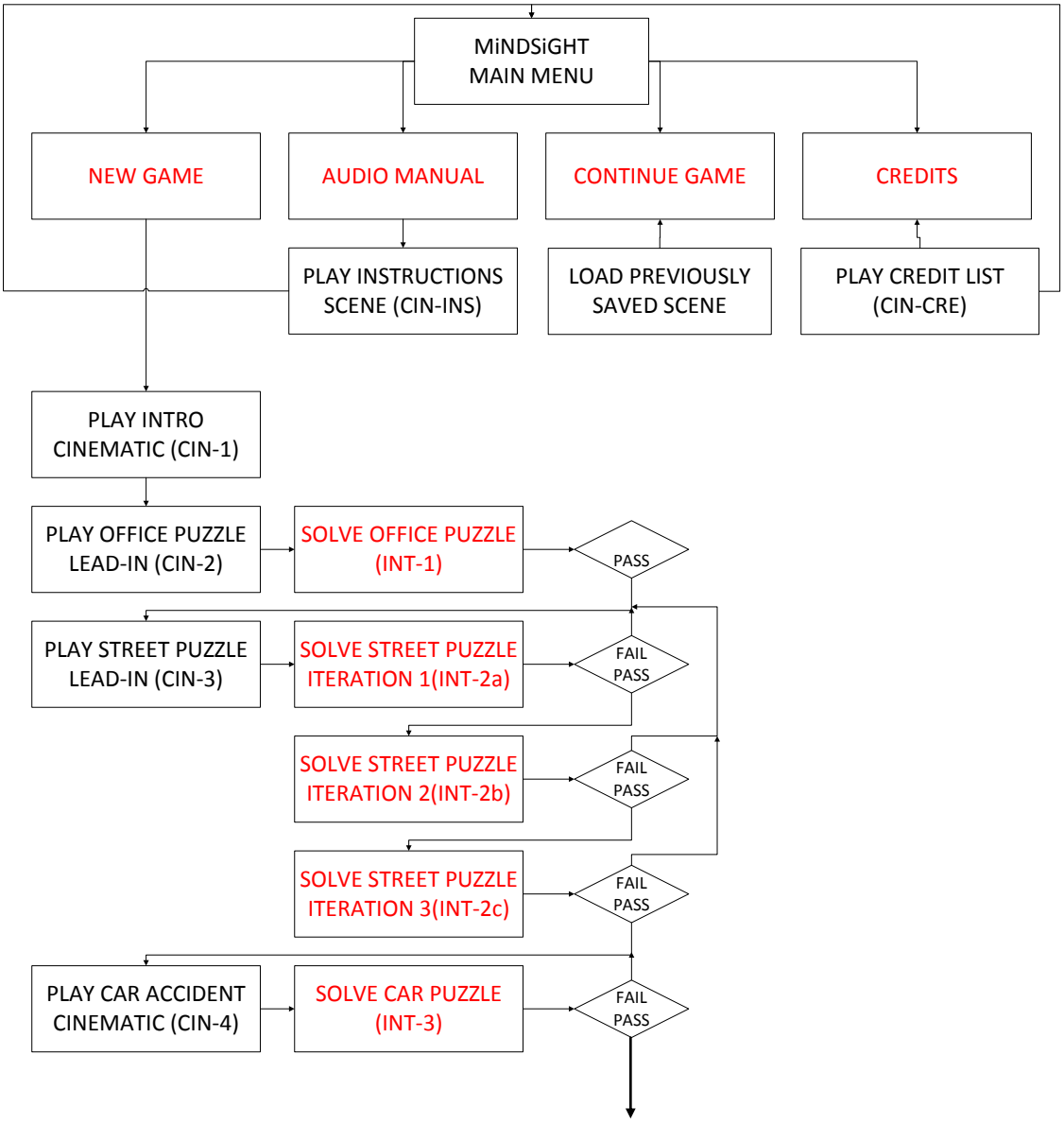
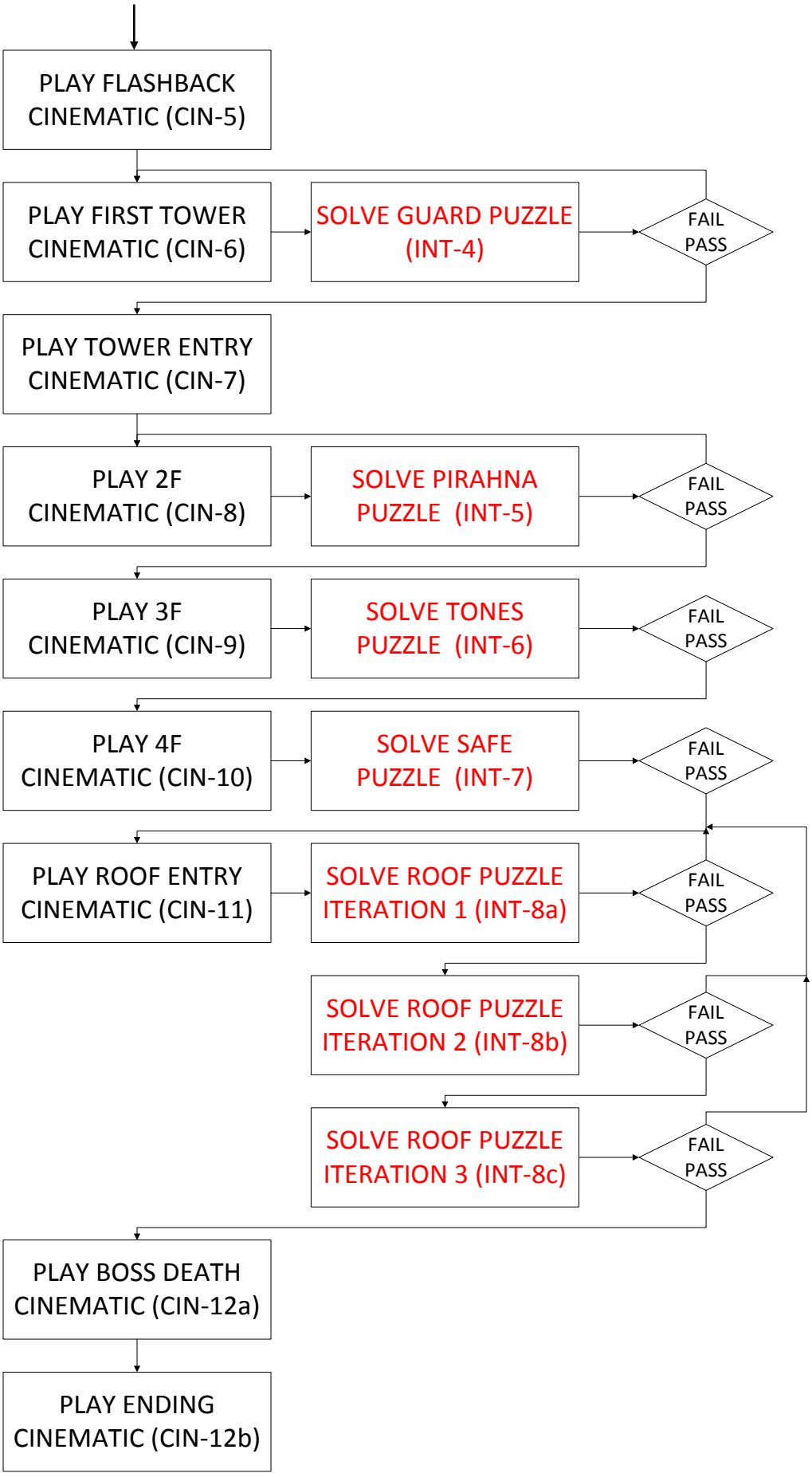


Diagram continued overleaf...


PLAY FLASHBACK CINEMATIC (CIN-5)
PLAY FIRST TOWER CINEMATIC (CIN-6)
SOLVE GUARD PUZZLE (INT-4)
FAIL
PASS
PLAY TOWER ENTRY CINEMATIC (CIN-7)
PLAY 2F CINEMATIC (CIN-8)
SOLVE PIRAHNA PUZZLE (INT-5)
FAIL
PASS
PLAY 3F CINEMATIC (CIN-9)
SOLVE TONES PUZZLE (INT-6)
FAIL
PASS
PLAY 4F CINEMATIC (CIN-10)
SOLVE SAFE PUZZLE (INT-7)
FAIL
PASS
PLAY ROOF ENTRY CINEMATIC (CIN-11)
SOLVE ROOF PUZZLE ITERATION 1 (INT-8a)
FAIL
PASS
SOLVE ROOF PUZZLE ITERATION 2 (INT-8b)
FAIL
PASS
SOLVE ROOF PUZZLE ITERATION 3 (INT-8c)
FAIL
PASS
PLAY BOSS DEATH CINEMATIC (CIN-12a)
PLAY ENDING CINEMATIC (CIN-12b)

Beyond the application's main menu, which provides separate user choices to begin a new game, listen to the game's audio manual, hear audio credits, or resume a previously saved game in progress, the game begins with its intro cinematic. All in-game cinematics are labelled **[CIN]** with an identification number, and all interactive scenes are labelled **[INT]** with a similar number, to allow easy identification of these scenes when implementing them into the program code. These interactive scenes are also coloured red to further separate them from the non-interactive scenes.

For the bulk of scene transitioning, the code is set up so that an audio file is played, then upon its completion, a **[bang]** is outputted so that the next file knows when to begin. A **[bang]** object in PureData is simply an object which begins a process – in this case, it is generated when a sound file ends, and it triggers the next file beginning, keeping a continual flow through the narrative.

Where interactivity is required, the **[bang]** which triggers the next scene occurs as a result of a user interaction. For example, in scene **[INT-3]** where Ethan needs to dive to the ground to avoid a car speeding at him, a timer is created in the code (lasting three seconds), and the game 'listens' for the **[bang]** on the correct direction. If the player tilts their device the correct way within that time limit, their success is registered and sent to the next process, which means that the scene relating to the player's success plays. If the timer counts down to zero from its three-second starting value, the game stops the player being able to register success, and sends the result of failure out of the process, which means that the scene relating to failure, and Ethan being run over by the car, will play. The code sample overleaf shows how such a process is laid out in PureData.

The input for this scene to begin is sent from **[r 17_output]** , which occurs when the scene introducing the car puzzle ends. At this point, the timer (the **[del 3000]** object, which initiates a three second delay) begins counting down. If the player sends in a left signal at **[r Left]** by tilting their iPhone to the left, that information passes through the **[spigot]** object and is allowed to pass to the output. If the player fails (i.e. if the timer counts down to zero), a zero is sent to the **[spigot]** object, effectively disabling it, meaning no data can be sent to the successful output. Data is then sent (via the **[s 18_fail]** object) to the relevant failure sound file, which plays, alerting the player to their failure.

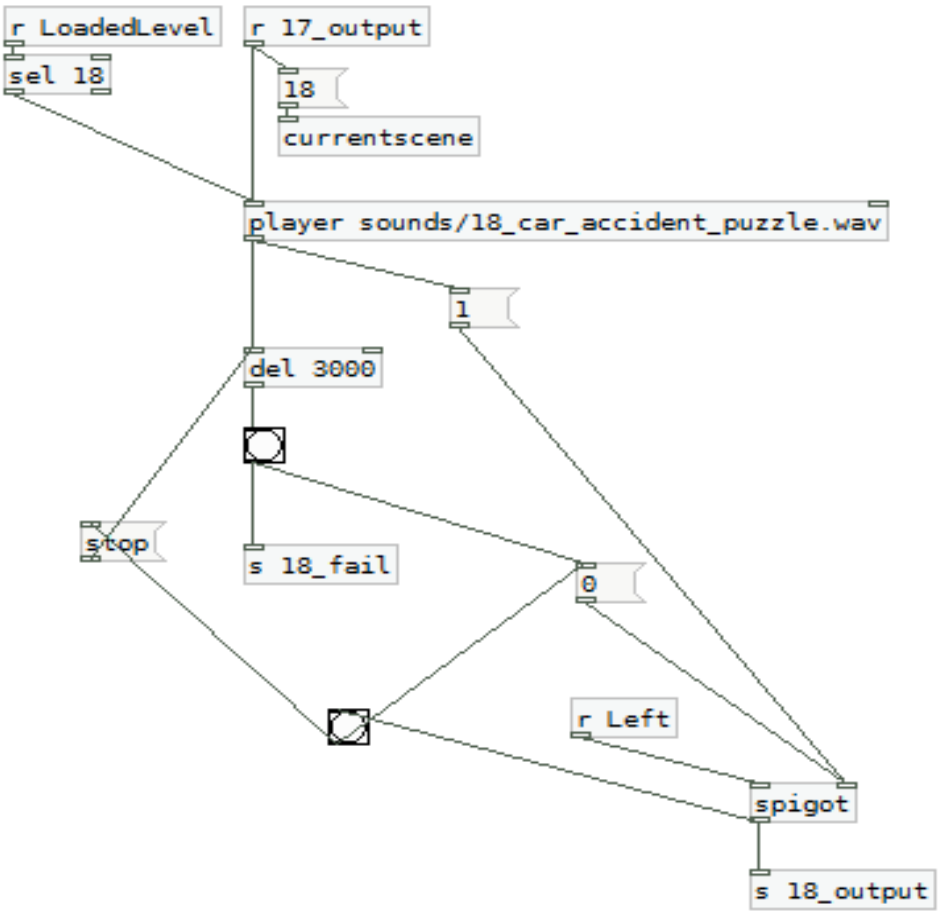


Figure 12: PureData Patch for Car Accident Puzzle.

While there are specific instances of variation in certain pieces of code which do not rely on timing, the majority of the scenes found in MiNDSiGHT operate using systems of **[bang]** and **[spigot]** to read the player's movements, and their timing, and logically determine whether or not they chose the correct direction, and within stipulated criteria, such as timing.

It should also be noted that the player's failures in the various puzzles the game poses do not allow branching beyond repetition, meaning that if the player fails a puzzle, they are allowed to retry it until they succeed. This constrained form of directionality with regard to narrative means that the player can be given a true sense that they are dictating exactly what Ethan Lewis should do, without the need to record unfeasible numbers of alternate scenes branching from failures. For example, puzzle **[INT-4]** poses the problem of how to gain access to the Teracorps building. The player is given the choice of fighting a guard posted at the entrance, or attempting to sneak into the building via the rear. In actuality, both of these choices result in the same outcome – if the player selects the fight, Ethan's sidekick Alix fights the guard and the pair drag him inside. If the player selects the evasive route, an additional scene plays where they are heard attempting to access the building from various other entrances, and failing, at which point they decide they have no choice but to fight the guard, and the same scene of Alix fighting him plays, resulting in the same conclusion. It is important not to become locked in what Manovich (cited by Ben Shaul, 2008) referred to as "an untamed, and apparently endless, unwinding of techniques".

This method of constrained decision-making provides an ideal balance between giving the end user freedom, but limiting the work required in development. Since MiNDSiGHT was created to demonstrate a concept, and on a limited time-scale, the storyline is not overly developed. If the concept

were explored further, with potential commercial funding and a more industry-standard time-frame was allocated, significantly deeper branching paths could be created, providing the player with far more replay value – extending the experience into multiple play-throughs, providing differing emotional cues and outcomes dependent on the user's choices. MiNDSiGHT adequately simulates this sensation, but does so within the confines of a research project largely focused on audio, not character development or narrative. However, as a device to prove the potential for such applications to have a viability in commercial markets, and be an advantageous investment for content providers, MiNDSiGHT achieves its goal admirably.

·VI·

MiNDSiGHT in Hindsight

Evaluation

Evaluating MiNDSiGHT

Since MiNDSiGHT was an entirely novel combination of concepts, the main issue that arose when attempting to evaluate the project was: how does one evaluate innovation? What comparators are available for a piece of software which coalesces drama and binaural audio, and allows it to operate on a device such as the iPhone? Two possibilities arose for comparative evaluation:

- Comparing MiNDSiGHT to existing radio dramas, and
- Comparing MiNDSiGHT to existing iPhone applications.

On top of these comparisons, the need to evaluate the work within its target audiences is also apparent. The two user bases MiNDSiGHT would be evaluated by were:

- Existing fans of interactive drama stories within video games, and
- Visually-impaired people interested in accessible entertainment.

After conducting these tests, the completed application would then be tested against the question set in the introduction: *can binaural recording techniques be employed effectively to create a fully immersive, and interactive audio-only entertainment experience?* The three initial criteria set out were **immersion**, **emotion**, and **portability**, and these criteria were then individually assessed and addressed to determine whether or not they were met.

By allowing both the intended audience and the developer to evaluate the project, an honest, realistic, and detailed evaluation of MiNDSiGHT can be guaranteed, which will determine whether or not the experiment can be deemed successful or not.

Comparisons and Competitors

A suitable method of evaluating any product is to compare its features, strengths, and weaknesses to those of competing products. MiNDSiGHT's two main characteristics are its 'radio-drama-like' narrative presentation, and the fact that it is an application on Apple's wildly popular App Store. These two characteristics are analysed in this section.

Comparison A: Radio Dramas

When drawing comparisons to famous (or perhaps infamous) radio dramas, it is difficult to avoid mentioning Orson Welles' 1938 production of The War of the Worlds, which actually proved immersive enough to convince listeners that an actual alien attack from Mars was occurring. Despite the probably increased sense of susceptibility to technology in pre-war America, the fact that this classic dramatisation proved so immersive to listeners is a testament to the efficacy of audio-only media to conjure emotion.

Another more modern example of radio drama is Radio 2's recent series 'The Adventures of Sexton Blake' (Perfectly Normal Productions, 2009). While the programme was presented in stereo, binaural recording techniques were not employed, and while the class of acting and writing featured far surpassed that of the MiNDSiGHT project, comparing the two side-by-side shows that while the quality of foley, voice direction, and music in Sexton Blake creates a stunning sense of immersion, it cannot be denied that the use of binaural technology could definitely have increased the levels by which the listener can be immersed in the action in real time, as opposed to feeling like an

outsider to the adventures. The obvious difference is that Sexton Blake is intentionally uninteractive, so debatably the less up-front aural presentation of the Radio 2 series fits its format more aptly.

While MiNDSiGHT, an amateur production released in the 21st century could never hope to attain as lofty immersivity as The War of the Worlds, a glance at the numerous YouTube videos of user reactions to Starkey's binaural virtual barbershop presentation (YouTube.com, 2009) proves the effect's startling realism to first-time listeners; MiNDSiGHT can offer the same effect, arguably in more gripping and immersive ways through the marriage of the technology and the dramatic narrative it conveys.

Comparison B: iPhone Applications

Through RjDj's in-app distribution system, the process of making MiNDSiGHT available to the wider public would be a very simple one. With over 2 billion downloads of over 85,000 applications so far, (Apple, 2009), the iPhone gaming market is burgeoning with incredible momentum, and understandably, during the six months that MiNDSiGHT has been in development, similar applications have launched. Two of those applications are the video-based decision adventure Hysteria (BulkyPix, 2009), and the binaural audio bird-hunting archery title Aves (Anderson, 2009), both of which aim to achieve similar goals to MiNDSiGHT (Hysteria mimicking MiNDSiGHT's decision-based gameplay mechanic and limited interactivity, and Aves mimicking the use of binaural audio techniques for a non-narrative-based positional shooting title), but differ in key areas.

Hysteria was released by French developer BulkyPix in April 2009, and tells a horror story involving the protagonist (played in first-person) escaping from a homicidal axe-wielding maniac through a deserted forest. The game features branching decision trees, and separate FMV scenes are played dependant on the user's choices of how to deal with the puzzles posed by the game. These puzzles are dealt with in a somewhat inorganic manner, actually stopping the gameplay to present on-screen dialogues which the player must read and decide upon. Reviews of the title on the App Store make reference to this detraction from the narrative; MiNDSiGHT's constant and uninterrupted stream of narrative ensures it is a more immersive alternative to the French project.

Aves is a three-dimensional bird-hunting game involving archery, using both realistic physics simulation and binaural audio (the developer has coined the term GHOST system for the audio engine). The game was released on October 5 2009, and reviews on the App Store criticise the experience as nothing more than a technical demo, demonstrating a concept. Reviews also state that the binaural audio makes the game too difficult. MiNDSiGHT was extensively tested by established gamers, and the results later in this chapter display clearly that they did not deem the game too difficult at all, and found its level of interaction suitable for navigating the narrative.

While MiNDSiGHT began development before the release of Hysteria or Aves, both products have been given neither a definitive thumbs up nor thumbs down from gamers on the App Store. MiNDSiGHT's niche has been carved very precisely with its blend of narrative, technical audio, and minor interaction to augment a gripping story, and neither Hysteria nor Aves compete in any direct manner, should MiNDSiGHT see commercial release.

The Opinions of Others

Two studies were carried out to determine the success and efficacy of MiNDSiGHT to immerse users in an audio-only world. Test A involved twenty gamers playing MiNDSiGHT to completion on their iPhone handsets, and completing a questionnaire ranking the experience. Test B involved a similar process, asking five lovers of interactive entertainment suffering from blindness or severely reduced eyesight to play MiNDSiGHT and complete an interview with the developer afterward to rate the experience.

Test A: The Gamer Test

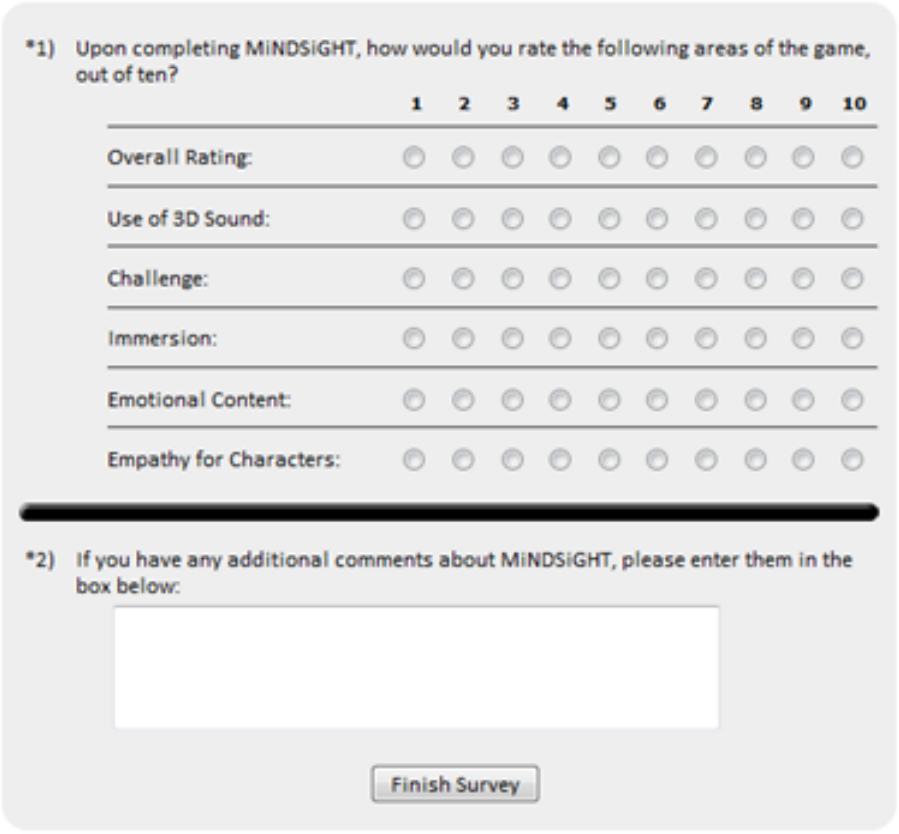


Figure 13: Online survey provided to players of MiNDSiGHT at GameFAQs.com.

GameFAQs.com is ranked within the top 300 websites in the world (Alexa, 2009), and its message boards are constantly abuzz with voracious gamers from all over the world with an insatiable appetite for interactive entertainment experiences. Twenty such gamers who owned iPhones were enlisted, provided with a link to download MiNDSiGHT via RjDj, and played the game to completion. They then completed a brief questionnaire based on their experiences (see Figure 13), ranking the game overall, its use of audio, challenge, immersion, emotional content, and empathy for the characters out of ten, and providing them with an area to enter additional comments on the game. The comments which came back were then tallied and Figure 14 below displays the results in a graphical format. The graph displays that the mean scores were in the higher ranges, with a score of 8 being the overall average across all criteria and score types.

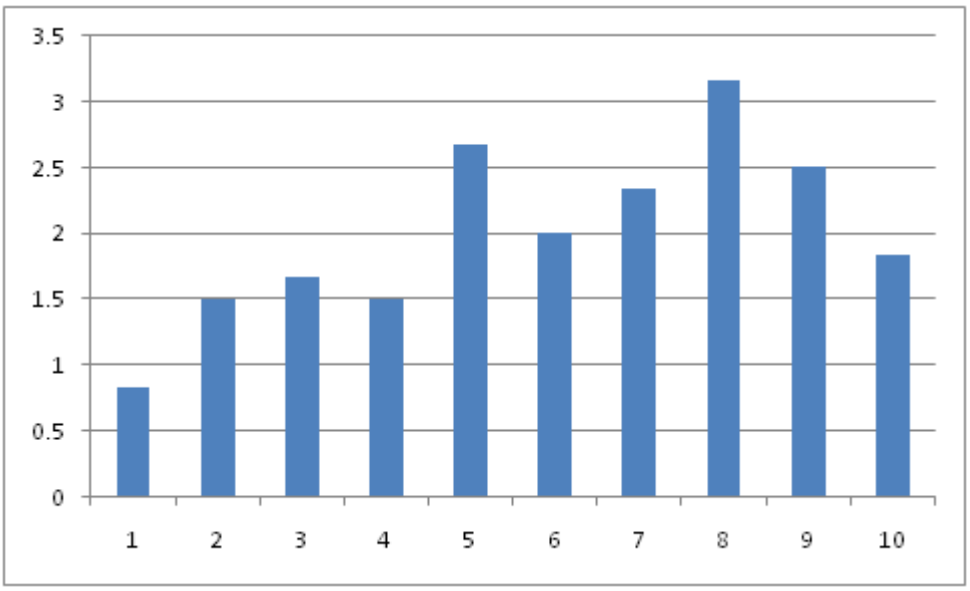


Figure 14: Graph demonstrating mean distribution of scores from one to ten over all criteria.

MiNDSiGHT was not met with universal acclaim from gamers – far from it, but the results of the survey (a more detailed view of which can be seen in Figure 15), demonstrate that specific areas where the game was pushed

especially hard did prove a success with the testers. The use of 3D sound was rated very highly, with seven of the twenty subjects rating it a perfect ten. The overall rating of the game peaked at seven out of ten, an admirable score considering the reduced professional quality and relative brevity of the script itself.

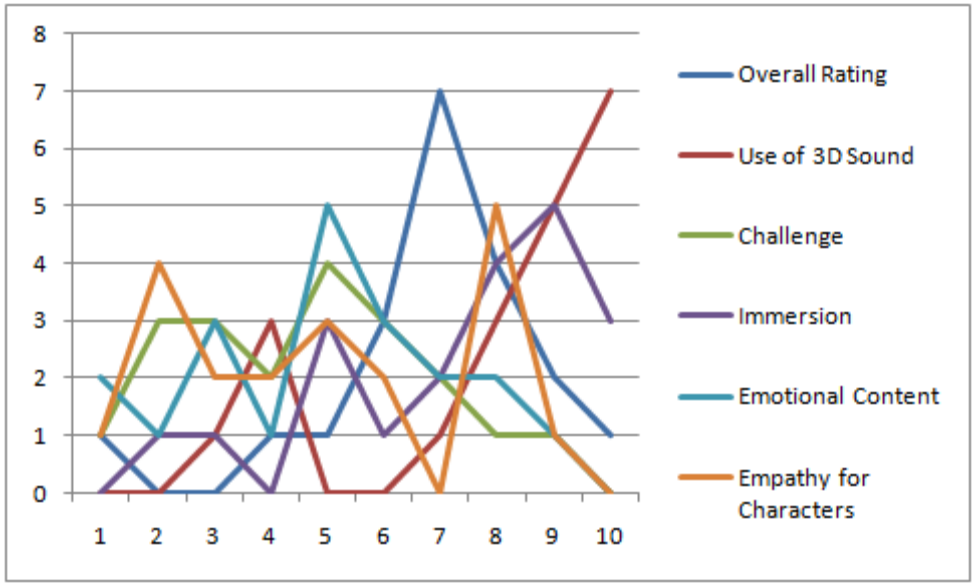


Figure 15: Graph showing individual scores out of ten for each individual criterion.

Test B: The Visually-Impaired Test

Through staff at Moorfields Eye Hospital in central London, four partially-sighted and one fully blind subjects were assembled with an interest in radio drama and at least a fundamental knowledge of interaction in entertainment. These five people were provided with an iPhone and MiNDSiGHT and permitted two hours each to experience the game. Afterward, each subject was briefly interviewed and asked to give opinions on the game, and more specifically on whether they felt it provided a viable entertainment medium to them, as visually-impaired members of society. The results of the

interviews were informative and provided insights that the gamer tests could not. Some particularly helpful comments included:

> *"After finishing MiNDSiGHT, I was left eagerly waiting for Ethan and Alix's next case. The use of stereo sound really helped immerse me in the game, and I felt like the game had been designed especially for me as a visually-impaired person."*
>
> *"This is a really great idea. I had heard the virtual barbershop recording before, and seeing the same type of sounds used in an interactive way where I was able to control the action myself was amazing. I'd definitely consider purchasing games of this type."*
>
> *"The game itself is excellent and most enjoyable, but a problem which would arise for me is that the iPhone is not equipped with voiceover menus, meaning that I would need somebody to actually load up the game before I could enjoy it."*

The latter comment in particular was something which had not been considered at all, and does indeed provide a limitation for the sightless actually getting into MiNDSiGHT from their iPhone's main menu screen. Unfortunately, there is no real way to combat this, as a fully touch screen interface like that of the iPhone provides no tactile response or aural confirmation of selected menu choices, meaning that blind players would need somebody to load the game for them.

Conducting tests with the two main demographics the game would be marketed to were it to be distributed allowed much more detailed conclusions to be drawn, and highlighted both the strengths and weaknesses of MiNDSiGHT as a whole.

Evaluating the Final Product

While comparing MiNDSiGHT to competitors and allowing external parties to assess its viability as an entertainment or narrative-delivery device, an evaluation from within is equally critical to ascertain which areas of the project met the aims established in the initial brief, and which areas failed to do so. The three initial targets set out in the learning agreement were that MiNDSiGHT should be **immersive**, **emotive**, and **portable**. The success of these three goals are evaluated below.

Target A: Immersion

How does one define successful immersion? Inspired by the morning commute, the intention from the outset was to be able to play the game on a crowded train, shut one's eyes, and forget the troubles of the day. This is the purpose of fiction; losing oneself in a fictitious world, temporarily moving oneself from reality into another experience. MiNDSiGHT was certainly subjected to this stress testing on countless train and bus journeys, and on more than one occasion resulted in the near-missing of the required stop.

The use of binaural audio proved the best choice for immersion. The ability to position items in an analogue format around the listener without relying on digital placement or cumbersome arrangements of multi-channel speakers meant that sounds could truly be heard exactly as they were captured. Alix speaking across the desk from Ethan in the game's introduction, then moving to his side, crouched beside him when he is at his most emotionally vulnerable proves the game's ability to position characters

in entirely realistic environments and to capture their movements from position to position with crystal-clear precision and finesse, even in subtle scenes such as the office scene. Where more dramatic scenes are concerned, such as the rooftop shootout at the game's climax, and the car accident puzzle, the immersive power of MiNDSiGHT really begins to emerge, and the urge to physically react to the approaching car or flying bullets threatening Ethan is palpable.

Target B: Emotion

MiNDSiGHT's story, while professionally scripted, is admittedly somewhat underdeveloped and minimal. During the course of the relatively brief narrative, it is somewhat challenging to develop a particularly emotive bond with the characters. Despite solid performances from the actors and a suitable script, MiNDSiGHT falls down to an extent in this field. However, as a concept to be presented to publishers with the intention of securing financial investment to create much more in-depth stories using the techniques MiNDSiGHT has pioneered, the minimal quantity of emotional content is sufficient to provide a window into the potential that the technology possesses.

The study carried out with gamers confirms this fact, with the average rating for emotive content being only a five out of ten, and empathy for characters attaining a large number of scores between three and five. The art of reproducing realism can only go so far if the realism being reproduced is limited in any way. Limitations in budget and timing meant that specific takes needed to be somewhat hurried, resulting in less-than-perfect

deliveries of specific lines including some jokes and some emotive lines. Martin's delivery of the line "If you hurt her, I swear it will be the last thing you ever do" was recorded over twenty times, but it is conventional in drama to record such emotionally-charged lines scores or even hundreds of times. While the emotive content of the script is not of a fully professional standard, given the use of actors and a writer working pro bono, it provides a functional springboard for future revisions of the technology to leap from.

Target C: Portability

MiNDSiGHT is fully portable. Literally all that is required to play is a phone weighing less than 150 grams, and a pair of stereo headphones. The target of portability has been unequivocally achieved. The use of a portable platform means that the game can simply be pulled from a pocket: anytime, and anywhere, and the use of binaural audio means that a simple pair of bundled iPod ear bud headphones are capable of catapulting the player into Ethan Lewis' world. The conventional need for a surround sound system to enjoy fully positional audio has been entirely negated, and MiNDSiGHT can easily and effortlessly be placed into the hands and ears of over ten million iPhone users worldwide.

The choice of using the iPhone platform was undoubtedly the right one, and the use of RjDj and PureData coding meant that the project could proceed without the need for unnecessary learning of new programming languages or obtaining Apple certification as a licensed developer. MiNDSiGHT really is an ideal App, quirky, immersive, enjoyable, and unlike anything players will have experienced before.

Final Thoughts

This final discussion aims to answer the question posed at the outset of this project: **can binaural recording techniques be employed effectively to create a fully immersive, and interactive audio-only entertainment experience?** The answer to this particular question is an emphatic and resounding *yes*.

The use of binaural microphones to record an audio-only drama has proved far more immersive than conventional recordings, where the foley and voices are digitally positioned within the one-dimensional stereo field. Binaural recording with a dummy head allows for not one, but full three-dimensional recordings on all axes surrounding the microphone, providing response to changes in not only position on the two-dimensional plane (as well as distance from the microphone), but also alterations in elevation which were put to superb use in the rooftop scene. Were the recording techniques employed effectively? *Yes*.

Was the experience immersive? Studies conducted by a broad array of test subjects confirmed through direct questioning that with an average score for immersion of nine out of ten, players genuinely felt engrossed in the plot, and the world it plays out in. Although brief and to an extent under-developed, the story of MiNDSiGHT is adequate and successfully demonstrates the inherent potential the technology possesses for allowing immersion in more developed interactive narratives.

Did the use of an audio-only environment with no stimulus beside aurality negate the experience in any way? One could ask the same question of a conventional novel: does the text-only format limit the narrative? The

answer to that question is obvious; humans are far more capable of mentally conjuring immersive environments when a seed is planted, rather than when the entire garden is delivered to them in a high definition video format. Advances in CGI will always be eclipsed by the power of the imagination, and MiNDSiGHT provides a sufficient bed for a narrative to be given a unique presence beyond the realm of conventional written fiction, while still retaining a mystique and openness that permits the listener to attach any visual imagery to the drama they deem fitting.

While time restraints and working around the hectic schedules of professional actors meant that not quite as much time was spent recording the dialogue of which MiNDSiGHT is comprised as would have been preferred, the quality of the voice acting remains solid and of a worthy standard, and the culminating ending scene, which was recorded at the chronological climax of the project, provides a stunning sense of drama which really engages with listeners and encourages them to use their reflexes, their brain, and their ears to engage with the fictitious environment.

The correct level of interactivity was added to the project. The user is granted options which allow them to remain in control of the challenges which befall Ethan and Alix, without trapping them in endless decision-making which, if overdone, would have served to detract from the naturally flowing nature of the narrative. Minimal interaction, tastefully and powerfully delivered in the correct places in the story, provide the user with a sense that Ethan's fate is in their hands, without feeling the need to hold his hand every step of the way.

MiNDSiGHT has proven that binaural audio is a worthy contender in the battle to create realistic and immersive audio which is all-encompassing,

allowing the listener to experience a world as if they were positioned in it directly. The physical and technological limitations of surround sound setups have impaired the delivery of audio in modern media both interactive and non-interactive, and this project's goal to reclaim some ground and relocate the footing of conventional stereo in the world of consumer audio has been unquestionably achieved. The project has positioned itself as a first step into a vibrant, exciting, and wholly untapped new medium of delivering quality entertainment in an exhilarating and captivating fashion: MiNDSiGHT has proven a success.

·Appendix A·

MiNDSiGHT
Script

Introduction to the Script

The dialogue, direction notes, and scripting seen on the following pages were written by Luke Charman, in cooperation with a freelance scriptwriter, Brooke Garner. The script has been formatted to the standard of a professional screenplay, and was distributed to and used by the actors when recording the scenes, as well as being used when recording and engineering the foley sound effects and diegetic and ambient sounds used to bring the words that follow to life in binaural stereo.

Specific requirements when writing a radio script (as laid out by Tony Palermo, 2005b) such as the positioning of page numbers were ignored to conform with the requirements for this document as a whole, but font, positioning, and direction conventions have all been adhered to, meaning that the actors instantly felt comfortable using the script.

Since the actors were given freedom to modify lines to better suit their takes on the actors themselves, specific lines in the following pages may not directly follow those in the audio. However, the overall meaning and direction each scene takes has remained fully intact.

MiNDSiGHT Script

SCRIPT – CIN1 – '*INTRODUCTION*'

INT: MEDIUM-SIZED OFFICE

The scene begins in the offices of ETHAN and ALIX. Alix is sitting in a chair in front of Ethan on the opposite side of his desk. The sound of a revolving ceiling fan can be heard overhead and the faint sound of a radio is heard to the extreme rear left of the room.

ALIX

Ethan, do you mind if I shoot off early today? I can't see us getting any kind of case today, can you? The city feels like it's been dry for weeks. Crime's clearly taken an extended holiday. I gotta say, I'm kinda jealous.

ETHAN

Don't worry, I'm sure something will come in soon. I've got a good feeling about today, and a man's gotta run with his intuition!

Alix

You'd think someone without functional eyes would have a slightly cooler sixth sense than relentless optimism.

Ethan

Don’t knock it ‘til you’ve tried it. C’mon, give the phone a good hard stare. It’ll ring, I promise.

Alix

[whispering] RING! RING! Come on, ring! What gives, Ethan? you promised!

The telephone on the desk begins to ring.

Ethan

I’ve never been one for gloating.. but hey.. intuition!

Ethan lifts the receiver from the phone and brings it to his right ear.

Ethan

Ethan Lewis’ office. Talk to me.

DEXTER

Ethan Lewis. Listen to me, and listen well. I have your sister Juliet here with me. She is safe, but you know only too well the kind of man I am – her safety is not guaranteed. If you have a desire to see her alive again, I suggest you meet me, at your earliest convenience. I'm waiting for you on the roof of Teracorps. Be here within two hours, or her head and shoulders will have a rather drastic falling out.

JULIET

[*through the telephone]* ETHAN! IT'S A TRAP!

HE KNOWS ABOUT MUM & DAD!

Ethan

Juliet! Juliet! If you hurt her, I swear it'll be the last thing you ever do.

Dexter

I have absolutely no interest in your pathetic whining, Ethan Lewis. Just be here unless you want her death on your conscience. I look forward to your arrival.

The line goes dead. Ethan slams down the receiver in frustration and anger.

Ethan

Damn it! Alix.. he's got Juliet. Dexter...he's got Juliet! Damn!

Ethan hurls the telephone and some clutter from the desk in anger, and throws back his chair in standing up abruptly. He moves across the room to the door, and tries to open it to find it locked. He then screams in frustration and anger, and we hear the door rattling further as the scene fades.

SCRIPT – CIN2 – '*KEY PUZZLE SETUP*'

INT: MEDIUM-SIZED OFFICE

The scene begins with Ethan sitting back at his desk, breathing heavily, but calmed from the end of the last scene. Alix, now sitting beside him to his left, has comforted him and calmed him down.

Alix

Ethan, we need to be careful. I know you. I know what's on your mind, and I understand, I really do.

Ethan

[*visibly shaken]* Alix I.. [*sighs]* We have to go, and we have to go now, that's all there is to it. To Hell with traps and risks, I can't let my sister die. How could I live with myself? I couldn't.. I.. I'd have failed my parents if I let anything happen to her.

Alix

[*thinking to herself]* Okay. I know there's no way I could persuade you otherwise. But Ethan.. there's no way I'm letting you do this alone. I'll be with you, your risks are my risks. Juliet means the world to me too, you know that.

Ethan

Alix, thank you. You don't know how much I appreciate it.

Ethan stands up, pushing his chair backwards and heavily placing his hands on the desk determinedly.

Ethan

Let's get going. Where did you put the office keys, Alix?

Alix

Umm.. this probably isn't the best time for this, but I'm honestly not sure. But, I can just use the locator. Listen up, the little speaker on 'em will start beeping.

Ethan

So we've just got to wait for the beep, right?

Alix

Yup!

SCRIPT – INT2 – *'FIND THE KEYS'*

INT: MEDIUM-SIZED OFFICE

The scene takes place in the office, as the previous scenes have. No background sounds have changed, the radio and fan are still operating.

Alix

Okay Eth, I'll beep it over and over. Just listen out, and if you hear it, let me know where they are. You know what my ears are like with high frequencies, I knew spending my teenage years at heavy metal shows would take its toll eventually.

Ethan

Okay, I got it, Ozzy.

Ethan solves the puzzle, by pressing right to indicate the direction of the beeping. Alix can be heard moving over to the same location as the beeping, and grabbing the ring of keys.

Alix

Good job, Sherlock. Let's hit the street.

Alix moves to the door and unlocks it with the keys.

Ethan

The game's afoot, Watson.

Alix

A what?

Ethan sighs, and the two can be heard walking out of the room and down the flight of stairs to the ground floor, then opening the door out to the street.

SCRIPT – CIN3 – *'STREET PUZZLE SETUP'*

EXT: CROWDED CITY STREET

The scene begins with Ethan and Alix walking along the street just outside the office. The streets are jam-packed with moving traffic, conversing and shouting pedestrians, and shops blaring music out of their doors. Ethan and Alix are finding it difficult to hear each other over the din.

Alix

Teracorps is almost a mile from here. It's insane out here, did they invent some second Christmas shopping period nobody told me about?

Ethan

Perhaps our nefarious foe has organised some massive troupe of mindless consumer bystanders to get in our damned way.

Alix

I wouldn't put it past him. Listen, we're going to need to be quick about this. I'll lead the way. Just follow the sound of my voice. If you're lagging behind, I'll stop and let you catch up, but it's going to be difficult enough to get through these crowds as it is, so keep your ears open and we'll manage, okay?

Ethan

Got it. I'll be fine.

Alix

Excellent. Let's roll, boss.

SCRIPT – INT2 – *'FOLLOW ALIX'*

EXT: CROWDED CITY STREET

The scene takes place on the same crowded street scene as the last. With each of the three iterations of the puzzle, some of the particular conversations and sounds from open shop doors have been moved around to give the illusion that Ethan has moved around in the world from one iteration to the next.

Alix

Ethan, here!

Ethan moves to the correct location, and the second iteration begins.

Ethan, this way!

Ethan moves to the correct location, and the third iteration begins.

Ethan, it's over here, come on!

If at any time Ethan makes a wrong move, Ethan says the following:

Ethan

Damn, I lost her. I'd better work my way back to the office and wait for her.

SCRIPT – CIN4 – *'CAR PUZZLE SETUP'*

EXT: LESS BUSY CITY STREET

The scene continues on the crowded streets, though the amount of noise and talking has diminished to an extent. Ethan and Alix are discussing the journey.

Alix

Okay, we're nearly there, Eth. It's just at the end of this street. Even at your pace, we should be there soon.

Ethan

Thank God, another moment in this urban death maze and I'd start to go insane.

Alix

You and me both, boss.

Suddenly, the sound of screeching tires appears far ahead and to the left of Ethan. The tires squeal and the car accelerates towards him. Alix screams out.

Alix

Ethan! Watch out! ETHAN!

SCRIPT – INT3 – *'DODGE THE CAR'*

EXT: LESS BUSY CITY STREET

The sound of the accelerating car and street noise, as well as Alix's last line of dialogue from the previous scene slow down and drop in pitch. The sound of an unknown voice (later revealed to be Ethan's mother SUSAN LEWIS) is heard whispering in his left ear.

SUSAN

Ethan, this way.

If Ethan presses the left key within the allotted time, the speed of the car and traffic returns quickly to its normal speed and pitch and Ethan's body can be heard hitting the pavement where he dove to avoid the car. Alix runs up to him and helps pull him to his feet, her footsteps audible on the concrete.

Alix

Ethan! Are you alright?

Ethan

[*groaning]* Sure, I'm fine, it's nothing.. whoa..

Ethan's body slumps and falls to the ground again, and Alix's voice can be heard calling his name again.

Alix

Ethannnnn!

If Ethan does not dodge the car and is hit by it, Ethan cries out in pain and scene returns to normal time and pitch. The car then screeches away behind Ethan.

Alix

Ethannnn, no!

SCRIPT – CIN5 – '*FLASHBACK*'

INT: COLUMBIAN MILITARY HOSPITAL

The scene takes place in Ethan's mind while he is unconscious. Soft strings play in the background to emphasise the 'dream' state, and ventilators and other hospital machinery beep and hum nearby as Ethan, a baby, lies on his back in a crib, breathing and gurgling softly, while his parents stand over him to his left and right, and discuss their son.

Susan

Tim, how can you be so calm about this? We were the only couple *stupid* enough to go through with a pregnancy on the base. You know what happened to all the others. They.. they changed.

TIM

Susan, dear, we came to work here for a reason. We're scientists, and we *knew* the risks and we still went ahead with this. We have to be strong, and we have to believe that Ethan can fight this. I know he can.. and you do too.

Susan

He's been exposed to more than any of the others, I've been here over a year and so have you. He's the only child that's ever been conceived on the base Tim – nobody else was this reckless. The radiation levels here are dangerous even to us. What chance does a newborn have? He isn't a guinea pig, he's our son.

Tim

Susan, don't you think I know that? I love him just as much as you do. All I'm saying is that we HAVE to stay positive about this. There's no concrete proof that says he'll even be affected at all. It's a chance, that's all. We have to remember that.

Susan

..you're right. We do. I'm sorry honey, this is just all so much to bear.

Susan can be heard moving round the crib to the same side as Tim, where the two are heard to kiss and embrace.

Tim

We're just going to need to be sure that we're there for him, no matter when and how he needs us. That we support him, that he grows up hearing *our* advice and understanding that he can be anything he wants to be.

Tim can be heard rustling through sheets of paper and documents, before reading a passage aloud to his wife.

"July 9th – subject's apparent ESP appears more prevalent with each week that passes. Subject appears able to determine the thought processes of even those outside the complex. We can only surmise the rate at which these.. puzzling abilities will intensify with age".

Susan

Tim, please. You're scaring me.

At this point, we hear the baby Ethan beginning to cry, apparently after sensing his mother's increased state of fear. Ethan's mother can be heard lifting him from the crib, and holding her beside her head, shushing him softly. The scene then fades and the flashback sequence ends.

SCRIPT – CIN6 – *'GUARD PUZZLE SET UP'*

EXT: SUBDUED CITY BACK STREET

The scene begins abruptly with the sound of passing cars and external sound, and Alix shouting to Ethan for him to wake up.

Alix

Ethan! ETHAN! Wake up!

Ethan

Ohh, God. Alix?

Alix

Yeah, it's me, Eth. Man, you really had me scared there for a sec, boss!

Ethan

How long was I out?

Alix

Well with your eyes it's sometimes hard to tell, but as for this little.. incident.. only ten minutes or so. Don't worry, the tower is right ahead, we're still in good time, but we need to get going, now!

Ethan

Oh, you're a charmer, Alix.

Alix

[*giggling]* Come on!

Alix and Ethan's footsteps are heard walking forward on pavement as they resume their conversation and draw closer to the tower.

Ethan

So how exactly are we going to get inside? I'm pretty sure Dexter will have the place locked up pretty tight. He'll have all sorts of games planned for us, I'm sure.

Alix

We're just going to have to be.. creative. Come on, you know me, would I steer you wrong?

Ethan

Was that a joke?

Alix

No!

Alix and Ethan's footsteps slow as they reach the outside of the tower. All pedestrian noise has faded out entirely at this point and the only sounds are of occasional passing cars.

Ethan

So, what have we got?

Alix

Okay, there's just one guard by the main door.. which is wide open. I guess their air conditioning must be broken?

Ethan

Thank heaven for inept repair servicemen! I guess that makes things a little simpler, huh? So what do you propose we do?

SCRIPT – INT4 – *'ENTRY DECISION'*

EXT: SUBDUED CITY BACK STREET

The scene begins where the last one left off, with Alix explaining the potential entry routes into the Teracorps tower.

Alix

We have two real choices here. This guard doesn't look too menacing, frankly. I think I can take him. It's pretty scary, but we both know I've been practising my hand to hand skills with the intention of using them, and now's as good a time as any to find out what I'm made of. Doesn't look like there's anyone else around, and I'm sure I could be very.. discreet about it! [*giggling]*

Ethan

And, do we have any other options?

Alix

There's always the evasive route. We could take a sneak around the tower, and see whether there's another way inside. These places usually have fire exits and delivery entrances all over them, but there's always the chance that they might have considered that to keep people like us out.

Ethan

So, you want me to decide?

Alix

Hey, you're the brains behind this operation. I'm just the eyes.. and the fists. I'm happy to take him out, or if you'd rather not risk arousing any other suspicion, we can go for a quick recon take around the building. Your call, boss!

Ethan is then given the choice. The **GUARD** *can be heard to Ethan's right, whistling to himself, indicating that a press of the right key will choose the option to fight, and left will choose the alternative route. If Ethan chooses the direct route, the scene plays out as follows:*

Alix giggles, and approaches the guard.

Alix

Hey, excuse me Sir, I was wondering whether you guys had a telephone I could use, my mobile's battery is completely tapped.

GUARD

Duhh.. I'm not sa'pposed t'a let anyone in here, girly. I'm just sa'pposed t'a guard this door, an' tha's what I'm g'anna do.

Alix

Ohh, that's too bad.

Alix pounces, and we hear the sound of her grabbing the guard, kneeing him in the stomach, throwing him to the ground, and kicking him once more. The guard can be heard groaning, and then goes silent.

He's out, boss! That was *too* much fun. Come on, quick, let's get him inside and hide him somewhere.

Ethan and Alix's footsteps can then be heard entering the building hurriedly, dragging the unconscious body of the guard with them.

If Ethan chooses the alternative route, the scene plays out as follows:

Alix

Gotcha. Right, come on, let's go!

The sound of footsteps and attempts to open various locked doors can be heard as they circle the building, eventually returning to their starting point.

Ethan

Well, that was a bust. Looks like we don't have a choice. You still think you can take this guy?

Alix

Looks like I don't have a choice, doesn't it?

The same sounds of Alix springing, grabbing, kneeing, throwing, and kicking the guard can be heard. The guard can be heard groaning, and then goes silent.

He's out, boss! Come on, quick, let's get him inside and hide him somewhere.

Ethan and Alix's footsteps can then be heard entering the building hurriedly, dragging the unconscious body of the guard with them.

SCRIPT – CIN7 – '*TOWER ENTRY*'

INT: TERACORPS TOWER LOBBY

The scene begins where the last one left off. Ethan and Alix are dragging the guard inside the tower. The ambient sound here is non-existent and the footsteps are along a large, echoing marbled floor. The sound of the guard being let go can be heard as he is dropped behind a desk out of sight.

Alix

I think we'd better leave him here. I can't see anyone here, it's quite eerie. Why would they just leave one hapless guy on the door, and why is it so empty inside?

Ethan

It's Dexter we're talking about here. He's not exactly known for his conventionality. He knew we were coming, and he knew we weren't one for a live remake of Die Hard. I'm not saying we're safe, but I've got a feeling he's playing with us.

Alix

Why can't we ever get into situations with *sane* bad guys?

Ethan

You know as well as I do – that's just not the M.O. of your average villain. You know that fine line between genius and insanity? This guy's basically wiped it out.

The sound of electrical static can be heard coming from behind the counter where the guard was just dragged.

What was that?

Alix

I don't know, what was it? Did you hear something? I didn't. [*begins to hum 'Iron Man'*] Curse you, Ozzy Osborne!

Ethan

Coming from behind the counter there. It sounded like a mobile phone or something. Would you take a look?

Alix

Yeah, sure. I don't know how I always forget your bat-genes, boss.

Alix checks behind the counter and picks up a walkie-talkie from inside one of the pockets of the unconscious guard.

You weren't far off – it's a walkie talkie. Not sure why it was crackling, it doesn't seem to be making any noise now.

Ethan

Maybe someone was checking in on Sleeping Beauty over there? I'd assume he's supposed to check in with someone at regular intervals. Y'know, so they can remind him to breathe and blink and stuff?

Alix

Normally I'd agree with you, but I'm not..

Alix is interrupted by the crackling static of the walkie talkie once again. The static doesn't last more than a few seconds, and soon, Dexter's voice can be heard over the airwaves.

Dexter

Ethan Lewis, at long last you've arrived. I trust you had no issue conquering your first challenge? If Lars could truly be *called* a challenge, with his.. limited capacities.

Ethan

Nice to know even psychopaths like you are equal opportunities employers, Dexter.

Alix

Dexter! You son of a bitch, where's Juliet? We're coming for her, and your head!

Dexter

My dear girl, such language is most unbecoming for a young thing such as yourself. Ethan's sister still lives. We are on the roof. In consideration of your reduced.. ocular abilities, I fitted the elevator buttons with Braille. I suggest you begin your ascent immediately. I await your arrival.

The walkie talkie begins outputting static again as Dexter ends the communication, returning silence to the lobby.

Alix

Come on, Ethan! The elevator's over here.

The two can be heard walking on the marble floor, crossing the wide room to the elevator. Alix can be heard pushing the button, and the doors open.

Ethan

After you.

Alix

Ever the gentleman.

Hey, what gives! The buttons are all stuck, and there's a kind of keypad thing. The only button that's lit up is for the second floor.

Ethan reaches and pulls a small note off the back wall of the elevator, which had been held in place with tape. He unfolds the note and hands it to Alix, who reads it aloud.

Having fun, children? You'll find this elevator quite out of service. In order to climb higher than the second storey of my impressive workplace, you will need to find the password. I'd recommend beginning your search on the second floor. The password you will find there will allow you to pass onward to the third floor. You will then need to find the password for the fourth floor, located upon the third. The path to the roof is located upon the fourth floor. I await your arrival.

[*sighing]*Let the games begin, Ethan. We'd better get up to the second floor, right?

Ethan

Agreed.

Alix pushes the button for the second floor and the elevator doors can be heard closing, as the elevator rises to the second floor.

SCRIPT – CIN8 – *'SECOND FLOOR'*

INT: TERACORPS TOWER SECOND FLOOR

The scene begins with the elevator doors opening and Ethan and Alix stepping out onto the second floor. They find themselves in another large room.

Alix

Okay, so he said the password for the elevator keypad was somewhere on this floor, right? We should start looking. I mean, uh.. I'll look, you..

Ethan

Y'know, a lesser man could take offence with these constant jokes at my expense.

Alix

Let's just see what we can find, hmm?

Ethan

Agreed.

Ethan and Alix begin moving around the room, Alix looking, and Ethan feeling his way around. After a short while, Ethan hears Alix cry out, and walks over to her.

Alix

Ethan! Get over here!

Ethan

What is it?

Alix

Looks like I've found our password. Also looks like it's sitting in this tank of water, guarded by three of the meanest looking fish I've ever seen in my life. Seriously, these guys make Jaws look like Nemo. Be thankful you can't see them, as I think they've currently deciding which one of us they'd rather pick clean.

Ethan

Well, we're going to have to get that password out of there somehow.

SCRIPT – INT5 – *'PIRANHA PUZZLE'*

INT: TERACORPS TOWER SECOND FLOOR

The scene begins exactly where the previous one left off, with Ethan and Alix discussing which one of them is going to reach into the tank to retrieve the password.

Ethan

Do we rock, paper, scissors for this?

Alix

No chance, boss! This one's on you. My crippling ichthyophobia seems to have set in, there's no way my hand's going anywhere near those things.

Ethan

[inhaling deeply] Okay, looks like it's on me.

Alix

It looks like they're swimming around in a big circle. Maybe that thing about the short-term memory is true? The paper's at the right side of the tank, so if you were to reach in when they're at the far left side, you might have a chance of keeping at least a COUPLE of your fingers.

Ethan

Hey, as long as I can still hold my coffee.

Alix

Good luck, boss.

Ethan pays close attention to the tank and the sound of the swimming fish can be heard sweeping from right to left and back again. At a specific interval at each time, the voice of Ethan's father can be heard whispering in his right ear the word "Now", indicating that this is the correct time to grab the password.

If Ethan chooses the right moment, he can be heard splashing his hand into the tank, grabbing the paper, and pulling it out rapidly.

You did it! Let's get moving to the third floor, pronto!

If Ethan chooses the wrong moment, the sound of the piranhas feasting upon his hand can be heard, and he cries out in pain, ending the scene.

SCRIPT – CIN9 – *'THIRD FLOOR'*

INT: TERACORPS TOWER THIRD FLOOR

The scene begins with the elevator doors opening and Ethan and Alix stepping into the elevator once again. Alix unfolds the password and is heard typing six numbers into the digital keypad. The elevator then begins to move.

Alix

Whoa! I didn't even need to push the Third Floor button. I hope it knows that's where we need to go!

Ethan

We'll soon see!

The elevator beeps, and opens on the third floor. Alix and Ethan step out onto the third floor and begin looking around for the next password.

Alix

Hmm, I wonder where he could've hidden this one.

Ethan

We'd better get to searching.

Ethan and Alix search the room, and Alix soon spots a small machine on a desk on the corner of the room. She calls Ethan over.

Alix

There's a START button here on the side. Should I push it?

Ethan

What's the worst that could happen?

Alix

Some kind of electrical current could turn me into a pile of ashes on the floor?

Ethan

Perhaps some kind of big metal spikes will start coming down from the ceiling, Temple-of-Doom style?

Alix

Okay, okay, Indy.

Ethan

Just push the button, Short Round.

Alix gulps, and pushes the button. Four audible tones are heard, around a second apart. The voice of Dexter is then heard through the machine.

SCRIPT – INT6 – '*TONES PUZZLE*'

INT: TERACORPS TOWER THIRD FLOOR

The scene begins directly after the end of the previous one, with no gap at all. Dexter's voice speaks through the speaker in the machine.

Dexter

> Congratulations on making it up here. I hope those finely-honed ears of yours are still firmly attached to your head, Ethan Lewis. I'd hate for something unpleasant to happen to my beautiful young captive. Regardless, those ears are about to be put to the test. There are four speakers in the four corners of this room. They will each play a tone. One of these tones is exactly twice that of another. Select that tone, and you win. Select the wrong tone? Well, let's just say that you and your ears will end up a considerably large distance apart. I await your arrival.

Dexter's voice cuts out, and four tones play, from the four directions surrounding Ethan. The tone on the left is one octave above the tone on the top. If Ethan selects left, the scene plays out as follows:

The machine spits out a sheet of paper from a thin slot in the side, which Alix hastily grabs, opens, and reads.

Alix

8-5-4-4-3-6. It's the next password! Come on, let's get back to the elevator!

If Ethan selects any other direction, a large explosion is heard, which abruptly ends the game without any further choice.

SCRIPT – CIN10 – '*FOURTH FLOOR*'

INT: TERACORPS TOWER FOURTH FLOOR

The scene begins with the elevator doors opening and Ethan and Alix stepping into the elevator once again. Alix unfolds the password and is heard typing six numbers into the digital keypad. The elevator then begins to move.

Alix

Well, here we go again. It says that that fourth floor is the top one, so I guess once these doors open, we might be in for a shock.

Ethan

I don't think he's quite through toying with us yet.

The elevator doors open on the fourth floor and Ethan and Alix step out onto the fourth floor. They find themselves in a similar room to the two previous ones.

Alix

Looks like we've got another puzzle to solve. There's nobody here, looks exactly the same as the last two floors. And.. that safe over there.

Alix notices the safe in a corner of the room, and a stethoscope laying on top of it. She walks over to it.

Eth, over here! There's a safe. And.. a stethoscope? Weird, what's he want me to do, check your pulse?

Ethan

Is it the normal, old-school kind of safe, or is it digital?

Alix

Nope, classic Columbo all the way. Looks like the code goes in in four directions, up, down, left, and right. Seems pretty old too, actually. Reckon I could grab this on the way out? I'm a sucker for a nice antique.

Ethan

Let's just focus on getting it open, okay? So, it's a rotary combination lock? The stethoscope.. I'll wear it, and hold it over the lock. If you turn the dial, slowly, I'll let you know what direction the latches click at, and we'll have this baby open in no time.

Alix

I never had you pegged as a safe-cracker, boss!

Ethan

A man's gotta keep some secrets, Alix. Now, let's do this.

SCRIPT – INT7 – *'SAFE PUZZLE'*

INT: TERACORPS TOWER FOURTH FLOOR

The scene begins as the continuation of the previous one. The puzzle begins here, with the stethoscope over Ethan's ears, and Alix about to begin turning the dial of the safe.

Alix

Okay, I'm going to turn the dial slowly. You'll hear it clicking whenever I turn it, and I'll let you know which direction it's pointing at, but on one of the directions, you should also hear the pin of the lock sliding away. I'll spin it a few times in case it's difficult to hear. After I'm finished, let me know which direction it came from, okay boss?

Ethan

Okay, I've got it. Piece of cake.

The sound of clicking is heard coming from the stethoscope on Ethan's ears, as well as Alix, outside of them, calling out the four directions with each click. Alix says each direction four times, and on each time she says "bottom", a faint click is heard. If Ethan selects the correct direction, the safe door springs open.

Alix

Yup, looks like we got it! What's inside?

Alix reaches around inside the safe, and pulls out a single key on a key-ring.

Alix

Aha! A key! And hey, there's only one door in this room, and we're certainly not going back into the elevator.

Ethan

Well then, it looks like this is it. Onward and upward!

If Ethan selects an incorrect direction, the scene continues as follows:

Alix

It won't open! Are you sure that was the right combination?

Ethan

Well, now that you mention it, I was kinda guessing. Sorry. Let's try that again.

Alix

Ethan! Come on, we both know your ears are more than capable of this. Pay attention.

The scene then repeats, with Alix giving the four directions again, and continue to repeat until Ethan gives the correct response.

SCRIPT – CIN11 – '*ROOF ENTRY*'

EXT: TERACORPS TOWER ROOFTOP

The sound of two sets of hurried footsteps climbing the staircase can be heard as our duo rush to the top of the flight. They can be heard bursting through a closed door at the top and emerge on a rooftop, the sounds of the busy street below them. Dexter is standing at the far end of the room, his arm around Juliet's neck, the other holding a gun to her temple. She is breathing heavily in fear. Ethan and Alix stand still, and Dexter speaks.

Dexter

At long last, the dynamic duo arrive. It's a pleasure to see that you are both alive and well. It appears you were able to conquer my traps with ease.

Ethan

Let my sister go, Dexter. It's all going to end here, and you're not taking her from me!

Dexter

You mean the way your parents were taken from you, Ethan Lewis? The way they were killed, in cold blood? It was I, Ethan. I killed them both. Your mother called our your name as I drove a bullet deep into her heart.

Ethan

My.. my parents? My parents aren't dead! They're still working underground at the facility in Columbia! You.. you bastard. I'll kill you!

Dexter

Ah yes, the 'quest for revenge' – how clichéd. I expected no less from you, Ethan Lewis. You know all about clichés, don't you? The superhero, with his superhuman powers. That's right, I'm fully aware of your.. shall we say.. unique condition? I'm also fully aware that your sister possesses the same exact trait. And that's why I have brought you both here. I intend to test those powers of yours – I believe your father called it.. MiNDSiGHT?

Ethan

I don't have a clue what you're talking about, you madman!

Dexter

Then I suggest you learn, Ethan Lewis. And, quickly.

Dexter pushes Juliet to the floor, cocks his gun audibly and points it at Ethan. Alix cries out to warn Ethan.

Alix

Ethan! He’s got his gun trained on you! Get out of the way, quickly!

Dexter

This is it, Ethan Lewis. Your final chapter has been told. You’re finished, don’t you.. see? I’m going to end this, here and now. I challenge you, and that heightened awareness of yours. Think you can outsmart me? Let’s see what this MiNDSiGHT of yours is capable of. It ends, now, Ethan Lewis!

SCRIPT – INT8 – *'ROOF PUZZLE'*

EXT: TERACORPS TOWER ROOFTOP

This puzzle begins as Dexter fires the first shot from his pistol at Ethan, from his right. Alix and Juliet are both heard screaming out Ethan's name. Time is heard slowing down and the bullet and all external noises slows with it, meaning the bullet can be heard approaching Ethan gradually from his right. The voice of Ethan's sister Juliet is heard whispering in his ear, via her MiNDSiGHT ability.

Juliet

Right, Ethan. RIGHT!

If Ethan presses right, he can be heard diving to the floor to his right, dodging the bullet. Dexter then can be heard from Ethan's right, and the process is repeated with Ethan needing to dive left. Finally, the shot comes from behind Ethan, and he needs to jump backwards to save himself. If Ethan fails any of these, he can be heard screaming out in pain as the bullet hits him, and the scene fades.

SCRIPT – CIN12a – *'BOSS DEATH'*

EXT: TERACORPS TOWER ROOFTOP

Ethan can be heard hitting the ground as he dives to avoid the final bullet. Time remains slowed down, and Juliet's voice can be heard all around Ethan, coming from a dozen places at once.

Juliet

Ethan. We have an incredible amount of mental strength. I will explain later – it's because of where we grew up, because of Columbia. This is our chance. You have to believe that you can do this. You have to believe that if we both focus our thoughts on Dexter, we can take him down. Just think Ethan, focus your mind. FOCUS!

Ethan

I can do this. You're going down, Dexter, once, and for all.

A sound effect plays, indicating that Ethan and Juliet's powers have combined, and Dexter cries out, being pushed backward by an unseen force. He can be heard struggling, and falling from the ledge, off the tower, plummeting to the ground below, screaming.

SCRIPT – CIN12b – *'ENDING SCENE'*

EXT: TERACORPS TOWER ROOFTOP

This scene picks up exactly after Dexter falls from the Teracorps tower. Juliet runs up to Ethan, and the two embrace.

Juliet

Oh, Ethan, we did it!

Ethan

We sure did, sis! I'm always looking out for you, remember?

Juliet

I knew you'd come. I just, knew. And now that you know about our.. abilities.. oh Ethan, this is going to change everything!

Ethan

I'm sure it will! Though I must admit, it's going to take some getting used to. I never thought much of psychics.

Alix

I’m so glad you’re okay, Juliet! I was pretty worried, but Eth kept so calm.. he was amazing, really.

Juliet

That’s my brother!

Ethan

So Alix, are you still worried the city’s dry of crime?

Alix

Hey, bring on the next super-villain boss. I’m ready when they are!

·Appendix B·

Floor Plans and Prop Lists

Floor Plans

In order to envision the environments that the characters of MiNDSiGHT would populate, specifications were drawn for each of those environments. The following pages are dedicated to those floor plans, detailing the positions of each audio prop used in each scene, and the positions (whether in motion or stationary) of the characters that inhabit those scenes.

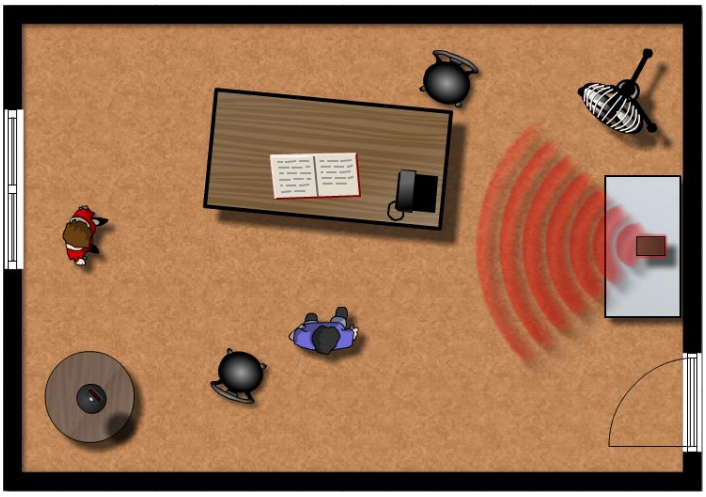
While these diagrams are not accurate to the last degree where dimensions and bearings are concerned, they give an idea of a visualisation of each scene to accompany the audio. Since specific elements of mixing were revised after this documentation went to press, certain prop locations may vary from their depiction here.



This first scene shows the layout of Ethan's office at the game's outset. Ethan is shown in the purple clothing and Alix in the red, sitting opposite him at

the desk. The radio is shown on a small table at the lower left of the room, the fan to the upper-right. The table on the right has the fob for the office keys on it, though it is not audible at this point. The door can be seen in the lower-right of the office.





This plan shows the room after Ethan has pushed over the desk. Alix moves around to his left and crouches beside him, to comfort him.

When Alix moves to the left of the room to begin pushing the button on the key locator, the sound originates (as represented by the red waves) from Ethan's right.



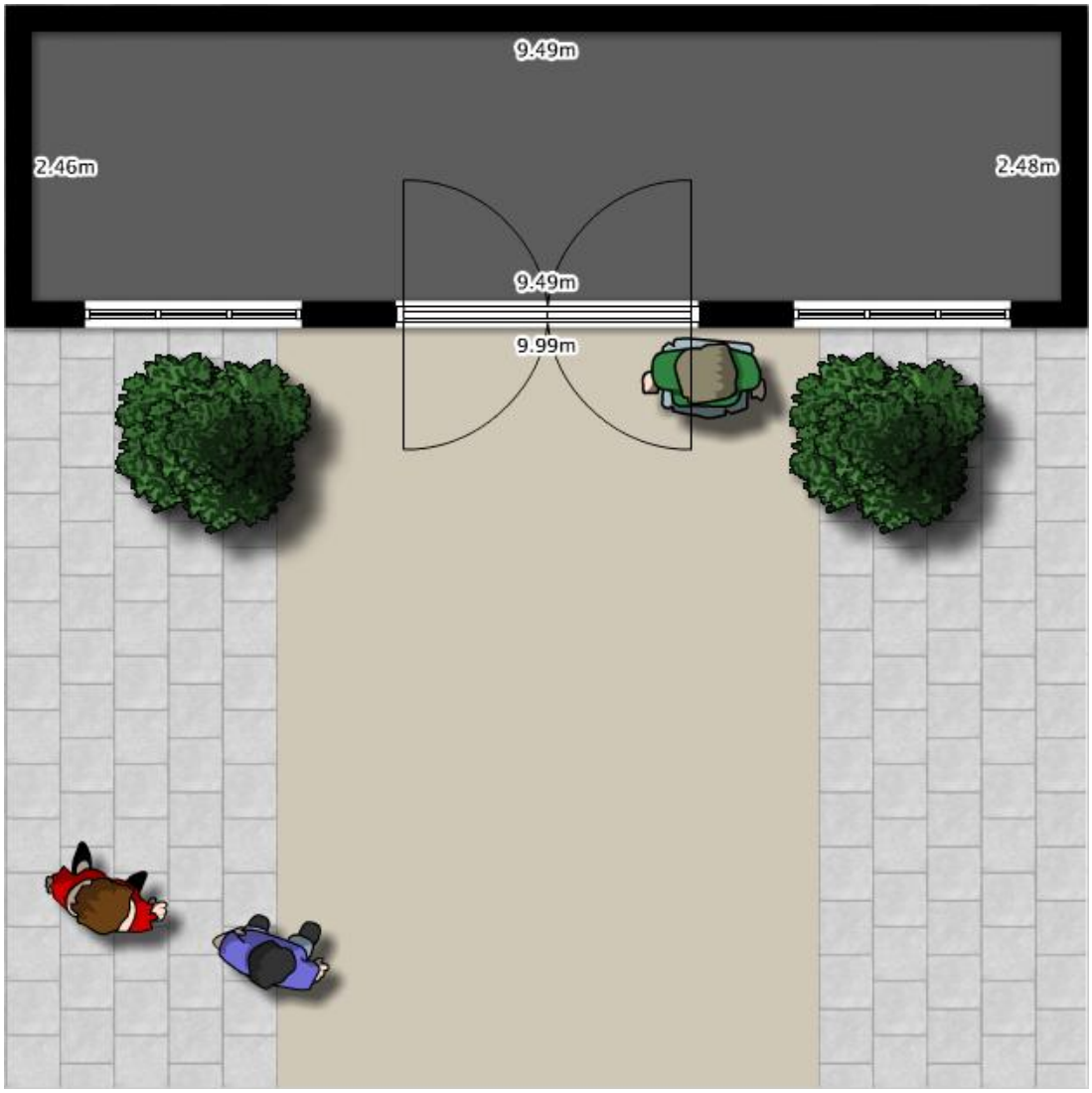
This plan shows the crowded street upon leaving the office. Numerous passers-by pepper the streets and cars pass in both directions. Alix is standing to the right of Ethan here.



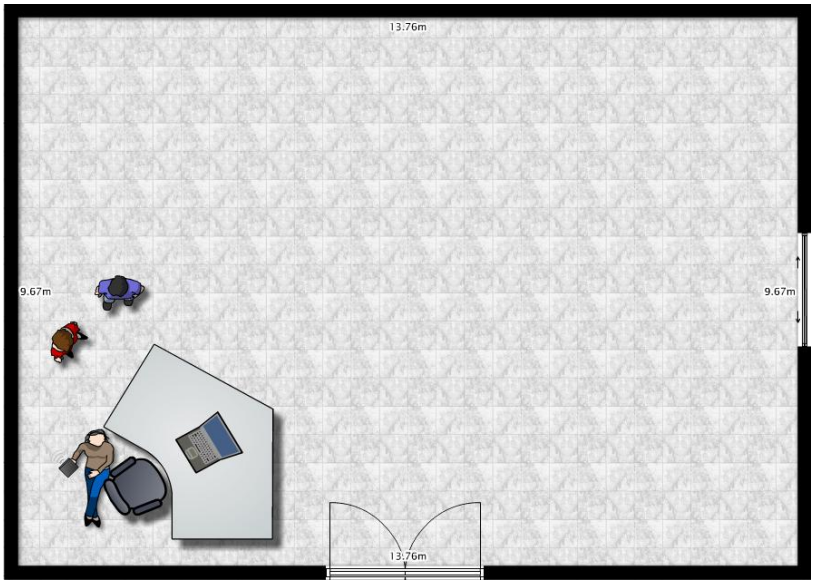
On a quieter street, a car swerves off the road from Ethan's front-left, and he must leap to his right to avoid it.



The flashback scene which occurs in a Columbian military hospital has baby Ethan in a crib with his mother to his left and his father to his right. They exchange lines in these positions, then Sarah moves around to join Tim at the right side of the crib. The machinery at various other points in the room creates ambience.



This scene shows Alix to Ethan's left as they stand outside Teracorps. The guard, Lars, is stationed to the right of the large double doors. Alix approaches and engages him in the central area.



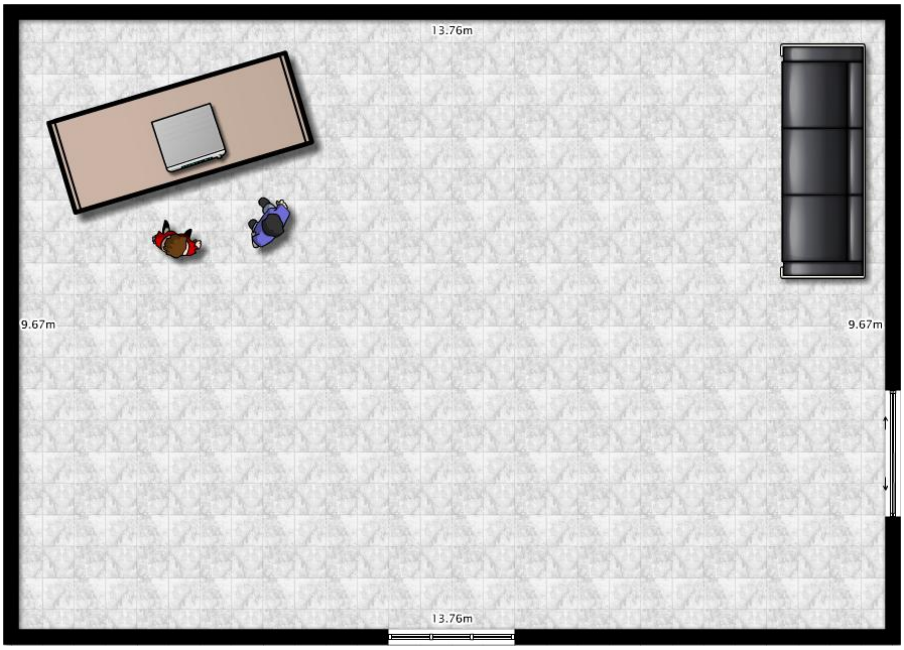
This is the first floor of the Teracorps building. Ethan and Alix drag the unconscious Lars behind the desk at the left. Note the size of the room, it is intended to be a large, echoed chamber with marbled floors.



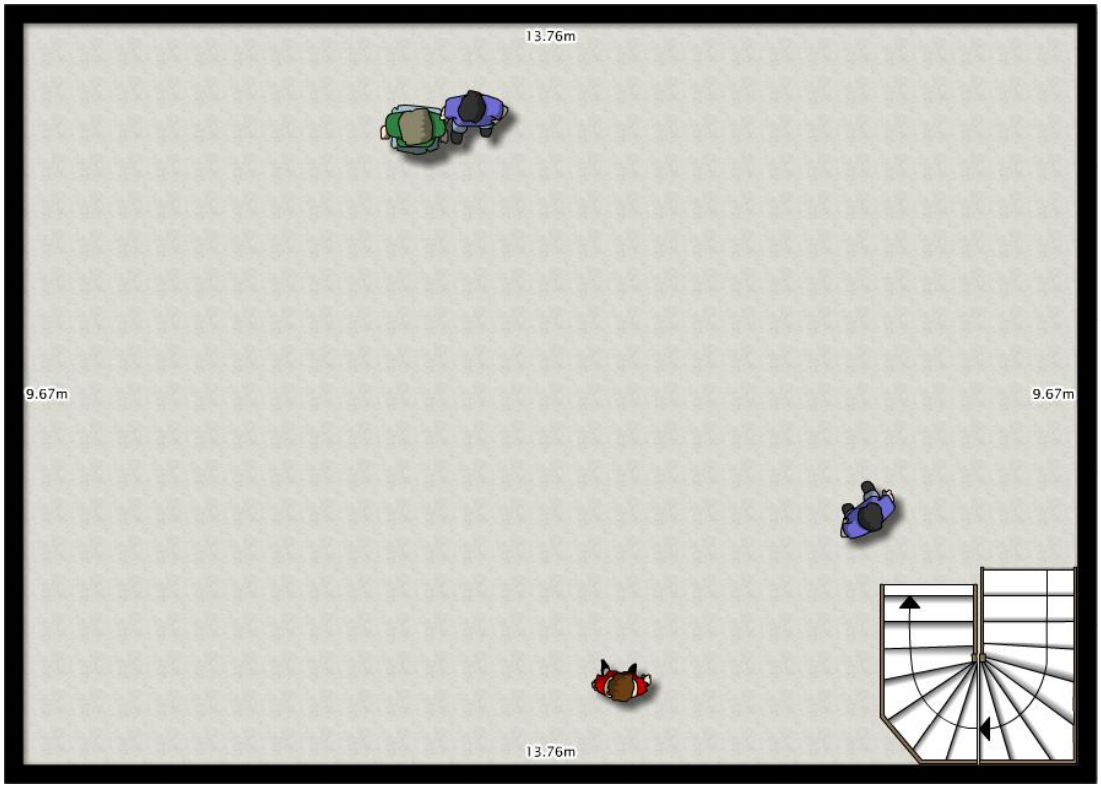
The second floor, much like the first, has the same marble flooring. The elevator at the right side of the room opens to allow Ethan and Alix to engage in the piranha puzzle by the tank at the upper-right of the large reverberating room.



This third floor room is the same size as the previous, and features a central table with the machine that tells Ethan what he needs to do here. The four speakers are positioned on the walls at the four compass directions above Ethan, near the ceiling.



The fourth floor features only a table in the corner, which has the safe and stethoscope on it.



The final scene takes place on the vast and wide rooftop of Teracorps. Ethan and Alix enter via the stairs at the lower-right, and Dexter is holding Juliet at the opposite side of the ledge.

Prop Spreadsheet

The following spreadsheet details the initial specifications for each of the audio props used to provide ambient or diegetic sounds to MiNDSiGHT.

Scene	Location	Type	Prop	Use
CIN1	Office	Ambient	Portable Radio	Ambient
		Ambient	Desk Fan	Ambient
		Prop	Desk Telephone	Dexter calls Ethan on this phone
		Prop	Desk	Tapping fingers, hurled to floor
		Prop	Door	Rattling Handle
		Floor	Carpeted	Items hurled to floor, footsteps
CIN2	Office	Ambient	Portable Radio	Ambient
		Ambient	Desk Fan	Ambient
		Prop	Desk	Ethan sits at this desk, puts his hands on it
		Prop	Chair	Pushed backwards when Ethan stands
INT1	Office	Ambient	Portable Radio	Ambient
		Ambient	Desk Fan	Ambient
		Prop	Key Locator	Beeps repeatedly until key is found
		Prop	Ring of Keys	Heard rattling when Alix grabs them
		Prop	Door	Unlocked by Alix
	Stairway	Ambient	Staircase	Climbed down by Alix and Ethan
		Prop	Main Door	Opens out to the streets
CIN3	Street	Ambient	Street/People	Busy street
INT2	Street	Ambient	Street/People	Busy street
CIN4	Street	Ambient	Street/People	Busy street, slightly quieter than before
		Prop	Car	Drives at Ethan
INT3	Street	Ambient	Street/People	Busy street, slightly quieter than before
		Prop	Car	Drives at Ethan
		Floor	Concrete	Ethan dives onto the concrete to avoid the car
		Prop	Car Hitting Ethan	Ethan is hit by the car
CIN5	Columbia	Ambient	Musical Strings	Quiet background music

Scene	Location	Type	Prop	Use
		Ambient	Ventilators	Hospital machinery beeping
		Prop	Baby Gurgling	Baby Ethan breathes
		Prop	Mobile	Child's toy
		Prop	Papers	Tim rustles through the pages of his journal
		Prop	Baby Crying	Ethan begins to cry
CIN6	Street	Ambient	Street	Ethan wakes up abruptly to traffic
		Floor	Pavement	Walking up to Teracorps
	Teracorps Ext.	Ambient	Quiet Street	Slight birdsong, distant traffic
INT4	Teracorps Ext.	Ambient	Quiet Street	Slight birdsong, distant traffic
		Action	Beating Guard	Alix kicks ass
		Action	Dragging Guard	Dragging guard along floor
		Floor	Concrete	Footsteps, and dragged guard
		Prop	Doors Opening	Various locked doors are tried
CIN7	Teracorps 1F	Ambient	Silence	Echoes of footsteps
		Floor	Marble	Lots of reverb on footsteps
		Action	Dragging Guard	Guard dragged behind counter
		Action	Dropping Guard	Guard dropped onto floor
		Prop	Walkie Static	Static crackling from walkie-talkie
		Prop	Elevator Button	Pushed by Alix
		Prop	Elevator Doors	Open to allow Alix and Ethan inside
		Prop	Taped Noted	Pulled off the elevator wall by Ethan
		Prop	Elevator Rising	Sound of elevator's ascent
CIN8	Teracorps 2F	Ambient	Silence	Echoes of footsteps
		Floor	Marble	Lots of reverb on footsteps
		Prop	Fishtank	Rustling waters of gnashing fish
INT5	Teracorps 2F	Prop	Fishtank	Rustling waters of gnashing fish
		Prop	Grabbed Note	Ethan grabs note successfully
		Prop	Eaten Fingers	The fish attack Ethan and eat his fingers
CIN9	Teracorps 3F	Ambient	Silence	Echoes of footsteps
		Floor	Marble	Lots of reverb on footsteps
		Prop	Elevator Doors	Open to allow Alix and Ethan inside
		Prop	Note Unfolding	Alix unfolds note to read it
		Prop	Beeping Keypad	Alix types numbers into keypad
		Prop	Tones	Four beeping musical notes

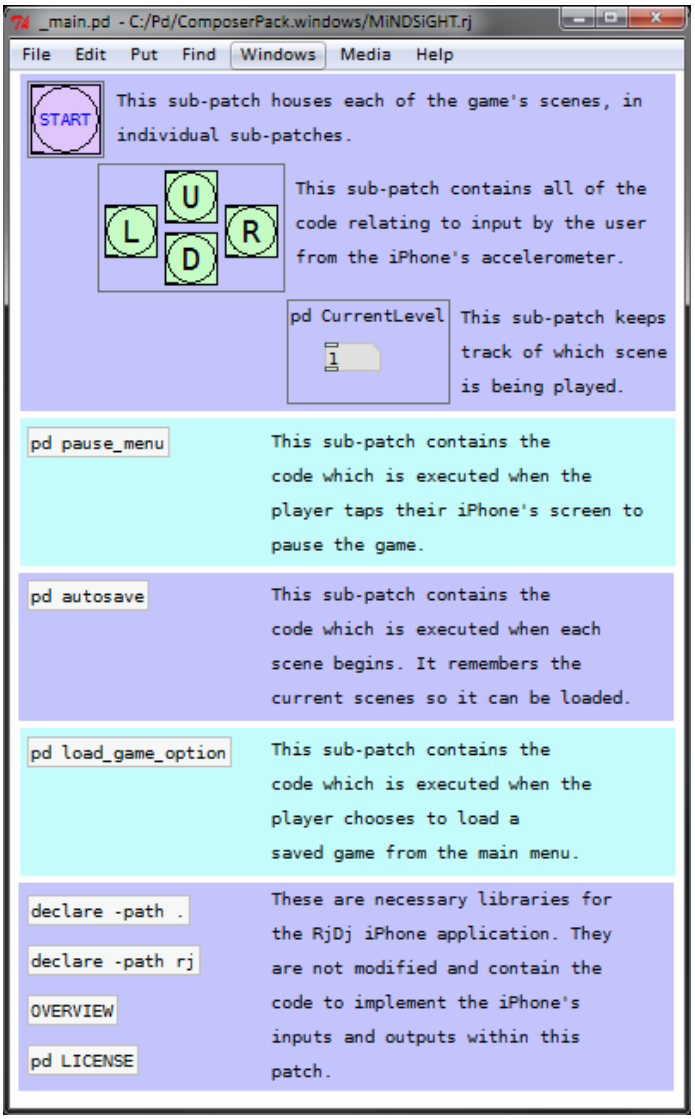
Scene	Location	Type	Prop	Use
INT6	Teracorps 3F	Prop	Tones	Four beeping musical notes
		Prop	Machine Printing	Machine spits out paper
		Prop	Explosion	Machine blows up, killing Alix and Ethan
CIN10	Teracorps 4F	Ambient	Silence	Echoes of footsteps
		Floor	Marble	Lots of reverb on footsteps
		Prop	Elevator Doors	Open to allow Alix and Ethan inside
		Prop	Beeping Keypad	Alix types numbers into keypad
INT7	Teracorps 4F	Prop	Stethoscope	Fitted onto Ethan's ears
		Prop	Safe Handle	Alix turns safe handle
		Prop	Pin Click	Occurs when the safe handle is positioned correctly
		Prop	Safe	Large safe
		Prop	Key	Removed from safe by Alix
CIN11	Teracorps Roof	Ambient	Rooftop	Distant traffic and people below, wind
		Floor	Metal staircase	Running up to roof
		Prop	Gun	Heard being cocked by Dexter
INT8	Teracorps Roof	Ambient	Rooftop	Distant traffic and people below, wind
		Prop	Gun	Fired at Ethan by Dexter
		Action	Ethan Dives	Ethan diving onto concrete floor to avoid bullet
		Action	Ethan Shot	Bullet hits Ethan and he screams
CIN12a	Teracorps Roof	Ambient	Rooftop	Distant traffic and people below, wind
		Prop	MiNDSiGHT Push	Sci-fi'ish noise pushes Dexter from roof edge

·Appendix C·

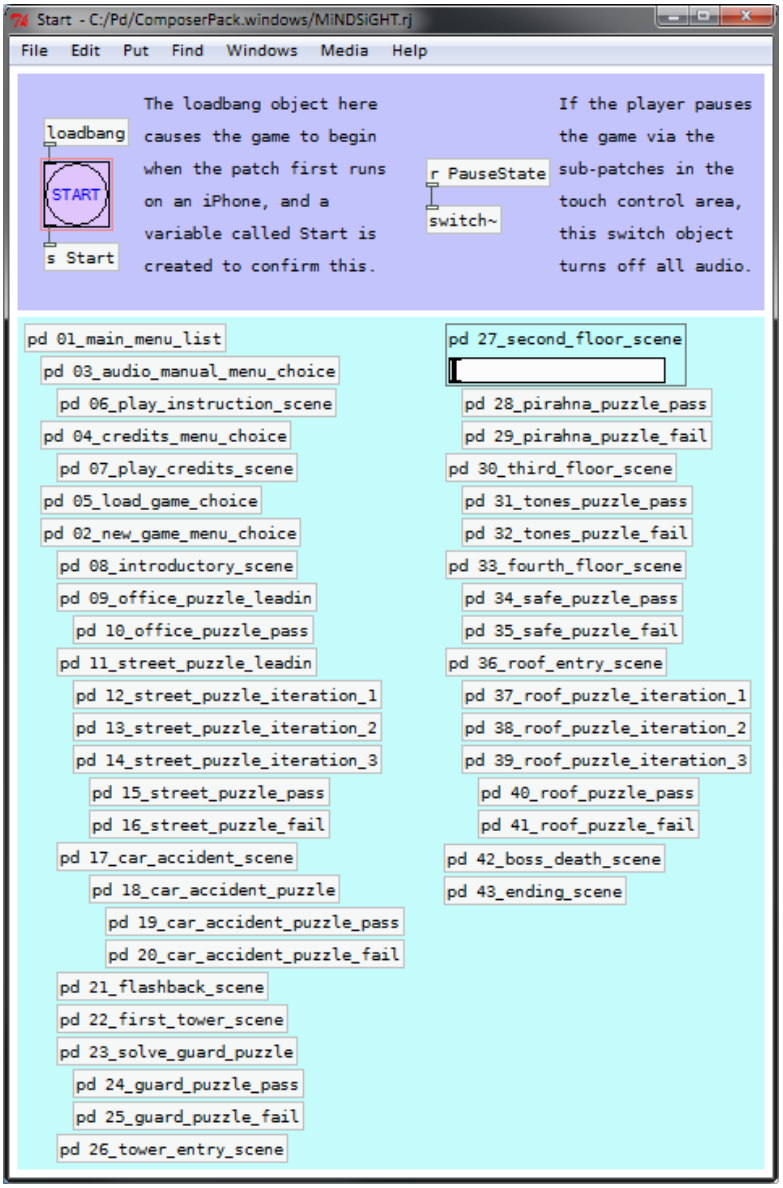
PureData Patches and Digital Documentation

Commented Code

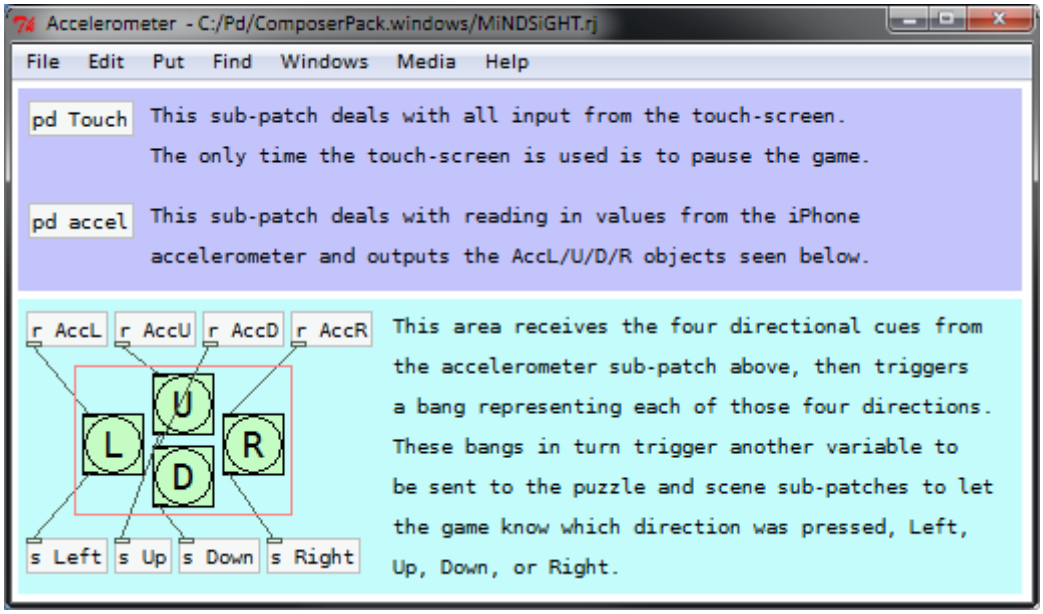
The following pages contain screenshots and explanative remarks of the key pieces of program code which make up MiNDSiGHT's technical side. Each piece of code is commented fully in the PureData patches which can be found on the disc accompanying this project.



The sheet above is **_main.pd** – the game's home. Every RjDj patch running on an iPhone must be named as such, and the patch is commented internally.

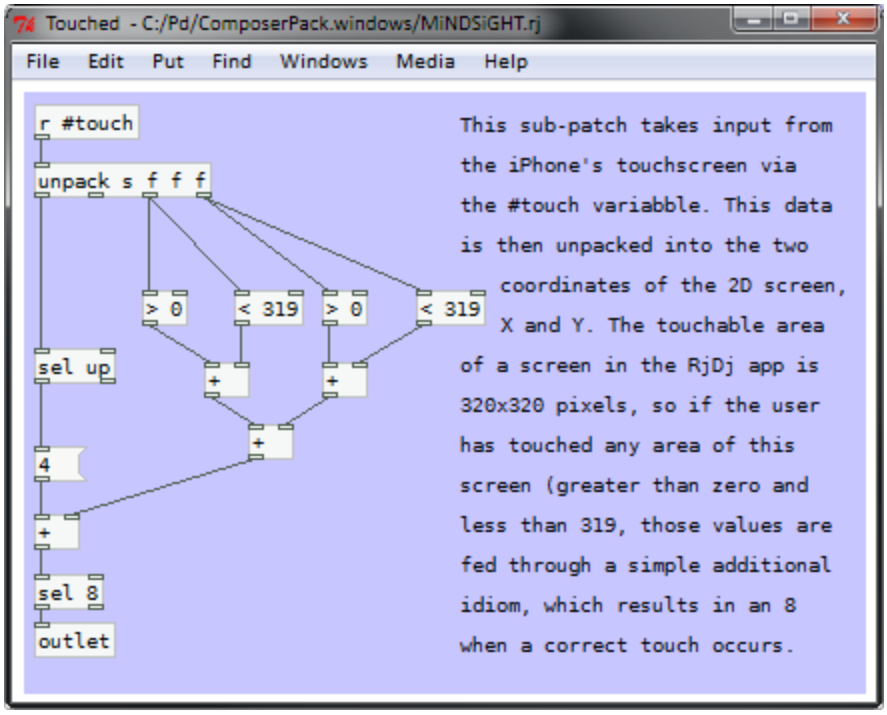


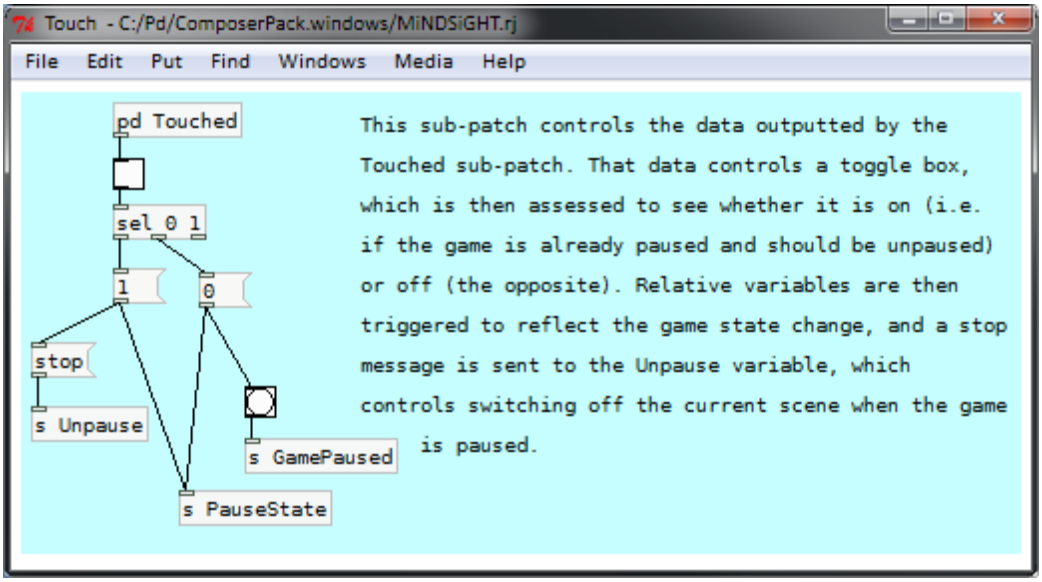
This sub-patch is located within the purple Start button on the main patch, and contains a separate sub-patch for each of the game's 43 scenes or eventualities. The visible slider for patch 27 (at the top of the second column) relates to the position of the moving fish in the piranha puzzle – for more detailed commentary regarding this, please consult that sub-patch within the **Pd** code on the accompanying DVD.



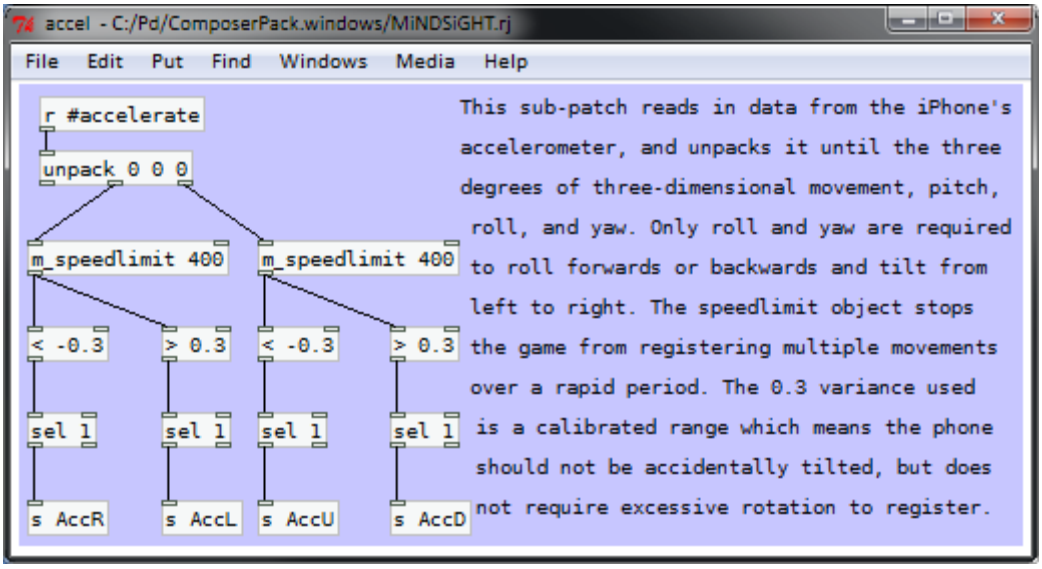
This sub-patch, located within the left/up/down/right buttons on the main patch, handles the control system. Separate sub-patches in the upper pane control the inputs from the touch-screen (for pausing/un-pausing the game), and the accelerometer (for controlling the game commands), and those commands are sent via the variables **AccL**, **AccU**, **AccD**, and **AccR** to the green buttons, which are then sent off to the scene sub-patches in the previous screenshot to control whether Ethan succeeds or fails in his trials and puzzles.

It is possible to play MiNDSiGHT without the iPhone by simply clicking with a mouse on these green U, L, D, and R buttons for Up, Left, Down, and Right, though this is not recommended.

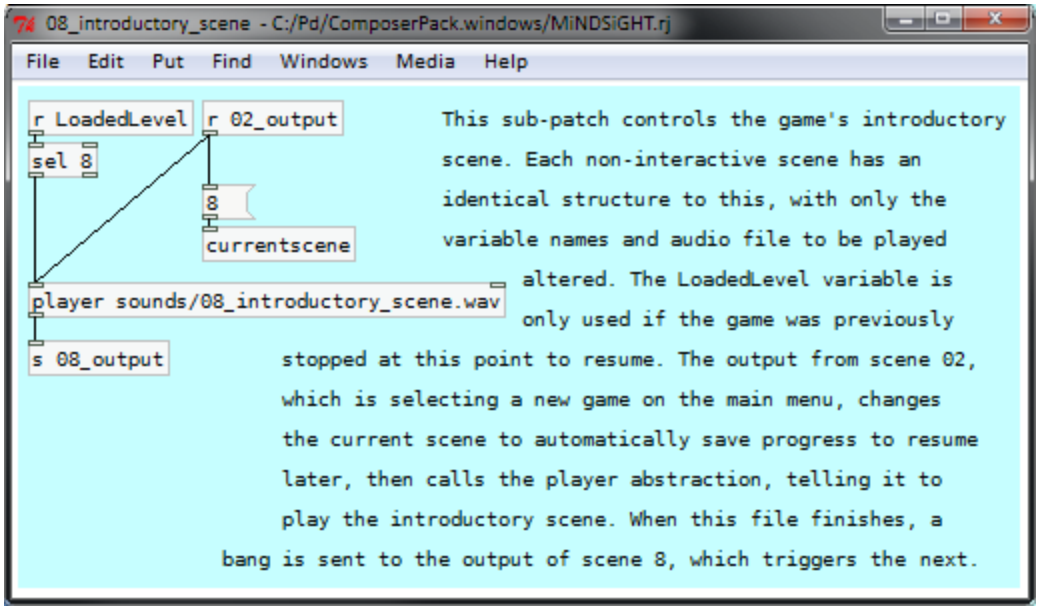




These two sub-patches control the touch-screen input, and pausing and un-pausing the game. Data is read from the iPhone's touch-screen via the top patch, then fed into the bottom one where the state is assessed, and if the game is already paused, it acts as an unpause, and if it is not paused, it acts as a pause – effectively it toggles the state.

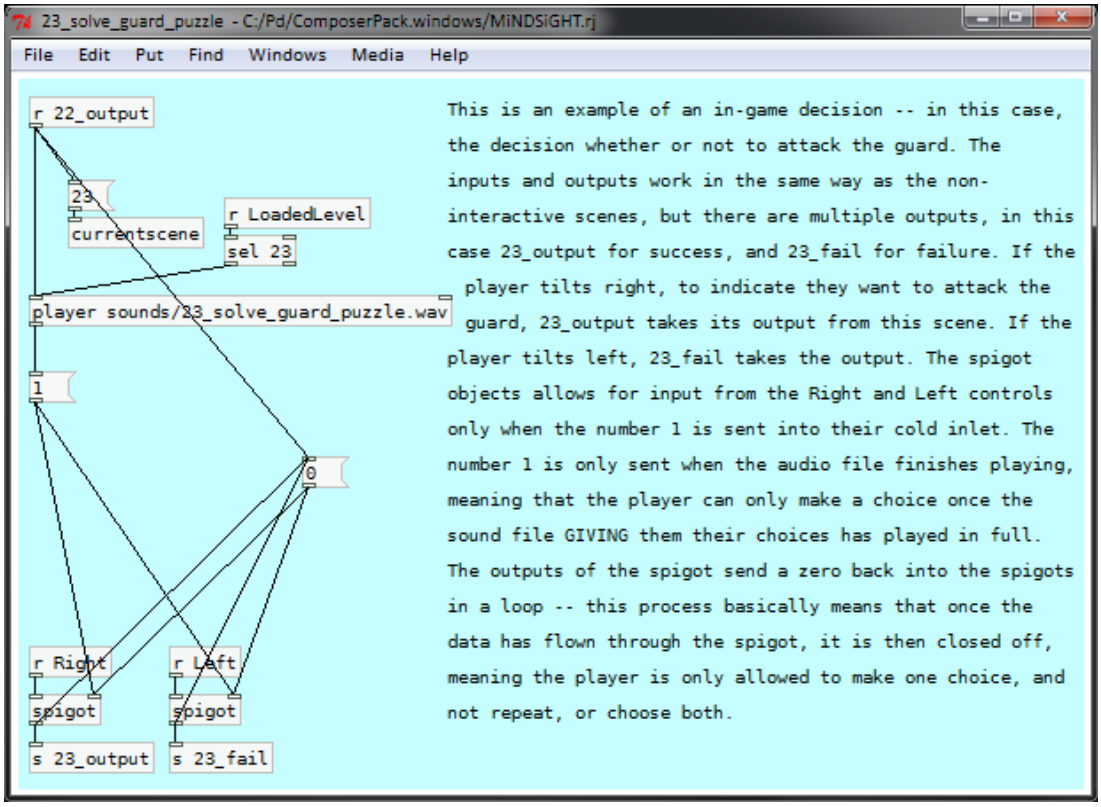


This sub-patch controls accelerometer input, which is taken, unpacked, and the relevant values are used to measure user movement upward, downward, left, or right. The **`#accelerate`** object outputs in radians, meaning that 0.3 (either positively or negatively for each of the two axes, which are zero when the phone is held upright) is a suitable range to register movement in those directions. Calibrating this value took a large amount of testing to find the ample rotational radius that was required to make a movement intentional, but not overbearing on the user's wrists. The resulting values are then sent out to the values read by the green directional buttons which control the actions Ethan takes in-game.

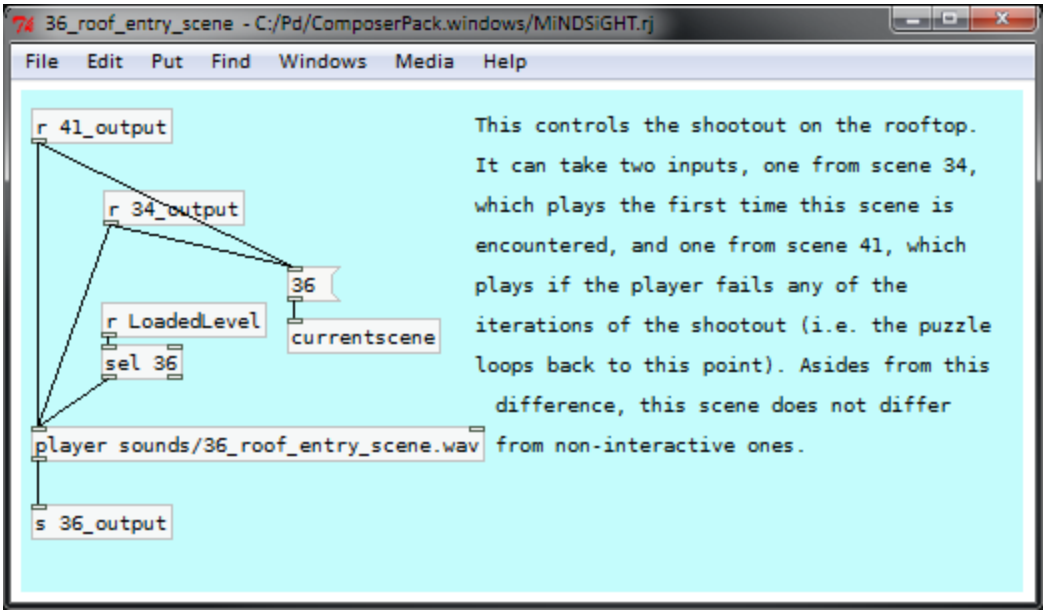


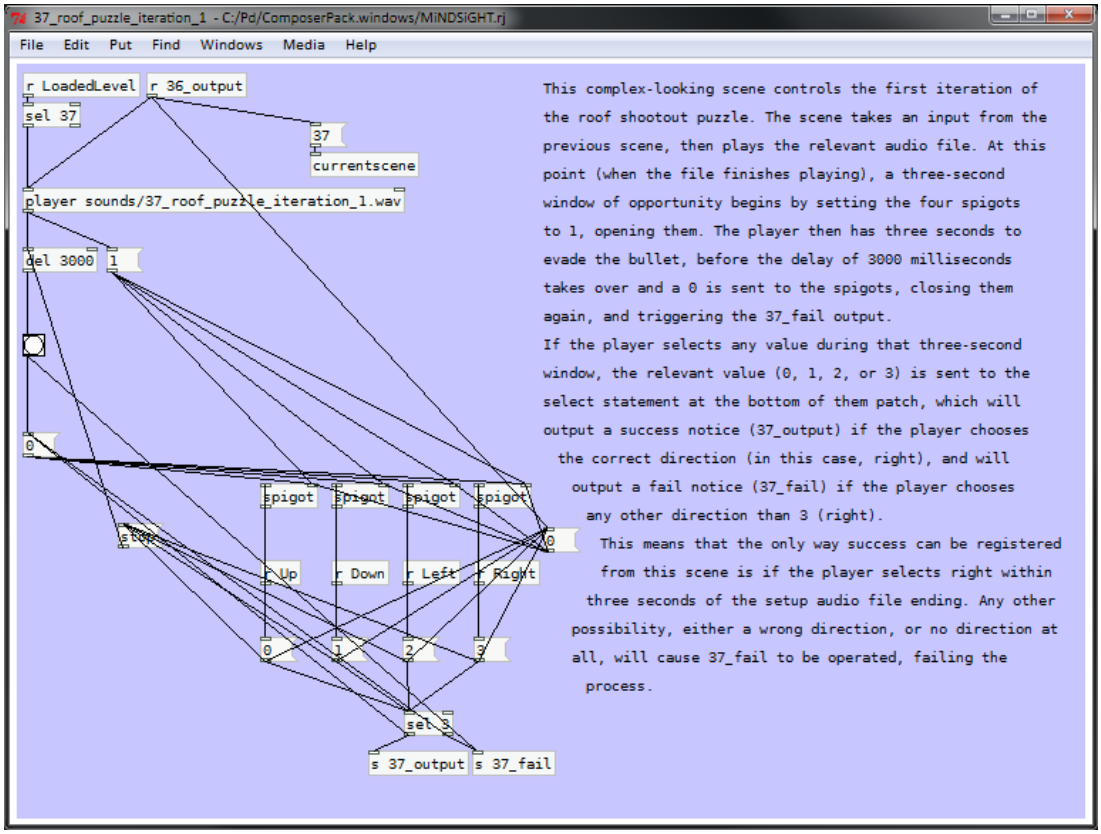
This sub-patch is a frequently-used idiom and controls all of the non-interactive scenes in the game. The sub-patch is sent an instruction to begin from the scene that precedes it, the game is auto-saved at this point, then the player abstraction plays the scene's audio. At the point where the scene's audio file ends, a bang is sent to this scene's output (at **`08_output`**). This scene's output is then picked up at the beginning (where **`02_output`** is in this example), and the next scene is instructed to play.

This means that when one scene finishes, the next will begin automatically, and thanks to the per-sample basis of PureData's audio engine, there will be no audible gap between one scene ending and another beginning, meaning a seamless transition from scene to scene.

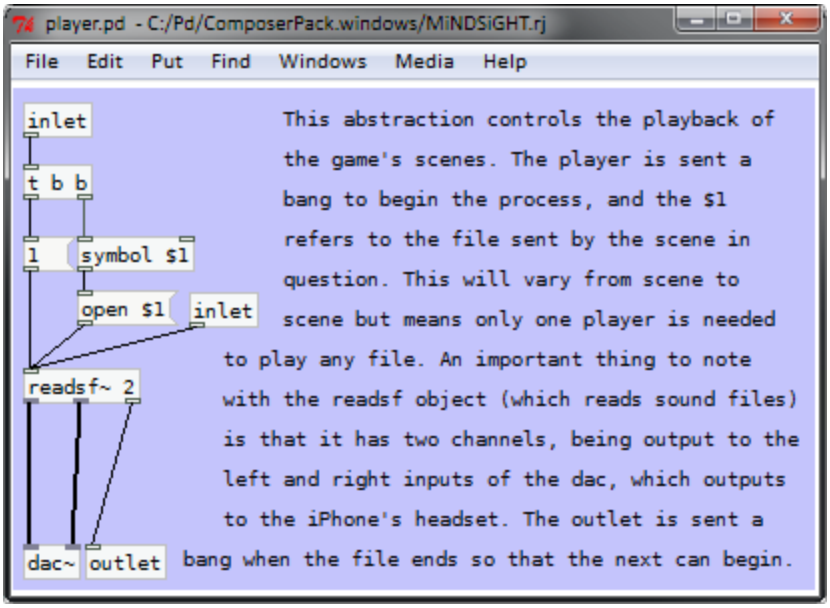


This sub-patch controls one of the game's decision-based puzzles. In addition to the detailed comments provided in the patch itself, it is important to note that the process takes one input, but provides two outputs. This means that one of two scenes could play next, depending on the decision made by the user.





These two sub-patches control an iterative puzzle. The puzzle is set up by the light blue patch at the top of the screen and the puzzle itself is controlled (and explained) in the lavender patch at the bottom.



This abstraction controls the playing back of the game's audio files, and sending the played audio to the iPhone's headset. A file is specified in the patch from where this abstraction is called, which is referred to in-patch as the **$1** variable. This file is opened and sent to the read sound file object, which is outputted with two separate channels, one left and one right, which are sent to the left and right inputs of the **dac~**, which plays back the audio on the iPhone.

The output to the right of the **dac~** sends a bang out of the player when the audio file ends. Since each file is a different length, this method means that the scenes are able to flow into one another regardless of their length, with the next scene being instructed to begin when this output is activated.

Digital Documentation

Included with this research project is a DVD-ROM containing the game itself, including all assets, and other errata and miscellanea relating to the development of MiNDSiGHT. This brief appendix explains the contents of each of the folders on the disc, and how to view the contents of MiNDSiGHT without needing to play the game on an iPhone.

- **Unteractive Playthrough** – this folder contains a playlist of audio files which play sequentially through MiNDSiGHT automatically, with no user input required at all. Please note that this option does not play any of the scenes where a puzzle is not completed successfully, and these files are found within a separate folder within this one called **Failed Puzzle Attempts**.
- **PureData Patches** – this folder contains the PureData patch, and all relevant sound files and libraries. This folder in its entirety is uploaded to the iPhone at which point it becomes playable in RjDj – no modification to this directory is required, and these exact files placed onto an iPhone will create a fully playable version of the game on the device. In order to view the code, please begin with **_main.pd** (which contains the vast majority of the code) and also be sure to examine all of the other patches in this directory.
- **Audio Files** – this folder contains labelled audio files for every scene of MiNDSiGHT, including every possibility at every puzzle.
- **Pictures** – this folder contains various photographs taken during MiNDSiGHT recording sessions. These photographs are not labelled.

·Appendix D·

Bibliography

Alexa (2009) *gamefaqs.com Site Info*. Available at: http://www.alexa.com/siteinfo/gamefaqs.com (Accessed 22 October 2009).

Anderson, J (2009) *Aves* [Digital Download]. Apple iPhone. Jeffrey Anderson. Available at: http://toucharcade.com/link/http://phobos.apple.com/WebObjects/MZStore.woa/wa/viewSoftware?id=321295493&mt=8 (Accessed 22 October 2009).

Apple (2009) *Apple's App Store Downloads Top Two Billion*. Available at: http://www.apple.com/pr/library/2009/09/28appstore.html (Accessed 22 October 2009).

Aulia, M (2009) *Cyber Snipa SONAR 5.1 Headset Review*. Available at: http://www.michaelaulia.com/blogs/cyber-snipa-sonar-5-1-headset-review.html (Accessed 13 September 2009).

Baken, R and **Orlikoff, R** (1987) *Clinical Measurement of Speech and Voice*. New York: Singular Publishing.

Baskent, D, and **Edwards, B** (2007) 'Simulating listener errors in using genetic algorithms for perceptual optimization' *Journal of the Acoustical Society of America*. Volume 121.

Ben Shaul, N (2008) *Hyper-Narrative Interactive Cinema: Problems and Solutions*. The Netherlands: Rodopi.

Blacker, I (1987) *The elements of screenwriting: a guide for film and television writers*. North Carolina: Scribner Publishers.

Blauert, J (1997) *Spatial Hearing: The Psychophysics of Human Sound Localization*. Massachusetts: MIT Press.

Bluth, D (1983) *Dragon's Lair* [Laserdisc]. Arcade. El Cajon, California: Cinematronics, Inc.

BulkyPix (2009) *Hysteria* [Digital Download]. Apple iPhone. Velizy, France: Vincent Dondaine. Available at: http://148apps.com/app/305911637/hide (Accessed 22 October 2009).

Charman, L (2008) 'The Major Fall and the Minor Lift', *SAE Institute Audio Engineering Diploma Written Assignment*. Unpublished.

CIPIC (2005) *The CIPIC Interface Laboratory Home Page*. Available at: http://interface.cipic.ucdavis.edu/ (Accessed 16 October 1009).

Crook, T (1998) *Principles of Writing Radio Drama*. Available at: http://www.irdp.co.uk/scripts.htm (Accessed 14 September 2009).

Crook, **T** (1999) *British Radio Drama*. Available at: http://www.irdp.co.uk/britrad.htm (Accessed: 12 September 2009).

Duda, **R** (2000) *An Introduction to Human Spatial Hearing*. Available at: http://www.docstoc.com/docs/2686978/AN-INTRODUCTION-TO-HUMAN-SPATIAL-HEARING-I-II-III (Accessed 10 October 2009).

Eckersley, **PP** (1941) *The Power Behind the Microphone*. London: Scientific Book Club.

Gamespot.com (2005) *PlayStation 3 announced for 2006*. Available at: http://uk.gamespot.com/news/2005/05/16/news_6124681.html (Accessed: 11 September 2009).

Gelfand, **S** (2009) *Essentials of Audiology*. New York: Thieme Publishers.

Hendler, **T** (2009) *Scary Music Is Scarier with Your Eyes Shut*. Available at: http://www.aftau.org/site/News2/1364361394?page=NewsArticle&id=10495&news_iv_ctrl=-1 (Accessed 20 September 2009).

Hess, **L** (1971) *The praying mantis: insect cannibal*. North Carolina: Scribner Publishers.

Hill, **P** (1997) *Front back confusion in systems for the production of virtual acoustic images*. Unpublished PhD thesis. University of Southampton.

Hofman, **P**, and **Van Opstal**, **J** (1997) 'Spectro-temporal factors in two-dimensional human sound localization', *Acoustical Society of America Digital Library*, Volume 130, Issue 5, pp. 2634-2648 [Online] Available at: http://scitation.aip.org/getabs/servlet/GetabsServlet?prog=normal&id=JASMAN000103000005002634000001&idtype=cvips&gifs=yes (Accessed 10 October 2009).

Hofman, **P**, **Van Riswick**, **G**, and **Van Opstal**, **J** (1998) 'Relearning Sound Localization with New Ears' *Nature Neuroscience* [Online] Available at: http://www.cns.nyu.edu/events/earclub/artfiles/Hofman1998.pdf (Accessed 9 October 2009).

Holson, **T** (2009) *TMH Labs Research*. Available at: http://www.tmhlabs.com/research/research.html (Accessed 13 October 2009).

Jesteadt, **W** (1997) *Modeling Sensorineural Hearing Loss*. Philadelphia: Lawrence Erlbaum Associates, Inc.

Knudsen, **E**, and **Konishi**, **M** (1979) 'Mechanisms of Sound Localization in the Barn Owl', *Journal of Comparative Psychology*, Issue 133, pp. 13-21 [Online] Available at:

http://www.springerlink.com/content/rx4217m58njn1733/ (Accessed 12 October 2009).

Lawson, **C** (2003) *The Cambridge Companion to the Orchestra*. Cambridge: Cambridge University Press.

Leung, **N**, **Braasch**, **J**, and **So**, **R** (2006) 'Optimizing the cost and performance of non-individualized binaural cues for surround sound simulations', *Physics and Astronomy Classification Scheme*, 43.66.PN [Online]. Available at: http://www.sea-acustica.es/Sevilla02/elegen006.pdf (Accessed 22 September 2009).

Lodge, **S** (2007) *ChooseCo Embarks on Its Own Adventure*. Available at: http://www.publishersweekly.com/article/CA6408126.html (Accessed: 12 September 2009).

Marks, **A**, and **Novak**, **J** (2009) *Game Development Essentials: Game Audio Development*. New York: Delmar / Cengage Learning.

Microphone-data.com (2009) *Neumann: KU-100*. Available at: http://www.microphone-data.com/microphone.asp?mic=528 (Accessed 18 October 2009).

Middlebrooks, **J**, and **Green**, **D** (1991) 'Sound Localization by Human Listeners' *Annual Review of Psychology*, Volume 42, pp. 135-159 [Online] Available at: http://arjournals.annualreviews.org/doi/abs/10.1146%2Fannurev.ps.42.020191.001031 (Accessed 13 October 2009).

Møller, **A** (2006) *Hearing: anatomy, physiology, and disorders of the auditory system*. Dallas, Texas: Envisage Academic Press,

Myers, **P** (1989) *Three-dimensional auditory display apparatus and method utilizing enhanced bionic emulation of human binaural sound localization*. US Patent Office no. 4817149 [Online]. Available at: http://www.patentstorm.us/patents/4817149.html (Accessed 22 September 2009).

Newsweek (1983) 'Dragon's Lair' 8 August, 1983.

Ngan, **K** (2005) *Optimizing front/back confusion rates in sound localization performance: cluster analyses and experimental studies*. Hong Kong: Hong Kong University of Science and Technology.

Palermo, **T** (2005a) *Scoring for Audio Theatre – Part 1*. Available at: http://ruyasonic.com/mus_score_1.htm (Accessed 27 September 2009).

Palermo, **T** (2005b) *How To Write Radio Drama Cues*. Available at: http://ruyasonic.com/wrt_cues.htm (Accessed 20 October 2009).

Perfectly Normal Productions (2009) The Adventures of Sexton Blake, BBC Radio 2, July 31.

Prodi, **N** and **Velecka**, **S** (2002) 'The evaluation of binaural playback systems for virtual sound fields' *Applied Acoustics*, Volume 64, pp. 147-161 [Online] Available at: http://acustica.ing.unife.it/staff/nicola/articoli%20NP/App%20Acoustic%20prodi%20velecka.pdf (Accessed 17 October 2009).

RjDj.me (2008) *RjDj: A Mind twisting hearing sensation*. Available at: http://www.rjdj.me (Accessed 21 September 2009).

Ross, **M** (2006) 'Are Binaural Hearing Aids Better?' *The Hearing Journal* Issue: March/April 2006 [Online] Available at: http://www.hearingresearch.org/Dr.Ross/Binaural-Better.htm (Accessed 19 October 2009).

Ruzicka, **J** (1999) *Starkey Labs, Inc. Unveils New Technology*. Available at: http://www.audiologyonline.com/news/news_detail.asp?news_id=6 (Accessed 18 October 2009).

Saitek (2009) *Cyborg 5.1 Headset*. Available at: http://www.saitek.com/uk/prod/cyborg5.htm (Accessed: 13 September 2009).

Sandel, **T**, **Teas**, **D**, **Feddersen**, **W**, and **Jeffrees**, **L** (1955) 'Localization of Sound from Single and Paired Sources' *Journal of the Acoustical Society of America*. Volume 27, pp. 842-852 [Online] Available at: http://scitation.aip.org/getabs/servlet/GetabsServlet?prog=normal&id=JASMAN0000270000500084200001&idtype=cvips&gifs=yes (Accessed 14 October 2009).

Searle, **C**, **Braida**, **L**, **Cuddy**, **D**, and **Davis**, **M** (1975) 'Binaural pinna disparity: another auditory localization cue' *Journal of the Acoustical Society of America*, Volume 57, pp. 448-455.

Searle, **C**, **Braida**, **L**, **Davis**, **M**, and **Colburn**, **H** (1976) 'Model for Auditory Localization' *Journal of the Acoustical Society of America*. Volume 60, pp. 1164-1175 [Online] Available at: http://scitation.aip.org/getabs/servlet/GetabsServlet?prog=normal&id=JASMAN0000600000500116400001&idtype=cvips&gifs=yes (Accessed 12 October 2009)

Siewert, **A** (2003) *Drama Team Handbook*. Illinois: InterVarsity Press.

Stecker, **R** (2005) *Aesthetics and the Philosophy of Art: An Introduction*. Boulder, Colorado: Rowman & Littlefield Publishers, Inc.

Stern, **D** (2004) *(5c) Coordinates*. Available at: http://www.iki.rssi.ru/mirrors/stern/stargaze/Scelcoor.htm (Accessed 10 October 2009).

Stevens, **S**, and **Newman**, **E** (1936) 'The localization of actual sources of sound' *American Journal of Psychology*, Volume 48, pp.297-306.

Stolfi, **J** (2009) Illustration of the Spherical coordinate system for 3D. Available at: http://en.wikipedia.org/wiki/File:Coord_system_SZ_0.svg (Accessed 10 October 2009).

Starkey Hearing Research Center (2004-2008) *Publications* [Online] Available at: http://starkeyresearch.com/our-research/publications.jsp (Accessed 19 October 2009).

Starkey Labs (2006) *nFusion Technology*. [Leaflet obtained through Starkey directly], 12 October 2009.

Strutt, **J** (1907) 'On our perception of sound direction' *Philosophy Magazine*, Volume 13, pp. 213-232.

TeamXbox.com (2005) *Xbox 360 Officially Announced*. Available at: http://news.teamxbox.com/xbox/8233/Xbox-360-Officially-Announced/ (Accessed: 11 September 2009).

Toonz.ca (2008) *The Bose L1 Encyclopedia FAQ*. Available at: http://toonz.ca/bose/wiki/images/7/7c/EqualLoudness400px-Lindos1.svg.png (Accessed 18 October 2009).

vescoFx (2009) *Hass Stereo Enhancing Delay Plugin*. Available at: http://www.vescofx.com/vfxHaas (Accessed 5 September 2009).

Young, **N** (2008) *iPerbole – An App Store history in quotes*. Available at: http://www.develop-online.net/features/546/iPerbole-An-App-Store-history-in-quotes (Accessed 22 September 2009).

YouTube.com (2009) *YouTube search results for 'Virtual Barbershop Reactions'*. Available at: http://www.youtube.com/results?search_query=virtual+barbershop+reactions&search_filter=0 (Accessed 22 October 2009).